Oil Shipping and Pirates *

—The Macroeconomic Effects of Oil Tanker Attacks—

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Abstract

This paper studies the macroeconomic effects of attacks on oil tankers. We identify oil-shipping-cost shocks by combining a narrative record of tanker attacks with high-frequency changes in tanker freight rates. We focus on attacks along key shipping routes in the Middle East and Southeast Asia. Focusing on the period since the onset of the U.S. shale oil boom, we find that these shocks lower economic activity and raise consumer prices in both the United States and the world economy. Crude production and oil prices show no significant immediate response, while inventories decline. These responses differ from those associated with conventional oil supply shocks and other shipping shocks. U.S. evidence points to an energy-trade channel, with increased congestion at Middle Eastern ports, lower crude imports and refinery inputs, leading to higher fuel prices. We also document that manufacturing orders and shipments fall immediately, consistent with precautionary adjustments in firms' purchases and investment plans. These findings, together with the contraction in global activity, suggest that oil-shipping disruptions can hurt both oil-importing and oil-exporting economies.

Keywords: Oil shipping, piracy, high-frequency identification, proxy SVAR, energy security

JEL Codes: E31, E32, F51, Q41, Q43

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1 Introduction

A large literature studies the macroeconomic effects of oil shocks (Hamilton, 2003; Kilian, 2009; Baumeister and Hamilton, 2019; Känzig, 2021), but less is known about disruptions to oil transportation between producing and consuming regions. Oil can be moved by land (via trucks and pipelines) and by sea (via tankers), which constitutes its main mode of transportation. For instance, in 2013, approximately 63% of global oil production moved by sea (U.S. Energy Information Administration, 2014). Compared to land transportation, these shipments are more vulnerable to piracy and several characteristics make them prime targets, as extensively discussed in Hacaga (2020). First, many transit through narrow chokepoints, notably the Strait of Hormuz, Bab el-Mandeb and the Strait of Malacca, where traffic is concentrated and opportunities to avoid attacks are limited. Second, their large size, limited and slow maneuverability, as well as their lack of defensive capabilities, make them easier to attack. Finally, the large volumes of oil they carry make tankers valuable targets for ransom demands and economic pressure through threats to oil deliveries.

Attacks on oil tankers make deliveries more costly and less reliable. This in turn can affect refinery activity and petroleum-product prices, with consequences for firms and households. Interestingly, these effects can arise even when oil production remains unchanged and world oil prices do not move. Domestic oil production can offset these disruptions only if it is able to replace the imported crude. However, crude oil differ in density and sulfur content, and refineries are usually configured to process particular grades, limiting the scope for that substitution. In the United States for instance, rising production of light crude has coexisted with continued imports of heavier grades on which refineries run (U.S. Energy Information Administration, 2022).

In this paper, we study the effects of attacks on oil tankers on the U.S. economy. We compile a narrative record of attacks on tankers transporting crude oil and heavy fuel oil. For each attack, we measure high-frequency variations in the cost of shipping this oil, proxied by the Baltic Dirty Tanker Index (BDTI). The idea is to allow time for attacks to be reported while limiting exposure to unrelated news or events in order to support exogeneity. Additionally, we proceed to a restrictive selection and exclude incidents coinciding with major oil-market or geopolitical developments. Relevance rests on the fact that tanker attacks affect freight rates through delays, diversions and increased shipping risk as discussed earlier. We aggregate the resulting surprises to the monthly frequency and use them as an external instrument for oil-shipping-cost shocks in a structural VAR for the United States, estimated over July 2009–February 2026 (Stock and Watson, 2012; Mertens and Ravn, 2013; Stock and Watson, 2018).

We find that attacks on oil tankers have sensible macroeconomic effects, although their magnitude is smaller than that of conventional oil shocks. Our identified shock, normalized to raise the BDTI by 10 percent on impact, lowers industrial production by approximately 0.32 percent immediately and raises consumer prices by 0.1 percent after five months. These effects emerge before any statistically significant increase in the benchmark oil price, whose response becomes significantly positive only at longer horizons. U.S. crude inventories decline persistently, reaching approximately 0.5 percent below baseline after five months. Oil-related geopolitical risk rises

on impact, but the increase immediately dissipates. We perform a battery of tests to assess the robustness of our results to instrument construction, event selection and model specification.

Importantly, our results differ from those associated with several established shocks in the literature. First, they contrast with the expansionary U.S. response to adverse oil-supply-news shocks documented by Känzig et al. (2026) for the sample we consider (i.e., the post-2010 period). Moreover, the delayed WTI response and declining U.S. crude inventories differ from the immediate oil-price increase and accumulation of global inventories following adverse oil-supply news (Känzig, 2021; Känzig et al., 2026). Second, our results contrast with the growing literature on geopolitical oil shocks, notably Verduzco-Bustos and Zanetti (2026) and Pinchetti (2026), who use geopolitical-risk indexes (Caldara and Iacoviello, 2022; Iacoviello and Tong, 2026) to study the transmission of geopolitical shocks through energy markets. We investigate a narrower but more precise source of disturbances, namely attacks on oil tankers, and exclude major geopolitical developments (e.g. wars) that these studies examine. Moreover, we find little effect on world crude production and a delayed increase in oil prices, in contrast to the production decline and sharp price increase documented by Verduzco-Bustos and Zanetti (2026). Our persistent inventory decline also differs from their initial drawdown followed by an accumulation. Third, our findings highlight the distinction between disruptions to oil transportation and broader shipping shocks. Känzig and Raghavan (2026) construct a dry-bulk shipping instrument using disruptions (groundings, collisions) to the Suez and Panama canals. Unlike theirs, our shock is not directly a congestion shock as congestion can be only a later consequence of attacks on tankers in our setting. Moreover, and consistent with their focus on a different freight market, their instrument has little explanatory power for BDTI innovations over the common sample. Finally, we assess whether broader maritime insecurity produces similar effects by constructing a complementary instrument using attacks on other vessel types. These attacks also raise tanker freight rates, but their estimated oil-market and macroeconomic effects are muted and generally statistically insignificant.

We investigate transmission through energy trade and manufacturing adjustment. Increased congestion at Middle Eastern ports is accompanied by declining crude imports from Saudi Arabia and lower U.S. refinery inputs, followed by rising fuel and consumer energy prices. Regional evidence supports this energy-trade channel, as U.S. districts more dependent on Persian Gulf crude experience larger increases in diesel and gasoline prices. Manufacturing orders and shipments fall immediately, consistent with a precautionary adjustment in firms' purchases and investment plans. Finally, we assess the external validity and quantitative importance of our findings using a VAR with global variables. We show that the contraction in industrial activity and rise in consumer prices extend beyond the United States.

Literature: First, our paper contributes to the literature studying the macroeconomic effects of oil shocks (Kilian, 2009; Kilian and Murphy, 2014; Baumeister and Hamilton, 2019; Känzig, 2021). We contribute by examining oil transportation, an understudied aspect of oil markets. We show that shocks originating in transportation produce a different pattern of responses from conventional oil shocks, but still have effects on the economy.

Second, our paper contributes to a growing literature on the economic effects of shipping and supply-chain disruptions (Carrière-Swallow et al., 2023; Bini, 2025; Känzig and Raghavan, 2026). To identify these disruptions, we draw on high-frequency methods developed in monetary economics and subsequently applied to energy and carbon markets (Nakamura and Steinsson, 2018; Känzig, 2021; Känzig, Forthcoming; Känzig et al., 2026; Colombo and Toni, 2026). The closest methodological comparison is Känzig and Raghavan (2026), who combine a narrative record of disruptions to Suez and Panama Canal traffic with high-frequency movements in the Baltic Dry Index. Their analysis focuses on disruptions to canal traffic, whereas we examine attacks on oil tankers that can raise transportation costs and risks even without congestion or restrictions on passage. Our analysis hence differs in its geographic coverage, the types of vessels we are looking at, as well as the macroeconomic outcomes and transmission channels examined. These differences allow us to address a separate question and to examine different channels. Our analysis also relates to Anderl and Nava (2026) who use state-dependent panel local projections to show that oil shocks have stronger inflationary effects when tanker shipping costs are high. We complement their analysis by identifying an example of disruptions originating in oil transportation and tracing their effects and transmission channels. We, however, do not focus on oil shocks.

Third, our paper contributes to research on the economic consequences of piracy and maritime insecurity. Besley et al. (2015) exploit variation in piracy intensity and geographic exposure and find that Somali piracy increased shipping costs on exposed routes by approximately 8 to 12 percent. Likewise, Burlando et al. (2015) estimate a 4.1 percent annual reduction in bulk-commodity trade through the Gulf of Aden over 2000–2010 due to piracy. These studes show that piracy have non-negligeable effects. We complement them by developing a new instrument for oil-shipping-cost shocks and by using a time-series framework to estimate their dynamic macroeconomic effects. Our event record also covers a longer and more recent period (2010–2024) and a broader set of oil-shipping corridors. More recent work examines how geopolitical tensions affect shipping networks. Bai et al. (2026) study disruptions to U.S.-bound container shipping. In one of their applications, they study the 2023-2024 Houthi attacks. We complement this study by focusing specifically on oil tanker attacks and oil markets and by using high-frequency causal identification.

Finally, our paper contributes to research on the macroeconomic effects of geopolitical tensions. This literature empirically relies on measures og geopolitical risk using news-based indexes put forward by Caldara and Iacoviello (2022) and Caldara et al. (2026). For instance, Verduzco-Bustos and Zanetti (2026) and Pinchetti (2026) use high-frequency information to identify geopolitical shocks affecting energy markets and study their macroeconomic transmission. Our focus on oil transportation highlights the role of non-state actors, such as the Houthis or pirates, whose attacks on tankers can disrupt trade and have global macroeconomic effects. Our findings also connect to the geoeconomics literature on international economic dependence (Clayton et al., 2026). In the case of the United States, we show how reliance on imported crude and maritime deliveries can expose an economy to disruptions abroad even when domestic oil production is substantial.

The remainder of the paper is organized as follows. Section 2 describes U.S. dependence on imported crude, the maritime routes supplying its refineries and the tanker freight market. Sec-

tion 3 presents the narrative record of tanker attacks and constructs the shipping-cost instrument. Section 4 sets out the econometric framework. Section 5 presents the U.S. macroeconomic effects, examines transmission channels and extends the analysis to global variables and assesses their quantitative importance. Section 6 reports the other-vessel placebo and robustness exercises. Section 7 concludes.

2 Oil Shipping and the United States

In this section, we discuss why U.S. refineries continue to import crude oil despite the United States being a net exporter of petroleum. We then show where these imports originate and how they reach the United States. We finally introduce the tanker freight market and the price index used in our identification strategy.

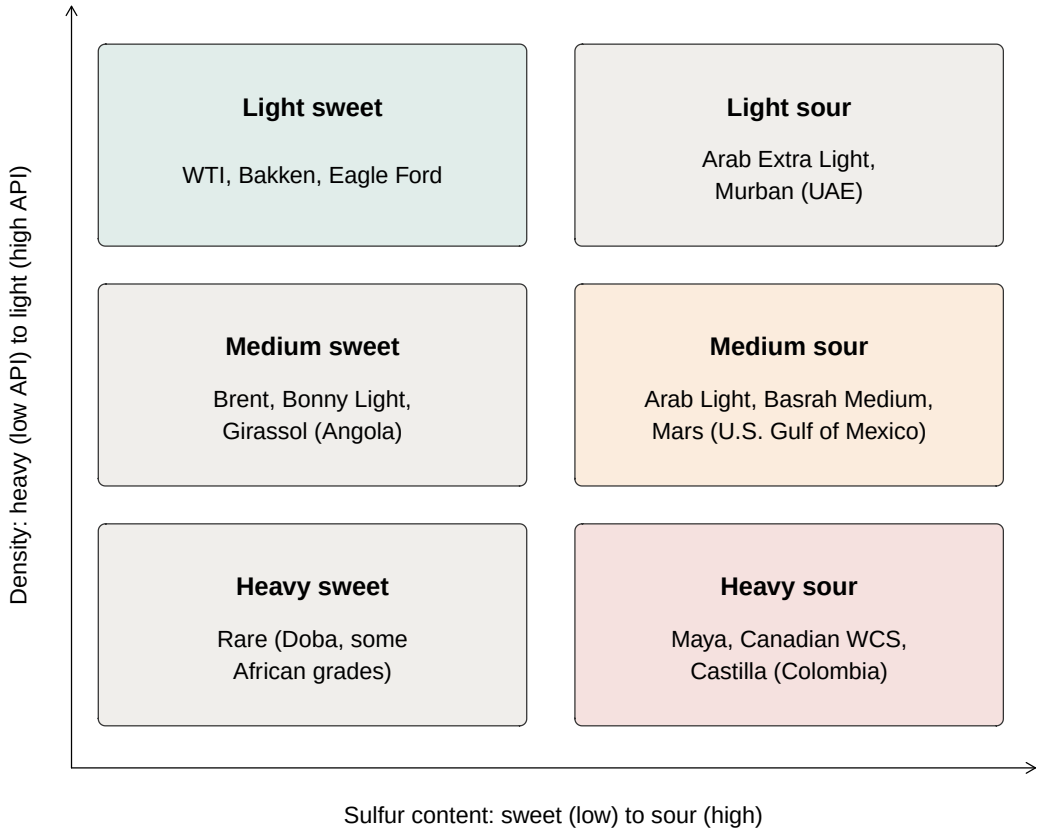
2.1 U.S. Refineries and Crude Oil Imports

Crude oil is far from being a homogeneous good. Indeed, as illustrated schematically in Figure 1, crude oil is in fact classified into different grades according to its density and sulfur content. It ranges from light to heavy according to its density and is described as sweet or sour depending on whether its sulfur content is low or high. These differences matter because refineries are configured to process particular mixtures of grades. For instance, many U.S. refineries have invested in equipment to process medium and heavy sour crude profitably (U.S. Energy Information Administration, 2019, 2023). However, panel (a) of Figure 2 shows that the United States produces mainly light crude oil. However, because refinery equipment is costly and slow to modify, refineries cannot quickly and cheaply adapt their equipment to process light crude oil efficiently. The United States therefore continues to import medium and heavy crude, which account for most of its crude imports, as shown in panel (b) of Figure 2. This mismatch helps explain why the country remains a crude oil importer despite being a net exporter of refined petroleum products. In 2025, its crude imports averaged 6.2 million barrels per day (U.S. Energy Information Administration, 2026c).

Moreover, grades have a geographic dimension as different regions supply different types of crude. For medium sour crude, U.S. refineries have relied primarily on the Middle East Gulf. Over 2009–2024, this grade accounted for about one sixth of total U.S. crude imports, with roughly three quarters supplied by the Gulf (Figure 3). This dependence persisted over time as the Gulf supplied more than half of U.S. medium sour imports in every year from 2009 through 2022 and remained an important source despite its declining share in 2023–2024 (Appendix Figure A1).[1] Heavy sour imports mainly come from Canada, Mexico, Venezuela and Colombia.

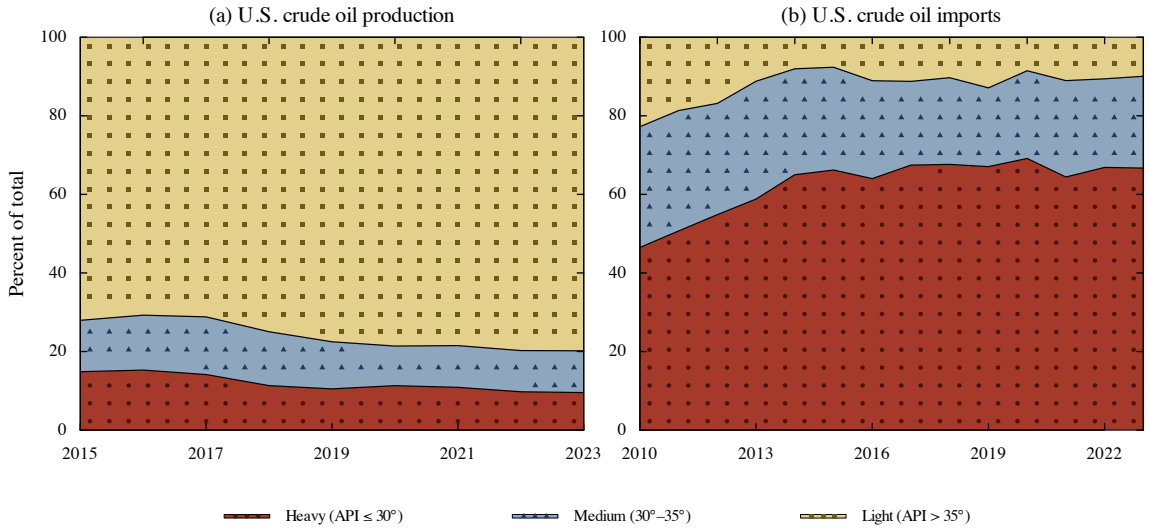
[1]These supplier relationships can also involve vertical integration. Saudi Aramco has wholly owned Motiva, which operates the Port Arthur refinery in Texas, since 2017, illustrating the ownership links between foreign oil producers and U.S. refining capacity.

Figure 1: Crude oil categories by density and sulfur content



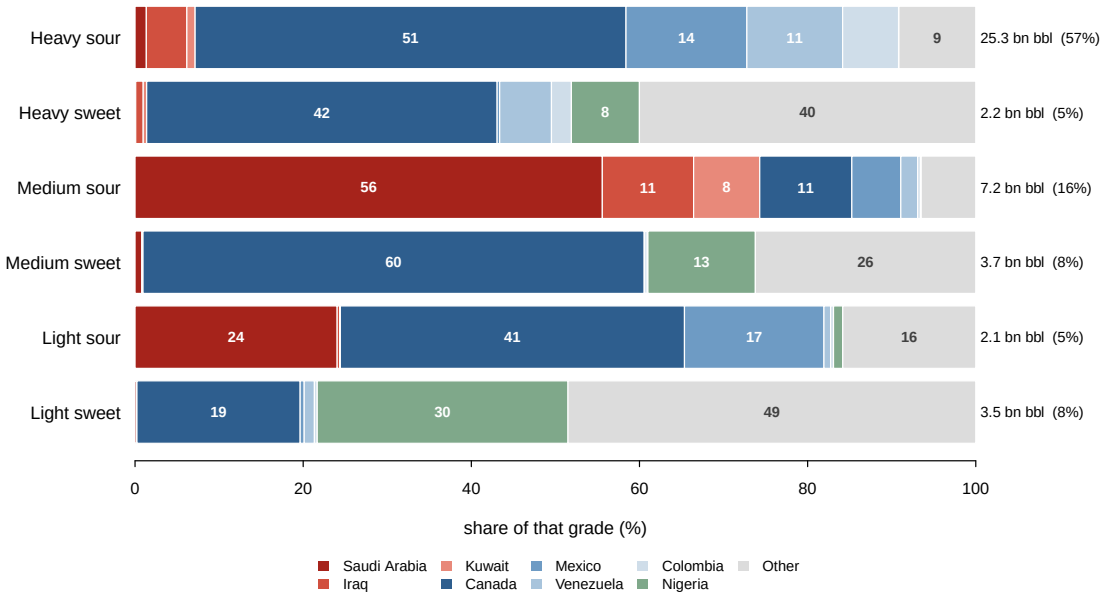
Notes: Schematic classification with illustrative crude grades. Higher API gravity denotes lower density; sweet and sour denote lower and higher sulfur content, respectively.

Figure 2: The composition of U.S. crude oil production and imports



Notes: Panel (a) reports the composition of crude oil and lease condensate production in the Lower 48 states by API gravity class since 2015. Panel (b) reports the composition of U.S. crude oil imports by API gravity class since 2010. Heavy: API $\leq$ 30°; medium: 30°–35°; light: API $>$ 35°. Sources: EIA production by API gravity (U.S. Energy Information Administration, 2026b) and imported-crude shares by API gravity (U.S. Energy Information Administration, 2026d).

Figure 3: U.S. crude oil imports by grade and country of origin, 2009–2024



Notes: Each bar reports the share of cumulative U.S. crude oil imports of a given grade supplied by each origin over 2009–2024. The label to the right of each bar reports the cumulative volume of that grade in billion barrels and its share in total imports. Grades are defined by API gravity (light: $>$ 35°; medium: 30°–35°; heavy: $<$ 30°) and sulfur content (sweet: $<$ 0.5 percent; sour: $\geq$ 0.5 percent). Source: EIA Form EIA-814, company-level imports (U.S. Energy Information Administration, 2026a).

2.2 The Market for Oil Tanker Freight

The geographic origin of these grades matters because it determines how they reach U.S. refineries. Canadian supplies arrive mainly overland, whereas imports from the Middle East Gulf and Latin American suppliers rely on maritime transportation, as illustrated in Figure 4. These seaborne deliveries depend on a global tanker fleet numbering more than 7,000 vessels in 2023, which differ in size and in the type of oil they transport. We focus on vessels carrying crude oil and heavy fuel oil, often referred to as "dirty" tankers.[2] Figure 5 illustrates the main vessel size classes and their capacity. Crude oil moves mainly on Aframax, Suezmax and Very Large Crude Carriers (VLCCs) and consequently, our analysis focuses primarily on these three classes. A typical VLCC is about 330 metres long, roughly the length of three football pitches, and can carry around two million barrels of crude oil (Bureau of Transportation Statistics, 2018). Vessel size also limits access to ports and waterways. For example, fully loaded large tankers may need to unload part of their oil before transiting the Suez Canal to meet its draft restrictions.

The price of transporting oil on these vessels varies across routes and is negotiated through brokers for each voyage in the spot market. The Baltic Exchange publishes daily assessments of these freight rates for major tanker routes worldwide, shown in Figure 6. The Baltic Exchange also combines route-level freight rates into indices for different markets: the Baltic Clean Tanker Index (BCTI) for lighter refined petroleum products, the Baltic Dry Index (BDI) for dry-bulk cargo, and the Baltic Dirty Tanker Index (BDTI) for crude oil and heavy fuel oil. We focus on the BDTI, which is constructed as a scaled weighted sum of route assessments,

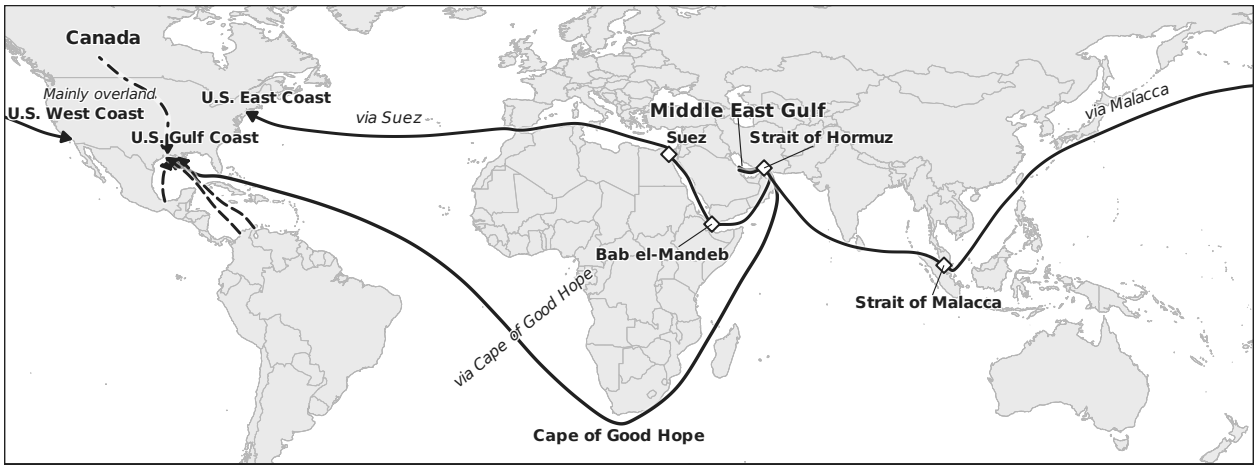
$$\text{BDTI}_t = c_t \sum_{r \in \mathcal{R}_t} w_{r,t} \, \text{WS}_{r,t}, \qquad \sum_{r \in \mathcal{R}_t} w_{r,t} = 1, \tag{1}$$

where $\text{WS}_{r,t}$ is the Worldscale freight-rate assessment for route r, $\mathcal{R}_t$ is the set of constituent routes, and c_t is the Baltic Exchange's scaling multiplier. Solid lines in Figure 6 identify routes included in the BDTI, while dashed lines indicate other assessed routes.

The geographic coverage of the BDTI raises the question of how well it captures disruptions along the corridors we study, mainly the straits of Hormuz, Bab el-Mandeb and Malacca. The index includes routes outside these corridors, such as TD28 within Europe, but excludes TD23, which connects the Middle East Gulf to the Mediterranean through Bab el-Mandeb and the Suez Canal. Section 3 shows that around attacks, BDTI changes correlate more strongly with freight-rate changes on exposed routes than on distant routesnthan on geographically distant routes. This finding also holds for TD23, which crosses the affected corridors but is not included in the BDTI. This geographic pattern supports interpreting changes in the index around attacks as reflecting conditions along the affected corridors. Moreover, by selecting attacks reported by Reuters and Lloyd's List, we aim to retain major incidents that are powerful enough to drive the BDTI, even if they occur on routes outside the index.

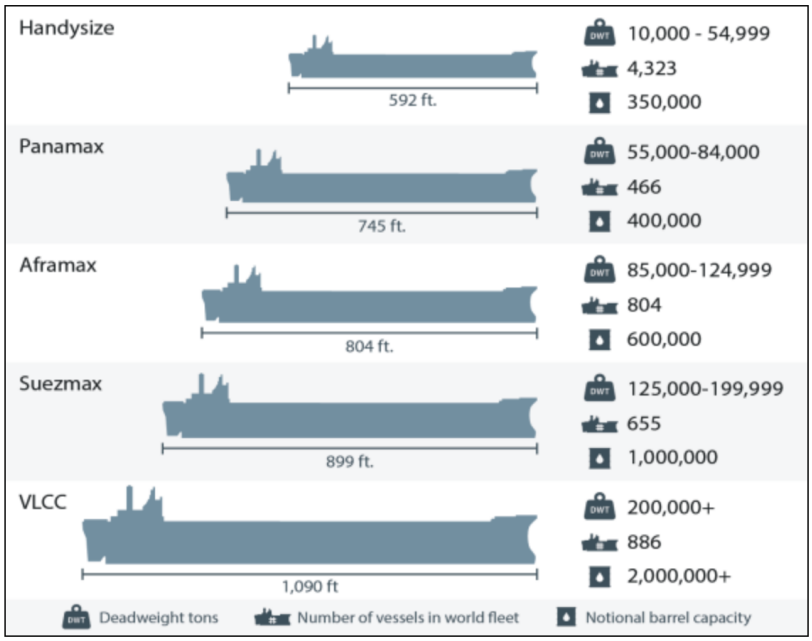
[2] "Clean" tankers carry lighter refined petroleum products and are generally smaller.

Figure 4: Selected U.S. crude-import routes by grade



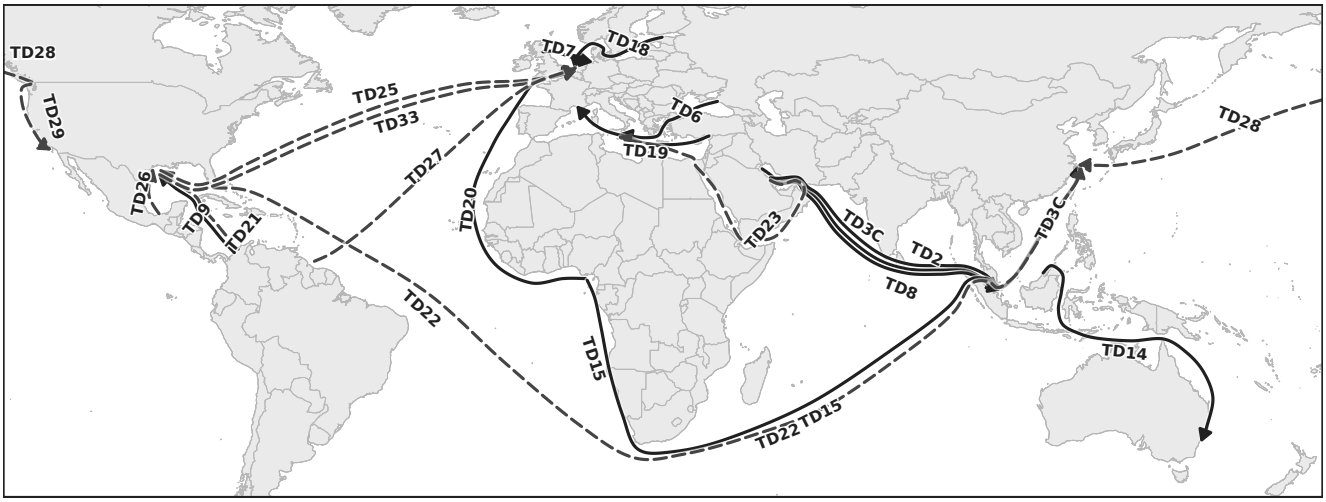
Notes: Representative routes for selected U.S. crude imports. Solid lines show medium sour shipments from the Middle East Gulf; dashed lines show heavy sour shipments from Mexico, Venezuela and Colombia. The dash-dotted line represents Canadian overland supply schematically, rather than an individual pipeline. Diamonds mark selected maritime chokepoints. Countries also supply other grades, and Canadian imports also include seaborne shipments. Routes and destinations are illustrative, not exhaustive; line widths do not represent volumes.

Figure 5: Oil tanker size ranges: world fleet as of 2023



Notes: For each size class, the figure reports the deadweight tonnage (dwt), the number of vessels in the world fleet and the notional barrel capacity, together with the typical vessel length. VLCC = very large crude carrier; Afra = average freight rate assessment. Because of different densities, crude oil and refined products yield different barrel capacities. A deadweight ton (dwt) measures the total weight a vessel can carry, including cargo, fuel, ballast water and stores. Source: Clarksons Tanker Register, 2023.

Figure 6: The routes behind the Baltic Dirty Tanker Index



Notes: Benchmark tanker routes assessed by the Baltic Exchange. Solid lines indicate routes included in the BDTI; dashed lines indicate other assessed routes. Labels identify the Baltic route codes. Routes are drawn schematically.

2.3 From Tanker Attacks to Economic Activity

Attacks on oil tankers matter because they can disrupt oil transportation through physical damage and increased risk, and both mechanisms can translate into higher freight rates. First, physical damage can interrupt a tanker's deliveries. Damaged vessels may remain at sea for extended periods, posing hazards to other ships. For instance, the tanker *Sounion* was still burning in the Red Sea in September 2024, several weeks after it was attacked in August (Figure 7). Moreover, potential oil spills and emergency operations can further disrupt surrounding maritime traffic, leading to delays and congestion (Bai et al., 2026). Second, attacks can increase the perceived risk of further incidents, raising insurance premiums and freight rates. As a consequence of this higher risk, shipping companies can also postpone voyages, divert vessels or temporarily avoid exposed passages. Deliveries can therefore become slower and less reliable even when shipping routes remain open and oil production facilities remain operational (Kilian et al., 2023).[3]

For importing economies, higher transportation costs and less reliable crude deliveries can affect both refinery activity and the cost of supplying petroleum products. Refiners can initially use their inventories to compensate for delayed and disrupted deliveries. However, if disruptions persist, their ability to maintain production will depend on alternative supplies that need to be compatible with their refinery configurations, as discussed earlier. Lower refinery output can reduce fuel availability and raise petroleum-product prices, which in turn raise costs for firms and households. Additionally, firms and households may also change their production and spending plans if they expect delivery problems, even before shortages occur.

[3]Reporting on the Strait of Hormuz in August 2026 described traffic remaining well below pre-war levels despite declarations that the strait was open, amid continuing attacks and restrictions. See, for instance, Mongilio (2026).

Figure 7: The oil tanker *Sounion* on fire in the Red Sea, September 2024



Notes: On 21 and 23 August 2024, the Greek-registered tanker *Sounion*, carrying about 150,000 tons of crude oil, was subjected to separate attacks by militants in the southern Red Sea. The photograph was taken on 14 September 2024. Source: EUNAVFOR Aspides, © European Union.

3 Identification Strategy

In this section, we explain how we construct our external instrument for oil-shipping-cost shocks. The central challenge is that the BDTI responds to several forces such as global oil demand and supply, port congestion or the availability of tanker capacity (Kilian et al., 2023). Therefore, we cannot attribute its movements to attacks on oil tankers alone. In order to isolate variation associated with tanker attacks, we combine narrative event selection with the BDTI revisions in a short window around each incident. We start by describing the procedure we use to select oil tanker attacks. We then explain how we aggregate the associated BDTI surprises to the monthly frequency and discuss their validity as an instrument for oil-shipping-cost shocks.

3.1 A Narrative Record of Attacks on Oil Tankers

Not all tanker attacks are suitable for identifying unexpected disruptions to oil transportation. Some occur during broader conflicts or coincide with news about oil production and demand, making their effects difficult to isolate. Others may be partly anticipated, particularly during an ongoing campaign. We therefore carefully screen candidate events using a narrative record of attacks reported by Reuters, particularly Reuters Shipping News, and Lloyd's List over 2010–2025. They provide financial and maritime news and are widely followed by market participants, so news of tanker attacks reported by these sources is likely to be quickly incorporated into freight rates.

We first identify candidate incidents through keyword searches and review each report individually to establish the incident date, vessel, location and available cargo information. We specifically apply the following screening:

- **Geographical selection.** We retain attacks occurring along selected oil-shipping corridors in the Middle East and Southeast Asia: the Red Sea, the straits of Hormuz and Bab el-Mandeb, the Gulf of Aden, the Arabian Sea, and the straits of Malacca and Singapore. As extensively discussed in Section 2, these are key routes for oil transportation where ships are exposed to attacks. We exclude incidents elsewhere, including within the Suez and Panama Canals, where such attacks are comparatively rare. An additional consideration for Suez is that limits on how deep vessels can sit in the water restrict the passage of fully loaded large crude carriers, as discussed in Section 2.2.

- **Vessel selection.** We target tankers transporting crude oil and heavy fuel oil, which are the cargoes most closely associated with the BDTI. We therefore exclude incidents involving identified cargoes of naphtha, gasoil, gasoline, methanol and other chemicals. We also exclude gas carriers and non-tanker vessels, whose freight markets are not represented in the BDTI. However, we do not require a vessel to be laden at the time of the attack, since an incident involving an eligible tanker can reveal risks to subsequent voyages.[4]

- **Incident selection.** We retain attacks and hijackings regardless of whether they cause physical damage, as well as attempts repelled by naval forces. These events can reveal risks to oil transportation even when the vessel experiences no damage. However, we exclude attacks on port infrastructure, as our focus is on attacks against tankers. We also exclude documented politically motivated seizures, announced rerouting and route closures. Seizures may be linked to diplomatic tensions or sanctions, while rerouting and closures may reflect risks already known to the market. It is therefore difficult to determine what new information these events provide. For the same reason, we exclude attacks conducted as part of declared interstate military operations. These exclusions reflect our deliberate focus on attacks against oil tankers, rather than a broader set of disruptions to oil transportation or geopolitical events in the Middle East. Although attacks on ports, military operations and politically motivated seizures may disrupt oil transportation, they may simultaneously alter oil supply expectations, financial conditions and the wider geopolitical landscape. Consequently, changes in the BDTI around these events will reflect several of these developments, making it harder to isolate the component associated with oil-shipping disruptions. Our narrower selection reduces this risk, although tanker attacks themselves may also affect geopolitical risk and financial markets.

- **Timing and competing news.** Finally, we exclude incidents that occur close to major oil-market, macroeconomic or geopolitical developments. We remove attacks occurring within three days of an OPEC announcement, because movements in the BDTI could reflect news

[4]Our sample includes one documented instance of a tanker that was not carrying oil at the time of the attack.

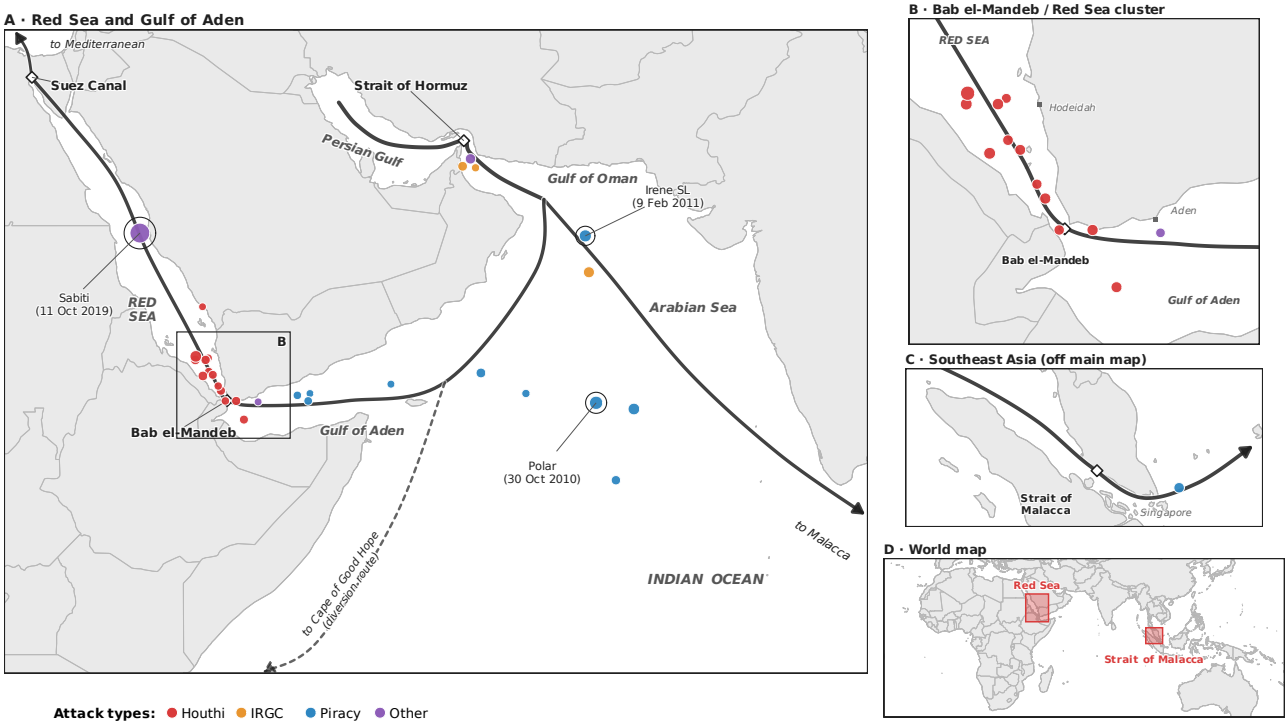
about oil supply. We also exclude events occurring around the onset of the Libyan civil war in 2011, Russia's annexation of Crimea in March 2014, Russia's full-scale invasion of Ukraine in February 2022, and the initial phase of the COVID-19 pandemic in March–May 2020. These exclusions reduce the risk of attributing movements in the BDTI caused by other shocks to attacks on oil tankers.

We use a large language model (LLM) to conduct additional searches on the events that passed our selection criteria. In particular, the LLM helped us gather information from external sources to verify each event's eligibility under the criteria presented above.

The final record contains a total of 30 reported oil tanker incidents between 2010 and 2024. Figure 8 shows their geographic concentration along the major shipping corridors mentioned earlier. The number of retained events reflects our rather restrictive selection criteria, which inevitably reduce the number of attacks we keep, as well as our focus on the 2010–2025 period, during which attacks cluster in distinct episodes. However, as extensively discussed, this is essential for isolating unexpected changes in shipping costs that can plausibly be attributed to oil tanker attacks.[5] Finally, these corridors also carry vessels transporting other cargoes, whose exposure to attacks provides a comparison with broader maritime insecurity. Indeed, attacks on these vessels may affect the BDTI by increasing concerns about route safety. We therefore construct an alternative instrument using attacks on other vessel types. Section 6 shows that these attacks also raise the BDTI, but their estimated macroeconomic effects are not statistically significant.

[5]Narrative identification can draw on relatively small sets of events. Mertens and Ravn (2013) construct tax instruments from 21 legislative changes. De Winne and Peersman (2016) identify 13 food-market episodes, subsequently used as an alternative instrument by Peersman (2022). Caldara et al. (2019) construct an oil-supply instrument from 15 episodes of exogenous production disruptions to estimate oil-market elasticities. Guerreiro et al. (2026) study the macroeconomic effects of temporary transfer payments using 22 narrative events in their baseline sample. Bloom (2009) considers 17 episodes of heightened stock-market volatility, while Antolín-Díaz and Rubio-Ramírez (2018) use narrative sign restrictions, first exploiting a single monetary-policy event and then extending the analysis to eight events.

Figure 8: Retained attacks on crude and heavy fuel oil tankers, 2010–2024



Notes: Location of the 30 reported incidents involving oil tankers in the narrative record, by type of attacker. Panel A shows the Red Sea, the Gulf of Aden and the Arabian Sea; Panel B enlarges the Bab el-Mandeb cluster; Panel C shows the retained incident near the Singapore Strait. Panel D presents the world map. Solid lines indicate tanker routes from the Persian Gulf toward the Mediterranean and Malacca; the dashed line indicates the diversion toward the Cape of Good Hope.

3.2 High-Frequency Shipping-Cost Surprises

For each retained attack, we measure the market reaction using the change in the log BDTI around the event. Let $t(d)$ denote the first Baltic trading day on or after the calendar date d of the attack. We define the shipping-cost surprise as

$$\text{SCS}_d \equiv \log P^{\text{BDTI}}_{t(d)+1} - \log P^{\text{BDTI}}_{t(d)-1}, \tag{2}$$

where P^{BDTI}_t denotes the index on Baltic trading day t. By comparing freight rates before and after an attack, we aim to capture the market's response to new information about transportation risks. Indeed, even during an ongoing campaign, the timing, location or severity of an individual attack may be unexpected. We measure this response over two trading days (equation 2), spanning the assessment before the event to the assessment after it.[6] This window allows time for the attack to be reported and incorporated into freight-rate assessments, while limiting exposure to unrelated macroeconomic, geopolitical and financial news.[7] Section 6.3 shows that our results are robust to

[6] Verduzco-Bustos and Zanetti (2026) use a two-trading-day window to study geopolitical oil shocks, while Känzig and Raghavan (2026) use a one-trading-day window to study changes in the BDI.

[7] Most events in our sample are reported the day after they occur.

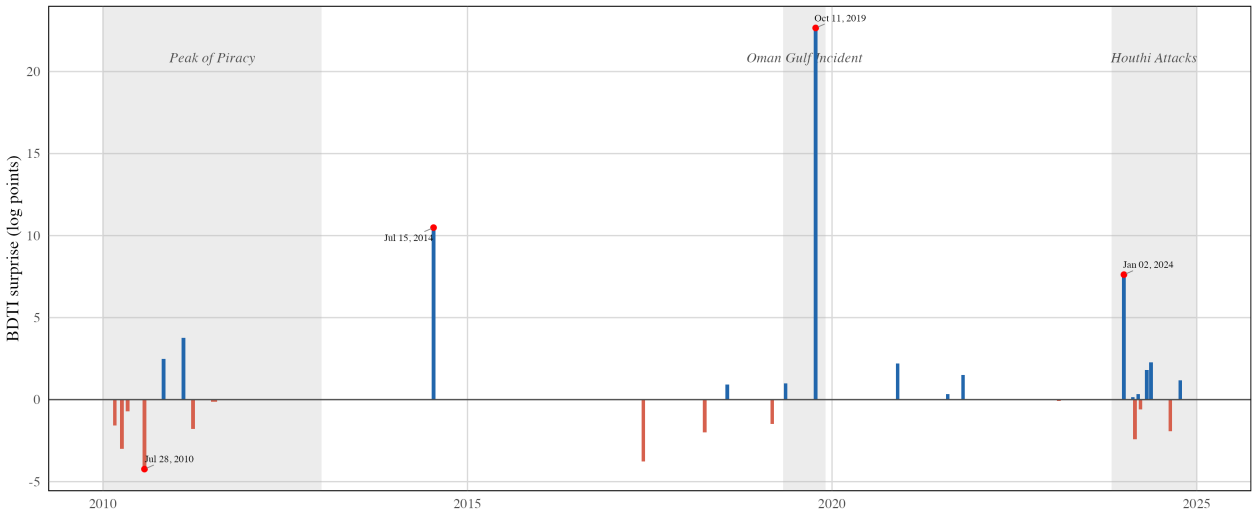
alternative event windows.

Figure 9 plots the resulting surprise series at the daily frequency. Several surprises exceed 5 percent. The largest one, 22.7 percent, corresponds to the attack on the Iranian tanker *Sabiti* off the Saudi coast on 11 October 2019, which caused oil to leak into the Red Sea near Jeddah. The second largest, at 10.5 percent, is associated with the 15 July 2014 incident involving the *Oriental Glory* in the Strait of Malacca. Section 6.4 shows that the main results are preserved when the three largest surprises are censored to zero, one at a time. The series also contains negative surprises. For instance, the explosion that damaged the tanker *M. Star* in the Strait of Hormuz on 28 July 2010 was associated with a decline of 4.2 percent in the BDTI over the event window. Negative surprises can arise when an event leads market participants to revise downward their assessment of shipping disruptions. Setting negative monthly instrument values to zero produces similar point estimates (Appendix Figure B1). However, we retain surprises of both signs to avoid selection bias.

Two natural concerns are that the surprises capture concurrent oil-market news or are predictable from information available before the attacks. Firstly, Section 3.3 reports no statistically significant correlations between the monthly instrument and established measures of oil-market and other macroeconomic shocks. Granger causality tests also find no significant predictive power from the lagged variables considered. Likewise, Section 5.1 extensively discusses how our results differ from those associated with conventional oil shocks. Secondly, Section 6.2 examines at the daily frequency whether information available before each attack predicts the surprises. None of the predictors is statistically significant. Moreover, removing the component associated with prior WTI movements or oil-related geopolitical risk leaves the main results broadly unchanged.

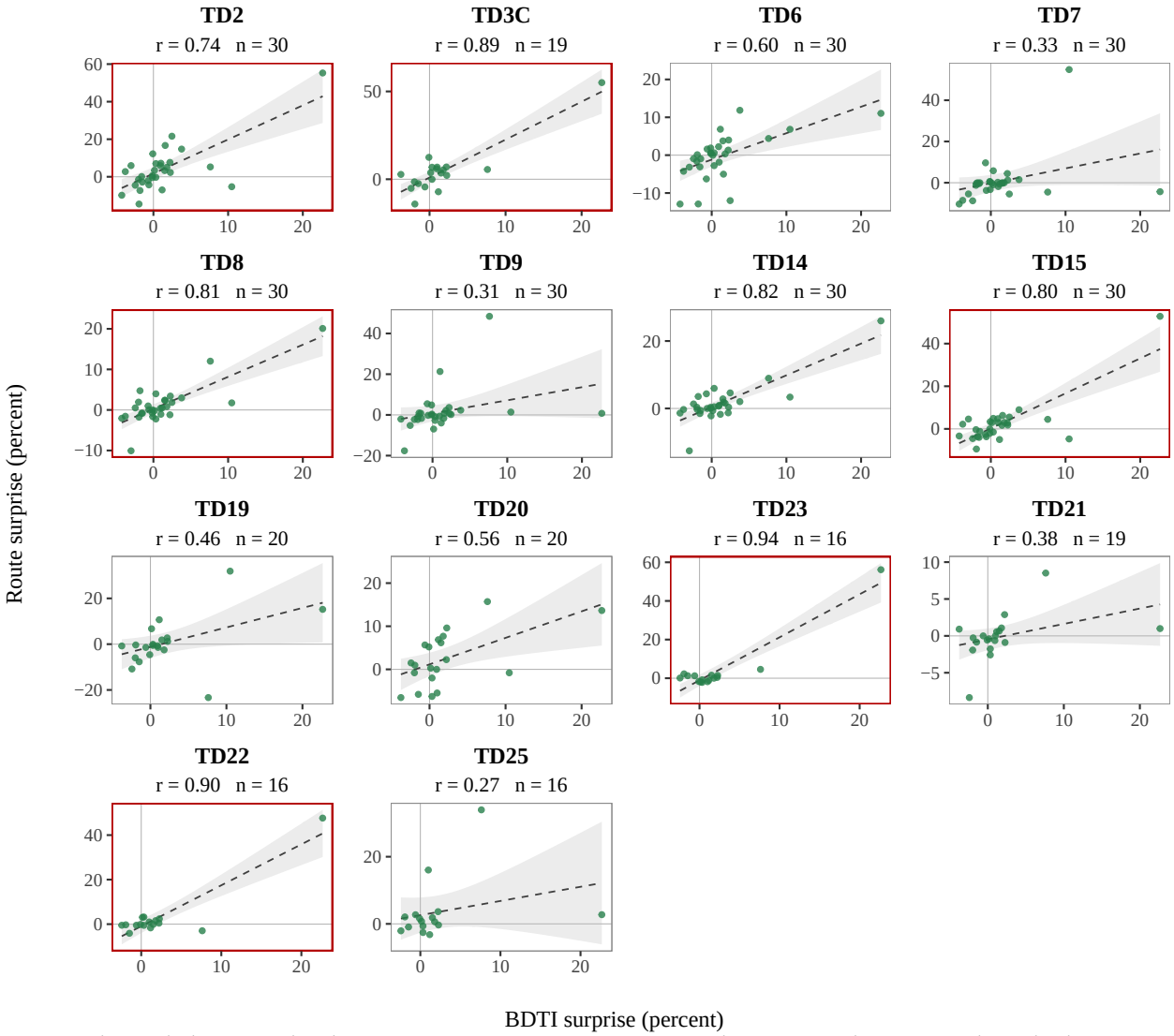
As mentioned in Section 2.3, a final concern is that the BDTI averages freight rates across routes whereas the attacks are geographically concentrated. An important verification is hence to check whether BDTI changes around attacks are more closely related to freight-rate changes on exposed routes than on others. For each Baltic route, we compute the same event-window revision, $\Delta \log P_d^r = \log P_{d+1}^r - \log P_{d-1}^r$, and correlate it with the BDTI surprise across events. Figure 10 shows strong correlations for routes crossing the affected corridors: TD2 and TD3C from the Gulf to Asia through Hormuz ($\rho = 0.74$ and 0.89), TD8 from Kuwait to Singapore ($\rho = 0.81$), and TD23 from the Gulf to the Mediterranean through Bab el-Mandeb ($\rho = 0.94$). These correlations are based on 30, 19, 30 and 16 events, respectively. Correlations with routes outside these corridors are much weaker and often statistically insignificant. This pattern suggests that BDTI changes around attacks reflect conditions along the exposed routes.

Figure 9: The shipping-cost surprise series



Notes: Daily shipping-cost surprises, computed as the revision in the log BDTI around each attack in the narrative record, equation (2). Blue bars are positive surprises, red bars negative surprises. Shaded areas mark three main episodes of piracy, namely the peak of Somali piracy (2010–2012), the Gulf of Oman incidents (2019) and the Houthi campaign (2023–2024). Labeled dots are the events discussed in the text.

Figure 10: BDTI surprises against route-level surprises



Notes: Each panel plots route-level surprises against BDTI surprises over the same attack-event windows, both measured as approximate percentage changes ($100\Delta \log P$). r is the Pearson correlation and n is the number of paired observations. Dashed lines show fitted linear relationships; shaded areas are 95 percent confidence bands for the fitted mean. Red borders identify routes with a documented direct link to the attacked corridors (TD2, TD3C, TD8, TD15, TD22 and TD23).

3.3 From Event Days to Months

Most of our outcomes of interest are available only at monthly frequency, whereas our shipping-cost surprises are measured around individual attack dates. Hence, we aggregate the daily surprises to a monthly series using linear-decay weights. Let T_t denote the number of calendar days in month t and $q(d)$ the calendar-day position of the effective event day d within that month. We construct

$$z_t = \sum_{d \in \mathcal{D}_t} \left(\frac{T_t - q(d) + 1}{T_t} \right) \mathrm{SCS}_d, \tag{3}$$

where $\mathcal{D}_t$ is the set of attack days in month t. An attack early in the month receives a larger weight than an attack near the end of the month because it affects average shipping conditions over a larger share of the month. This weighting accounts for the use of monthly average BDTI values in the VAR (Kilian, 2024; Känzig and Raghavan, 2026). Months without a qualifying attack are assigned exactly zero. The main impulse responses remain similar when daily surprises are summed within each month without weighting as discussed in Section 6.

Diagnostics. We perform the checks proposed by Ramey (2016) to assess whether the monthly instrument behaves like a shock. First, it reproduces the narrative. The largest values correspond to the piracy peak of 2010–2011, to the Gulf of Oman incidents of 2019 and to the Red Sea attacks of late 2023 and early 2024. Second, we find no statistically significant serial correlation in the monthly instrument (Appendix Figure B2). Third, it is not forecastable. Appendix Table B1.1 reports Granger causality tests using the variables of the VAR, oil-market and uncertainty measures, and short-term interest rates. Appendix Table B1.2 reports the corresponding tests for global variables. The six-lag tests find no statistically significant predictive power for these variables. Fourth, it is orthogonal to the structural shocks identified in the literature. Appendix Table B2 reports its correlations with oil supply, oil demand and oil price expectation shocks (Baumeister and Kilian, 2017; Baumeister and Hamilton, 2019; Caldara et al., 2019; Känzig, 2021), natural gas shocks (Alessandri and Gazzani, 2025; Colombo and Toni, 2026), monetary policy shocks (Bauer and Swanson, 2023), supply chain shocks (Känzig and Raghavan, 2026; Bini, 2025), and a battery of uncertainty and geopolitical risk measures (Baker et al., 2016; Iacoviello and Tong, 2026; Pinchetti, 2026).

4 Econometric Framework

In this section, we explain how we use the shipping-cost surprises constructed in the previous section to estimate the effects of oil transportation shocks on the U.S. economy. We first present the SVAR-IV framework and its identifying assumptions. We then describe the baseline specification of our VAR and finally the methods used to study additional outcomes or settings.

4.1 SVAR-IV

As extensively discussed in Section 3, our shipping-cost surprises capture only part of the variation in oil-transportation conditions. We therefore cannot treat our surprises as a direct measure of the structural oil-shipping-cost shock. Instead, we follow the external-instrument approach to structural VAR identification (Stock and Watson, 2012; Mertens and Ravn, 2013; Stock and Watson, 2018) and use our surprise series as an instrument to identify oil-shipping-cost shocks. This requires the series to be correlated with these shocks (relevance) but uncorrelated with the other structural shocks in the VAR (exogeneity). Formally, let $\mathbf{y}_t$ be an $n \times 1$ vector of monthly variables describing the tanker market, the U.S. oil market and the U.S. economy, after accounting for deterministic

terms. We assume that $\mathbf{y}_t$ admits the structural vector moving-average (SVMA) representation

$$\mathbf{y}_t = \mathbf{B}(L)\mathbf{S}\boldsymbol{\varepsilon}_t, \qquad \mathbf{B}(L) = \mathbf{I} + \mathbf{B}_1 L + \mathbf{B}_2 L^2 + \dots, \tag{4}$$

where $\boldsymbol{\varepsilon}_t$ is an $n \times 1$ vector of serially uncorrelated structural shocks with zero mean and identity covariance matrix, $\mathbf{S}$ is a nonsingular structural impact matrix, and L is the lag operator, $L^i x_t = x_{t-i}$. We assume that $\mathbf{B}(L)$ is square summable, so that $\mathbf{y}_t$ is second-order stationary.

Under the baseline assumption that this structural moving-average representation is invertible, we can write

$$\mathbf{A}(L)\mathbf{y}_t = \mathbf{u}_t, \qquad \mathbf{A}(L) = \mathbf{B}(L)^{-1} = \mathbf{I} - \mathbf{A}_1 L - \mathbf{A}_2 L^2 - \dots, \tag{5}$$

where the reduced-form innovations satisfy

$$\mathbf{u}_t = \mathbf{S}\boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\Sigma} \equiv \mathbb{E}[\mathbf{u}_t \mathbf{u}_t'] = \mathbf{S}\mathbf{S}'. \tag{6}$$

We approximate this representation by a VAR of order p and estimate its coefficients and residual covariance matrix. Our objective is to identify the impact vector of the oil-shipping-cost shock. Without loss of generality, we order this shock first in $\boldsymbol{\varepsilon}_t$, so that the object of interest is the first column of $\mathbf{S}$, denoted $\mathbf{s}_1$.

Let z_t denote the proxy for $\varepsilon_{1,t}$ (we use the notation z_t for $Z_t - \text{proj}(Z_t \mid \mathbf{W}_t)$, where Z_t is the raw instrument, $\mathbf{W}_t$ is the history of $\mathbf{y}_t$, and $\text{proj}(x \mid w)$ denotes the linear projection of x on w). The proxy SVAR imposes the identifying assumptions

$$\mathbb{E}[z_t \varepsilon_{1,t}] = \phi \neq 0, \tag{7}$$

$$\mathbb{E}[z_t \varepsilon_{j,t}] = 0, \qquad j = 2, \dots, n. \tag{8}$$

Condition (7) is relevance: attack-induced freight-rate revisions must be correlated with the structural shipping-cost shock. The first-stage statistic reported in Section 5.1 supports instrument relevance. Condition (8) is exogeneity: the instrument must be contemporaneously orthogonal to all other structural shocks. It is not testable, but the narrative selection, the narrow event window and the diagnostics of the previous section are designed to make it credible. If the identifying assumptions hold, it follows that

$$\mathbb{E}[z_t \mathbf{u}_t] = \phi\, \mathbf{s}_1, \tag{9}$$

so that $\mathbf{s}_1$ is identified up to scale as

$$\mathbf{s}_1 \propto \frac{\mathbb{E}[z_t \mathbf{u}_t]}{\mathbb{E}[z_t u_{1,t}]}, \tag{10}$$

which we estimate by two-stage least squares, regressing the reduced-form residuals $\hat{\mathbf{u}}_t$ on $\hat{u}_{1,t}$ using z_t as an instrument. Identification of the impact effects does not require invertibility; invertibility is needed only for the dynamics beyond impact, and we test it below.

Normalization. In order to facilitate comparison with the oil-shock literature (Baumeister and Hamilton, 2019; Känzig, 2021; Känzig et al., 2026) and the shipping-cost shock of Känzig and

Raghavan (2026), we normalize our shock to raise the BDTI by 10 percent on impact. This magnitude lies in the upper tail of the observed event-window surprises and illustrates the macroeconomic effects of a sharp rise in shipping costs following an oil tanker attack. We believe this scenario is relevant for policymakers assessing exposure to substantial shipping disruptions, including those associated with the ongoing Houthi campaign in the Red Sea.

Inference. We use the residual-based moving block bootstrap of Jentsch and Lunsford (2019). We use 5,000 replications and a block length of 19, applying a small-sample bias correction before constructing the bootstrap distribution (Hall, 1992; Känzig, Forthcoming). We report 68 and 90 percent confidence bands.

Invertibility. The baseline proxy-SVAR assumes that the shipping shock can be recovered from current and past values of the variables included in the model or from the reduced-form VAR innovations. We assess this assumption by testing whether future VAR innovations help explain the instrument after conditioning on contemporaneous innovations, following the procedure from Forni et al. (2023). Under invertibility and the instrument-validity conditions, future innovations should provide no additional information. We perform this test with eight leads and do not reject invertibility ($p = 0.321$). As an additional robustness check, we estimate an internal-instrument VAR, which includes z_t as an endogenous variable ordered first in a recursive identification scheme (Plagborg-Møller and Wolf, 2021). To limit the number of parameters in this larger model, we use three lags rather than the six lags of the baseline proxy-SVAR. The responses confirm the main qualitative findings as industrial production contracts immediately, crude inventories decline, and consumer prices rise (Appendix Figure C1). The WTI price also increases, although its response occurs earlier than in the baseline, with no significant effect on impact. Overall, this exercise supports the robustness of our main conclusions to relaxing the invertibility assumption. We also examine whether explicitly controlling for financial conditions alters the estimated responses. Indeed, as in Känzig (2021), oil inventories are the only forward-looking variable in our system, so agents may condition on financial-market information that our VAR does not otherwise span. To solve that issue, we follow Mori and Peersman (2026), who include financial variables jointly in the VAR to enlarge the model's information set and mitigate potential invertibility concerns. We add, one at a time, the three-month Treasury bill rate, the S&P 500 and the oil volatility index (OVX), which capture short-term interest rates, equity prices, and expected oil-price volatility, respectively. The main response patterns remain similar as can be seen in Appendix Figures C2, C3 and C4.

4.2 Empirical Specification

The baseline VAR contains seven monthly variables covering tanker freight rates, the U.S. oil market and the broader economy. The BDTI measures tanker freight rates, and oil-related geopolitical risk is measured using the index of Iacoviello and Tong (2026). Crude oil inventories, crude oil production, industrial production and consumer prices are seasonally adjusted using standard

procedures.[8] Appendix Table A1 describes the data and their sources.

Following Känzig (2021), we measure the real oil price using the WTI spot price deflated by the U.S. CPI. All variables enter in log levels (Sims et al., 1990). The lag order is $p = 6$, and the deterministic terms include a constant and a linear trend.[9]

$$\mathbf{y}_t = \begin{bmatrix} \log \text{BDTI}_t & \log \text{Inventories}_t & \log \text{Production}_t & \log \text{WTI}_t^{real} & \log \text{GPR}_t^{oil} & \log \text{IP}_t & \log \text{CPI}_t \end{bmatrix}'.$$

The data span July 2009 to February 2026. We start in July 2009 for three main reasons. First, it excludes the most acute phase of the global financial crisis, during which unusually large financial disturbances would complicate identification. Second, it excludes most of the 2006–2008 commodity boom, during which tanker rates were driven partly by the surge in Chinese demand for oil. Finally, with six monthly lags, the effective VAR estimation sample begins in January 2010, coinciding with the start of the shale expansion and its many implications for the U.S. economy, as discussed by Känzig et al. (2026). This, in particular, avoids having to account for an important structural break. We decided to end our sample before the March 2026 closure of the Strait of Hormuz, so that this exceptional episode does not affect the estimates.

Finally, to limit the influence of the large disturbances of the pandemic, the baseline includes a heteroskedasticity adjustment over March–August 2020 (Hamilton, Forthcoming). The covariance matrix of the reduced-form innovations is allowed to be scaled by a factor $\delta > 0$ during the pandemic window, $\mathbf{\Omega}_{\text{COVID}} = \delta \mathbf{\Omega}$, with δ estimated by iterated generalized least squares, while the VAR coefficients are common across regimes. Observations in the pandemic window therefore receive less weight in the estimation of the common dynamics. Section 6 shows that replacing the adjustment by pandemic dummies leaves the results unchanged.

4.3 Additional Outcomes

In order to investigate transmission mechanisms, we need the responses of variables that are not in the baseline VAR. For monthly variables, we follow Gertler and Karadi (2015) and augment the baseline VAR with one additional variable at a time, re-estimating the full model and the bootstrap, and report the response of the added variable. This preserves the identification of the baseline while avoiding the loss of degrees of freedom that a much larger VAR would entail.

For outcomes available over shorter samples, we use local projections (Jordà, 2005). With fewer observations, estimating a separate response at each horizon can produce imprecise estimates, particularly at longer horizons where fewer observations remain available. We therefore use the smooth local projections of Barnichon and Brownlees (2019), which pool information across horizons and penalize excessive fluctuations in the estimated response. We first recover the oil-

[8] Crude inventories and production are adjusted using X-13ARIMA-SEATS; industrial production and the CPI are obtained in seasonally adjusted form from their source.

[9] We follow Känzig (Forthcoming) in choosing six lags for a monthly VAR of this size; Section 6 shows that the results are robust to twelve lags and to a Bayesian VAR-IV with Minnesota-type shrinkage.

shipping-cost shock from the monthly VAR as

$$\hat{\varepsilon}_{1,t} = \mathbf{s}_1' \boldsymbol{\Sigma}^{-1} \hat{\mathbf{u}}_t,$$

where $\mathbf{s}_1$ is the identified impact vector, $\boldsymbol{\Sigma}$ is the covariance matrix of the reduced-form innovations, and $\hat{\mathbf{u}}_t$ denotes these innovations (Stock and Watson, 2018). Appendix Figure B3 plots the recovered shock and the variables included in the baseline VAR. Before estimation, we residualize each outcome on a COVID-19 dummy covering March–August 2020. We denote the resulting series by $y_{j,t}$ and estimate

$$y_{j,t+h} = \alpha_{j,h} + \beta_{j,h} \hat{\varepsilon}_{1,t} + \sum_{\ell=1}^{6} \gamma_{j,h,\ell}\, y_{j,t-\ell} + u_{j,t+h}, \qquad h = 0, \ldots, 12. \tag{11}$$

We include six lags of the dependent variable to control for its past dynamics, matching the lag length of the baseline VAR. Following Barnichon and Brownlees (2019), rather than estimating $\beta_{j,h}$ independently at each horizon, we represent its response profile using cubic B-spline basis functions:

$$\beta_{j,h} \approx \sum_{m=1}^{M} b_{j,m}\, B_m(h), \qquad h = 0, \ldots, 12, \tag{12}$$

where $B_m(\cdot)$ are the basis functions and $b_{j,m}$ are coefficients estimated by penalized least squares. The penalty discourages excessive fluctuations across horizons, balancing flexibility against precision.[10] Its strength is selected by cross-validation. Responses are rescaled to correspond to the same approximately 10-percent impact increase in the BDTI as in the baseline. Confidence bands use heteroskedasticity- and autocorrelation-consistent (HAC) standard errors.

Finally, we use panel local projections to examine whether refinery responses vary with regional dependence on Persian Gulf crude. We interact the shock recovered from the baseline VAR with regional exposure, defined as the average ratio of Persian Gulf crude imports to refinery crude inputs over the twelve months preceding the shock. The specification includes regional fixed effects, exposure itself and a pandemic dummy. Section 5.2.1 presents the specification and results in greater detail.[11]

5 The Macroeconomic Effects of Attacks on Oil Tankers

We first present the baseline responses for the U.S. economy. We then investigate transmission through energy trade and manufacturing adjustment. Finally, we extend the analysis to global aggregates and assess the contribution of oil-shipping-cost shocks to their fluctuations.

[10] We use a third-order difference penalty, which shrinks the response profile toward a quadratic function of the horizon.

[11] Ongoing work examines weekly and daily responses to better identify the timing of adjustment to tanker attacks.

5.1 Baseline results

Figure 11 shows the impulse responses to our identified oil-shipping-cost shock, normalized to raise the BDTI by 10 percent on impact. In each panel, the solid line is the point estimate and the dark and light shaded areas are the 68 and 90 percent confidence bands. The first-stage regression yields an R^2 of 7.94 percent and a heteroskedasticity-robust F-statistic of 71.04. These diagnostics support the relevance of the instrument. [12] [13]

Effects on oil-shipping cost. Oil tanker attacks significantly raise the BDTI for several months, with the response gradually returning toward zero after roughly six months. This persistence is consistent with both disruptions to navigation and continued concerns about route safety following attacks, which we discussed extensively in Section 2.3.

Effects on oil variables. The real WTI price does not respond significantly on impact or during the first months, and its response becomes statistically significant only at long horizons. Crude inventories decline immediately and persistently, reaching -0.5 percent after five months. This is consistent with refiners using stored crude to cushion disrupted deliveries. However, domestic production of crude oil does not expand to offset the disruption in oil shipping. This is consistent with the limited substitutability between domestic and imported crude grades discussed in Section 2.1. The delayed WTI response and declining U.S. inventories contrast with the immediate oil-price increase and accumulation of global inventories following an oil-supply-news shock (Känzig, 2021). Moreover, the persistent decline in inventories differs from the short-lived decline following the structural oil-supply shock of Baumeister and Hamilton (2019). Augmenting the baseline VAR with their shock leaves the responses unchanged (Appendix Figure D1). Finally, oil-related geopolitical risk rises on impact, but the initial increase dissipates almost immediately. This is reassuring, as the persistent macroeconomic effects are not accompanied by a sustained increase in measured geopolitical risk. This can also be seen by the response of defense and aerospace equipment that does not increase and is mainly flat, motivating the fact that our effect does not come from a broader landscape of increased geopolitical tensions.

Effects on macroeconomic variables. The identified oil-shipping-cost shock has stagflationary effects as it reduces U.S. economic activity on impact and raises consumer prices at the same time.

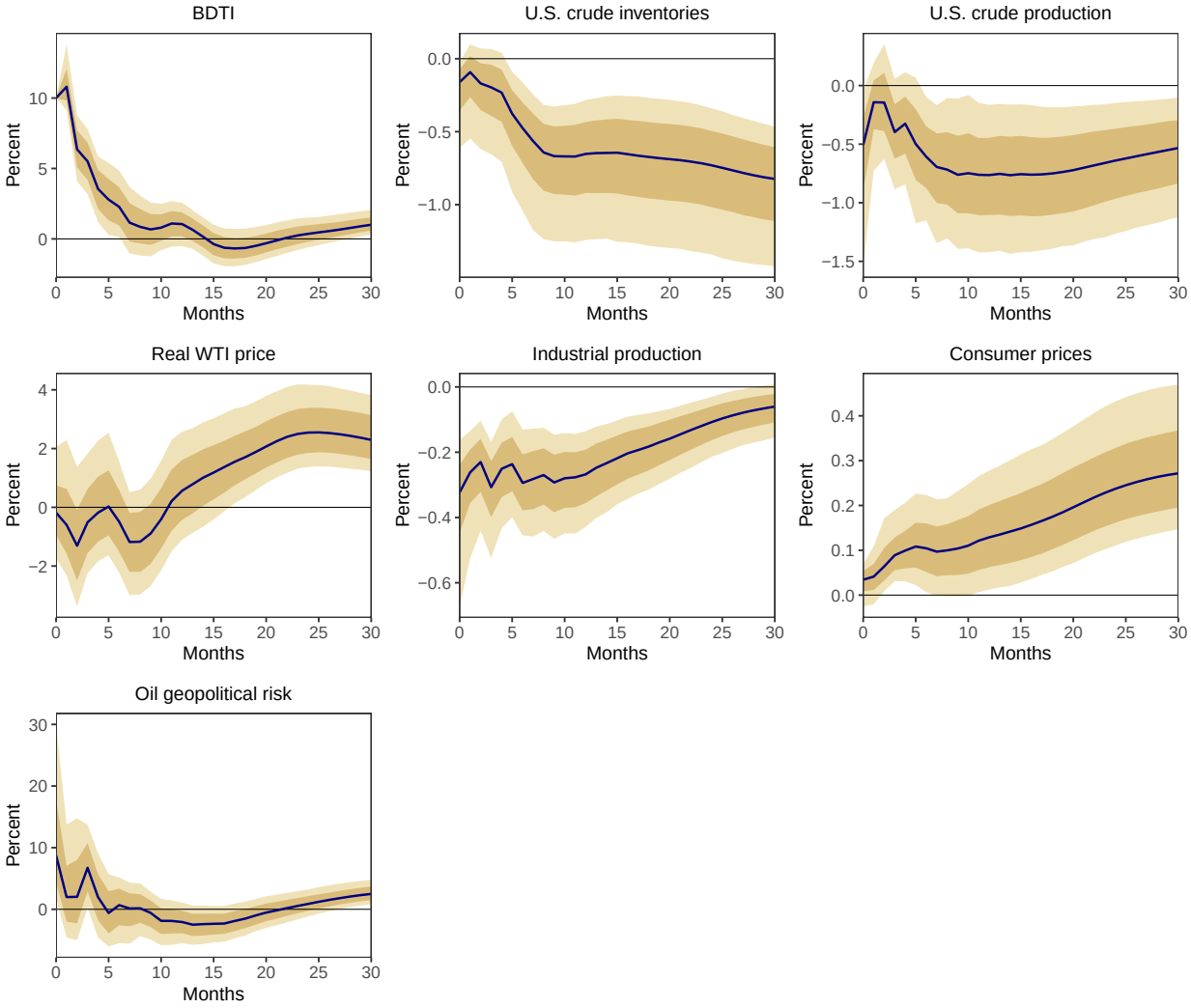
[12] Large first-stage F-statistics also appear in other macroeconomic applications. Examples include Menkhoff et al. (2021) (47.9–540.5 across specifications), Miescu and Mumtaz (2024) (174), Miranda-Agrippino and Ricco (2021) (67.61 for the extended Romer–Romer instrument), Forni et al. (2025) (49.58 using the Baumeister–Hamilton instrument) and Nekarda and Ramey (2011) (above 100). Finally, in a closely related shipping application, Känzig and Raghavan (2026) report a robust F-statistic of 33.41.

[13] The high first-stage F-statistic may partly reflect a few large surprises. Section 6.4 examines this possibility by excluding the three largest monthly surprises one at a time. The main response patterns remain broadly unchanged, showing that our results are not drive by one particular event. Following Bini (2025), we replace each event-level surprise with $+1$ if positive and -1 if negative, then sum these signs within each month. This removes the influence of event magnitudes while retaining their timing and direction. Reassuringly, the point estimates preserve our baseline results, although the rise in consumer prices is more delayed.

Indeed, industrial production falls by approximately 0.32 percent on impact before gradually returning toward baseline. CPI rises gradually, reaching approximately 0.1 percent above baseline after five months. This contraction in activity contrasts with the expansionary U.S. response to adverse oil-supply-news shocks during the post-2010 period documented by Känzig et al. (2026). Additionally, the estimated output and price responses are smaller than those reported by Känzig and Raghavan (2026), although they are using a different freight measure (Baltic Dry Index) and a different sample period (1970–2022), which limits direct comparisons. Their shock, normalized to raise dry-bulk freight rates by 10 percent, produces an approximately 0.5 percent decline in industrial production and a 0.4 percent increase in consumer prices. To make things more comparable, we proceed to do two additional exercises. Firstly, we show that our instrument indeed captures different variation in shipping costs. The shipping-cost instrument of Känzig and Raghavan (2026) has little explanatory power for BDTI innovations over our common sample, whereas our tanker-attack instrument remains strongly relevant.[14] This is consistent with the differences in events and freight markets discussed in Section 3. Second, we add their estimated dry-bulk shipping shock to our SVAR-IV. Since their series ends in December 2022, we estimate both models over this shorter sample. The responses remain broadly unchanged, although confidence bands are wider in the augmented model (Appendix Figure D2).

[14] Replacing our instrument with theirs in the baseline VAR yields a heteroskedasticity-robust first-stage F-statistic of 0.02, compared with 62.97 for our instrument over the same sample ending in December 2021. Our monthly instrument also has a small and statistically insignificant correlation with their estimated supply-chain shock ($\rho = -0.066$, $p = 0.401$; Appendix Table B2).

Figure 11: Impulse responses to an oil-shipping-cost shock



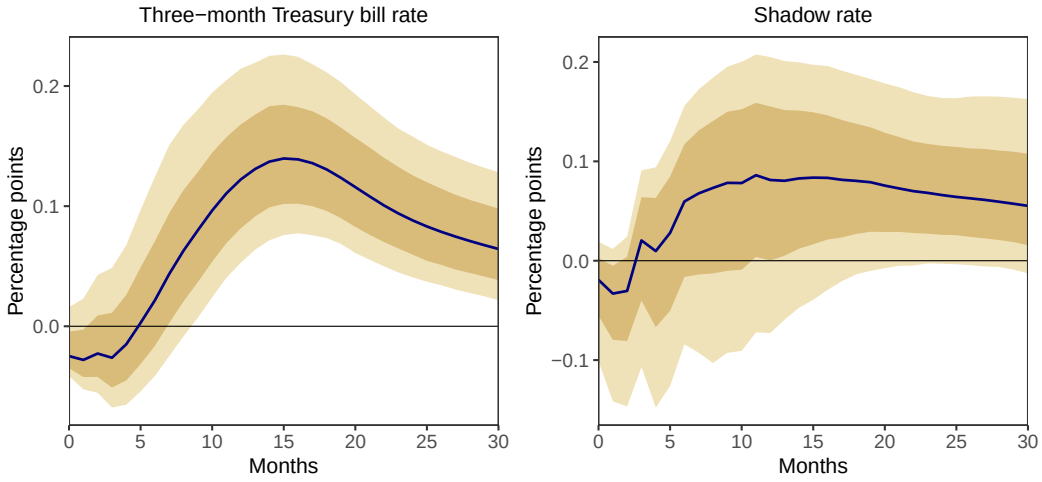
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The blue lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

Monetary policy response. We augment the baseline VAR with the three-month Treasury bill rate to examine the response of monetary policy.[15] It rises by approximately 0.12 percentage points after one year (Figure 12). However, our sample includes periods when U.S. policy rates were constrained by the zero lower bound (2010–2015 and 2020–2022). Following Miyamoto et al. (2026), who show that the monetary policy regime matters for the transmission of oil-supply-news shocks, we augment the VAR separately with the shadow rate of Wu and Xia (2016). Its response is similar to that of the three-month Treasury bill rate, although less precisely estimated. Both responses are consistent with monetary tightening in response to inflationary pressures. Additionally, Appendix Figures C2 and D3 report the full set of impulse responses, showing that the main results

[15]We choose the three-month Treasury bill rate as it represents a good balance between monetary policy stance and the zero lower bound.

are unchanged in both specifications.

Figure 12: Monetary policy response



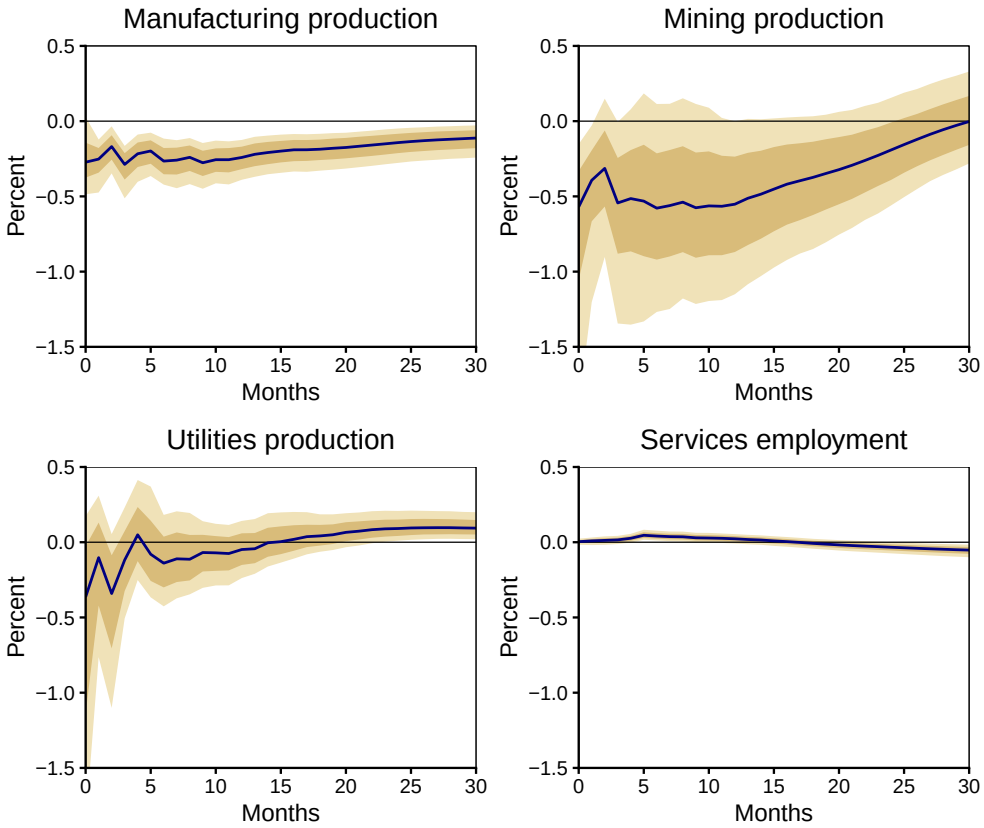
Notes: Responses to an oil-shipping-cost shock normalized to increase the BDTI by 10 percent on impact. Each rate is added separately to the baseline VAR. The blue lines are point estimates, and the dark and light yellow shaded areas are 68 and 90 percent confidence bands, respectively. Responses are measured in percentage points. The shadow-rate specification uses the series of Wu and Xia (2016) through June 2023, its last available observation, and the effective federal funds rate thereafter, when the policy rate was no longer constrained by the zero lower bound.

Sectoral responses. We next investigate how the response varies across sectors. Figure 13 shows immediate declines in manufacturing and mining production, with a persistent effect for manufacturing. Utilities production also declines, although the response is less precisely estimated, while services employment shows little response (below 0.1 percent and positive). This pattern is consistent with disruptions to oil transportation affecting goods-producing sectors more directly.

Within manufacturing, durable-goods production falls by approximately 0.45 percent on impact, compared with 0.11 percent for nondurable goods (Figure 14). The larger estimated decline in durable goods is consistent with the fact that their purchases can be postponed. The more detailed responses in Figure 15 show immediate declines in machinery, computers and electronics, plastics and rubber, and paper, indicating that the contraction extends beyond petroleum refining.

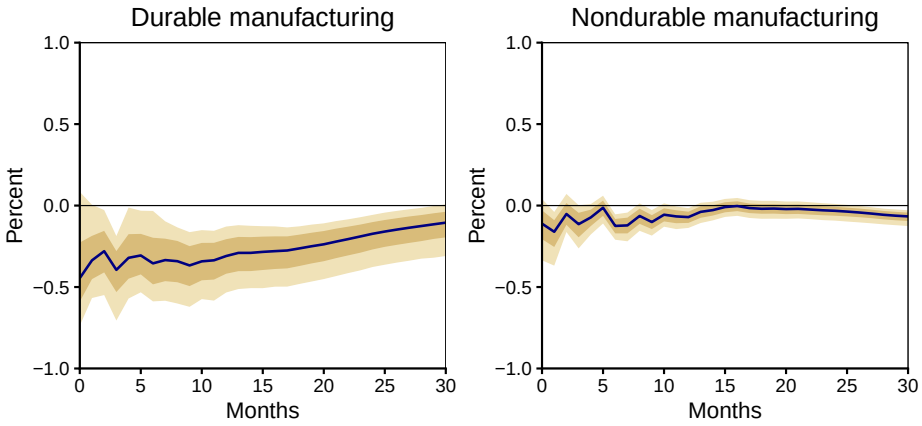
Finally, Figure 16 shows declines across all four product groups. The impact declines are statistically significant at the 90 percent level for consumer goods and intermediate products, and at the 68 percent level for business equipment and construction supplies. Together, these results are consistent with oil-shipping disruptions affecting a broad range of industries through the cost and availability of petroleum inputs and transportation.

Figure 13: Sectoral responses: broad sectors



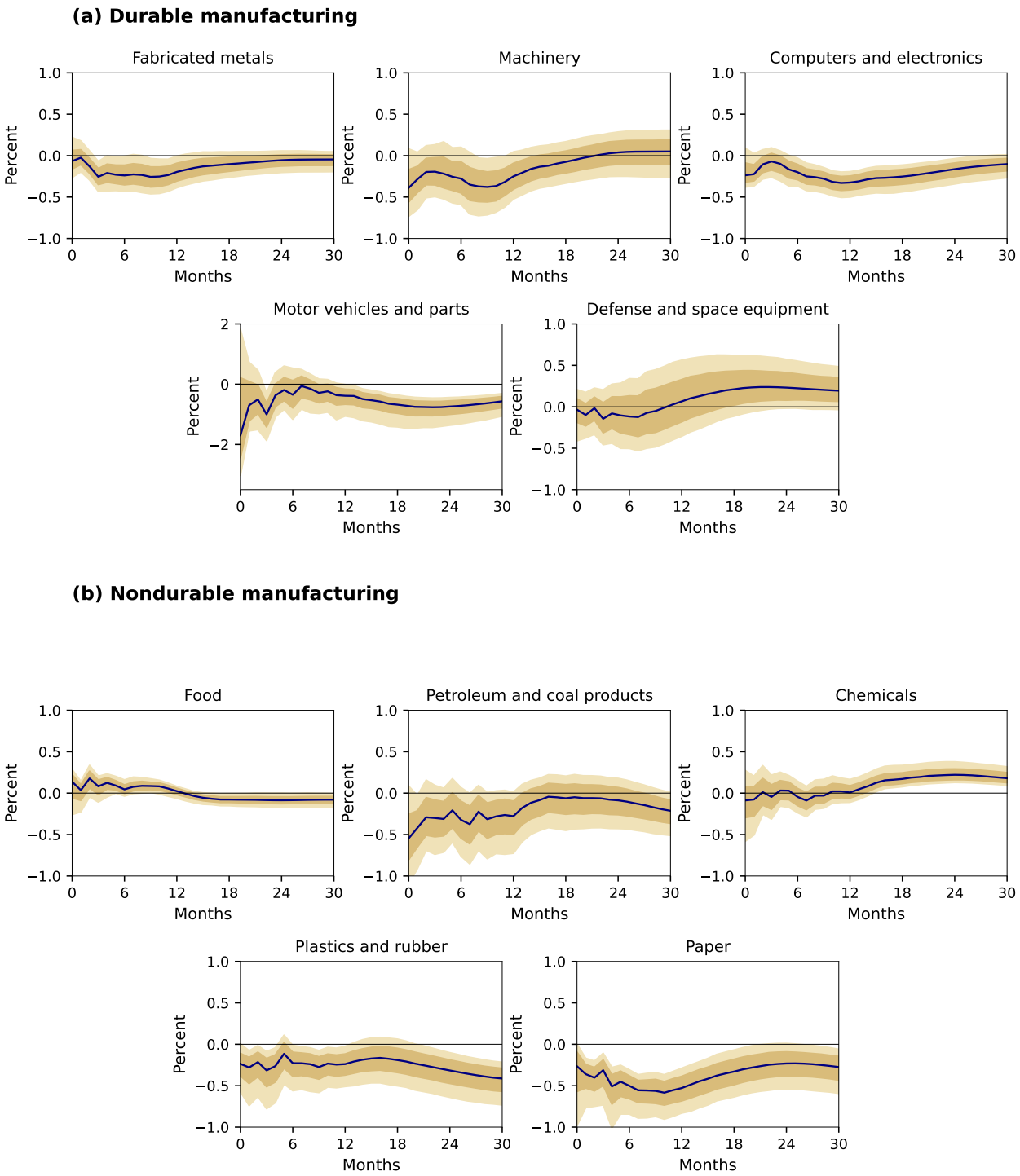
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Each outcome is added separately to the baseline VAR. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

Figure 14: Durable and nondurable manufacturing



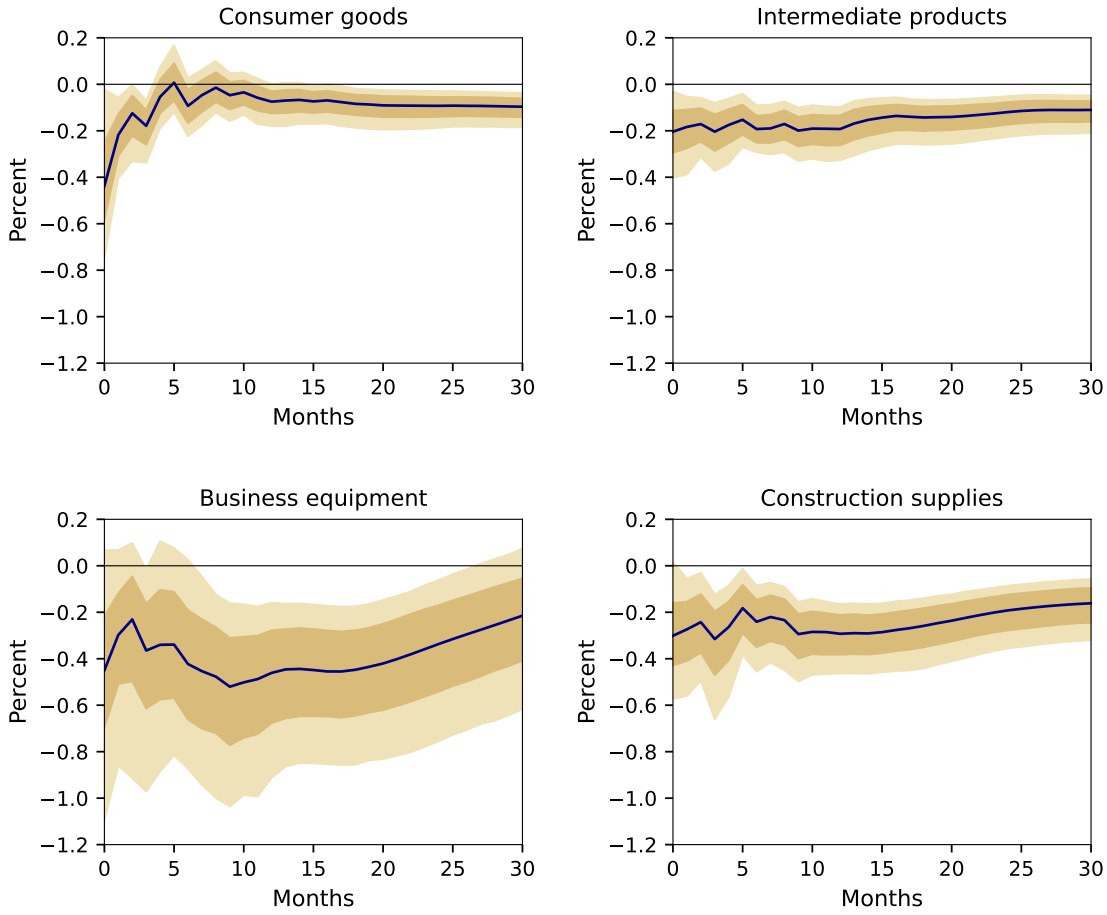
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Each outcome is added separately to the baseline VAR. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

Figure 15: Responses across manufacturing industries



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by approximately 10 percent on impact. Responses are obtained by augmenting the baseline VAR with each variable of interest separately. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. For readability, the motor-vehicles-and-parts panel uses a wider vertical scale (-3.5 to 2 percent); all other panels use -1 to 1 percent.

Figure 16: Sectoral responses: production by product group



Notes: Responses to an oil-shipping-cost shock normalized to raise the BDTI by 10 percent on impact. Each outcome is added separately to the baseline VAR. Dark and light bands show pointwise 68 and 90 percent bootstrap confidence intervals. CREA series are seasonally adjusted production measures at constant prices, with the Federal Reserve identified as the original source. Responses are in approximate percent deviations.

5.2 Transmission channels

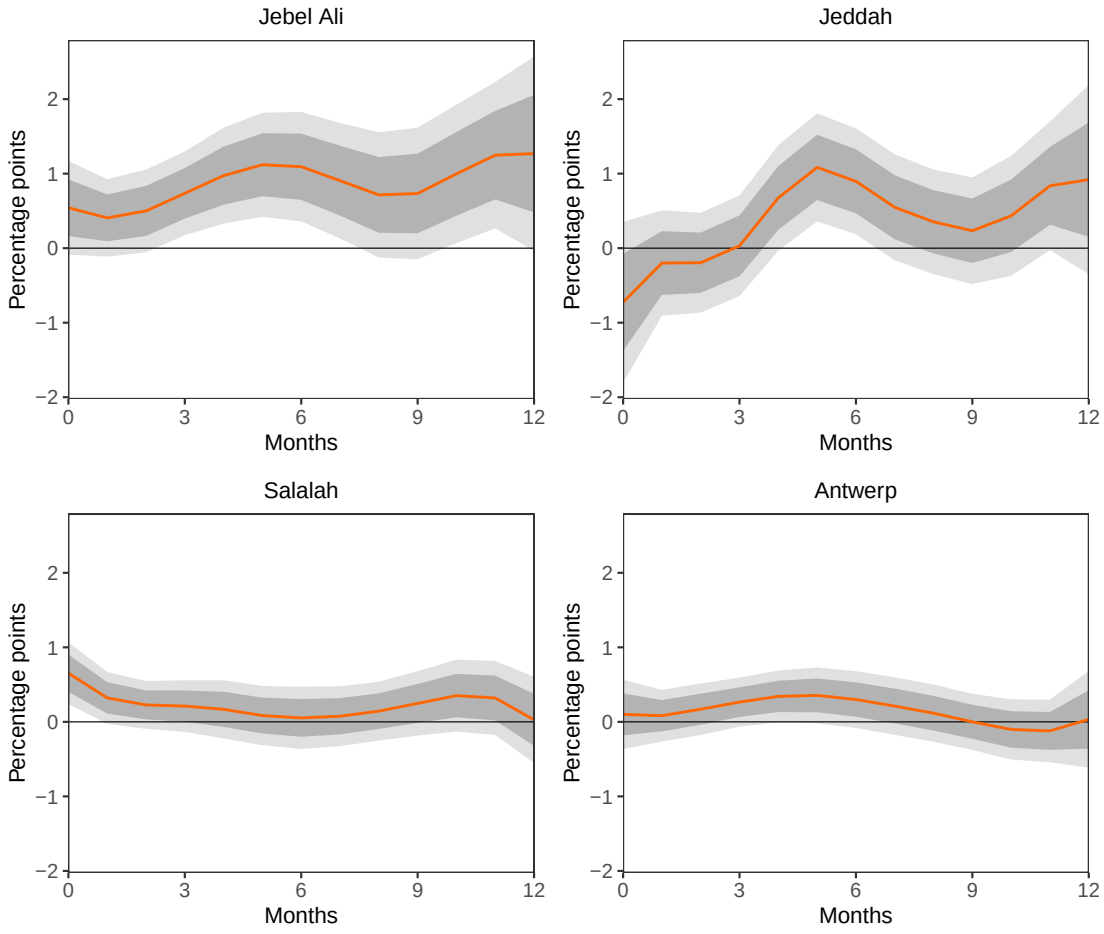
5.2.1 Energy trade

We now investigate different transmission channels. We start by studying the mechanisms through which disruptions to oil transportation affect the availability and cost of energy.

Congestion at Middle Eastern ports. We start by examining whether tanker attacks are followed by increased congestion at ports near the shipping corridors where the attacks in our sample occur. To examine this geographic pattern, we consider the following three major ports: Jebel Ali, in the Persian Gulf and accessed through the Strait of Hormuz; Jeddah, on the Red Sea; and Salalah, on Oman's Arabian Sea coast. Salalah provides a regional comparison outside the two chokepoints of the Strait of Hormuz and Bab el-Mandeb. Additionally, we consider Antwerp in Belgium, which provides a geographically distant comparison. We use the monthly port-level congestion measures

constructed by Bai et al. (2026) from Automatic Identification System (AIS) data and available since 2016. They capture the share of container-ship visits experiencing delays. Although they cover container ships rather than tankers, they still allow us to assess whether tanker attacks are followed by broader disruptions to port traffic. To test that, we estimate the smooth local projections in equation (11), using the structural shock recovered from our baseline VAR. The results are presented in Figure 17 and show larger and more persistent increases in congestion at Jeddah and Jebel Ali. The response at Salalah is smaller and concentrated around impact, while Antwerp's response is modest and statistically indistinguishable from zero. This geographic pattern is hence consistent with greater disruption at ports near the exposed shipping corridors. Appendix Figures D4 and D5 report the corresponding responses of waiting hours and delayed visits. Finally, complementary evidence from U.S. ports points to increases in tanker and liquid-bulk vessel dwell times at some horizons following the shock (Appendix Figure D6). These measures provide additional evidence of disruptions to tanker operations, although their coverage is limited to January 2019–June 2023.

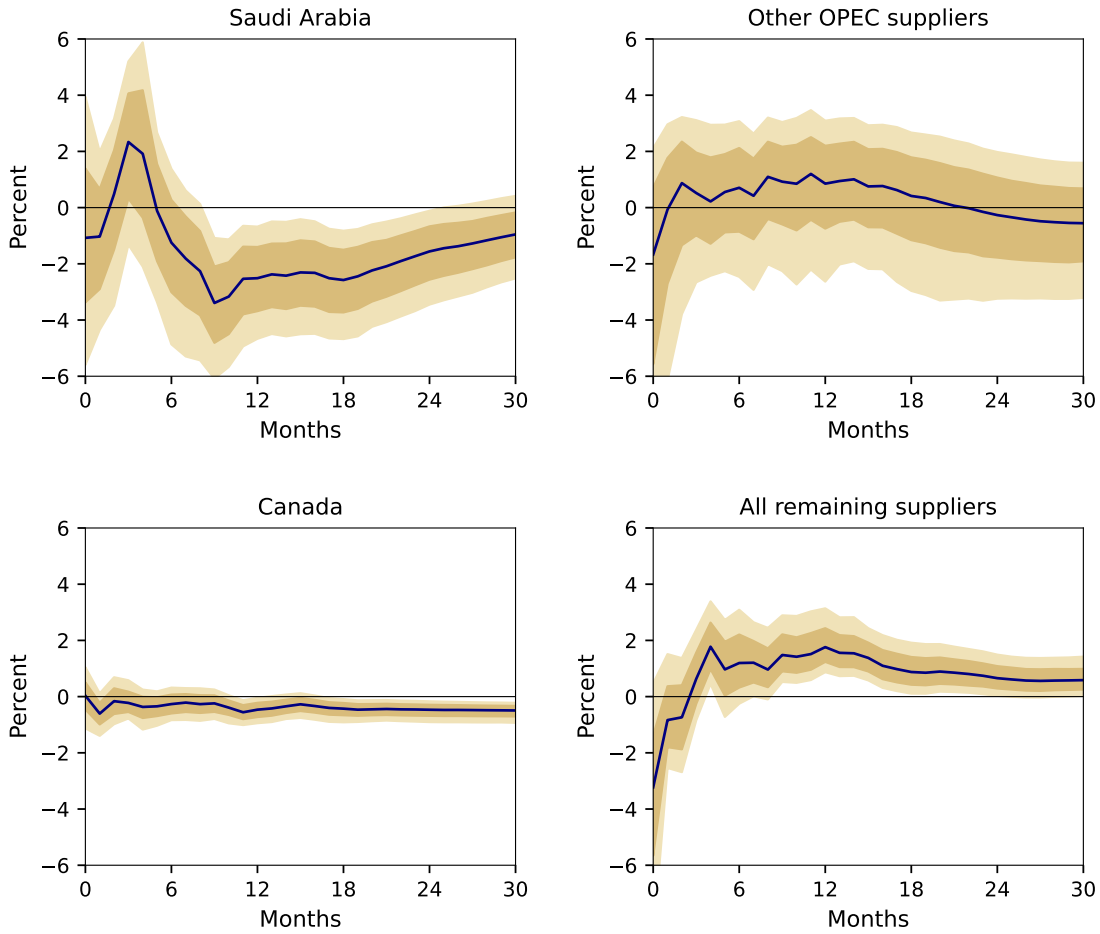
Figure 17: The energy-trade channel: congestion at Middle East ports



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are estimated by smooth local projections (11) on the structural shock extracted from the baseline SVAR-IV. The orange lines are point estimates, and the dark and light grey shaded areas are 68 and 90 percent confidence bands, respectively.

U.S. crude oil imports. We examine whether the higher freight rates and disruptions to maritime traffic documented above are accompanied by changes in U.S. crude imports across suppliers. Figure 18 reports responses for four mutually exclusive groups, namely Saudi Arabia, other OPEC suppliers, Canada, and all remaining suppliers. Imports from Saudi Arabia decline persistently, falling by approximately 2 percent after six months. The gradual response is consistent with disruptions taking time to affect deliveries, as tankers already beyond the affected corridors continue to arrive and subsequent shipments face delays or diversions. Imports from Canada show a small, statistically insignificant decline, providing no evidence of an offsetting increase in Canadian supplies. Likewise, the response of imports from other OPEC suppliers is also not statistically distinguishable from zero. Imports from remaining suppliers initially decline, but subsequently recover and rise temporarily. These responses suggest that substitution across suppliers is not immediate and may develop over time, consistent with differences in crude grades and refinery requirements limiting such adjustment.

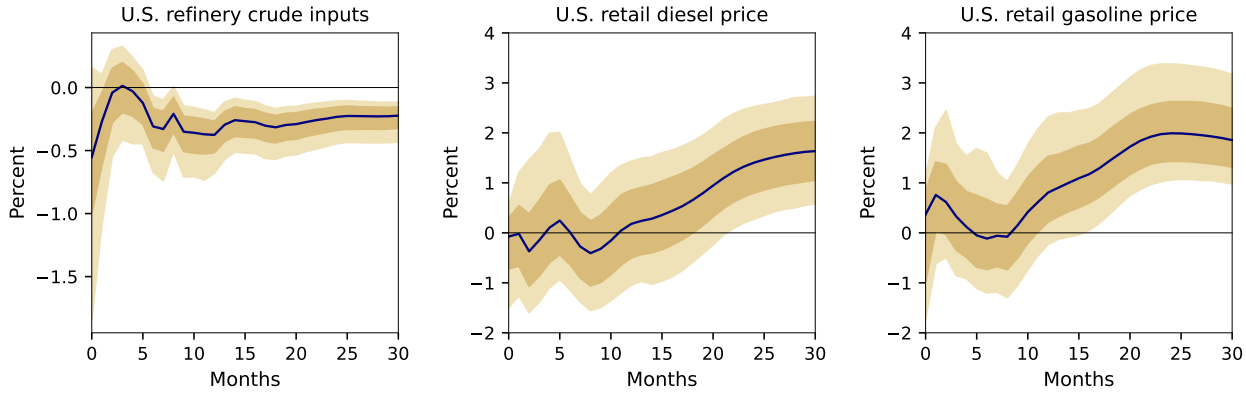
Figure 18: U.S. crude oil imports



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are obtained by augmenting the baseline VAR with each variable of interest separately. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. Other OPEC suppliers exclude Saudi Arabia; all remaining suppliers exclude OPEC and Canada.

Refinery activity and petroleum-product prices. We now examine whether disruptions to crude-oil deliveries are transmitted to refinery activity and downstream fuel prices. Figure 19 reports the responses of crude-oil inputs at U.S. refineries and blenders and of retail diesel and gasoline prices. Refinery inputs decline on impact, with the contraction becoming more precisely estimated around six months after the shock. This pattern is consistent with inventories initially cushioning refinery activity against disruptions to incoming deliveries. The lagged increase in retail diesel and gasoline prices is consistent with this gradual adjustment in refinery activity.

Figure 19: Refinery activity and fuel prices



Notes: Responses to an oil-shipping-cost shock normalized to raise the BDTI by 10 percent on impact. Each outcome is added separately to the baseline VAR. Dark and light bands show pointwise 68 and 90 percent bootstrap confidence intervals. Retail fuel prices are nominal; refinery inputs measure physical volumes.

However, these aggregate responses may mask regional differences, as refineries vary in their dependence on crude delivered from the Persian Gulf. We therefore examine whether regions with greater exposure experience larger adjustments in refinery activity and fuel prices following an oil-shipping-cost shock. For each of the five U.S. petroleum districts (PADDs), we measure monthly exposure, denoted by $E_{p,t}$, as Persian Gulf crude imports divided by refinery crude inputs, with both quantities expressed in barrels per day. Our baseline exposure measure ($\bar{E}_{p,t-1}$) is the average of this ratio over the twelve months preceding the shock (months $t-12$ through $t-1$). This measure captures recent dependence on Persian Gulf crude, smooths monthly fluctuations in deliveries and refinery operations, and uses only information available before the shock.

To study how responses vary with exposure, we interact this measure with the oil-shipping-cost shock recovered from the baseline VAR, $\widehat{\varepsilon}_t$. For each outcome and horizon h, we estimate the following panel local projection by ordinary least squares (Jordà, 2005):

$$\begin{aligned} y_{p,t+h} - y_{p,t-1} = \alpha_{p,h} + \beta_h \widehat{\varepsilon}_t + \gamma_h \widehat{\varepsilon}_t \bar{E}_{p,t-1} \\ + \delta_h \bar{E}_{p,t-1} + \theta_h D_t^{\text{Covid}} + u_{p,t+h}, \end{aligned} \tag{13}$$

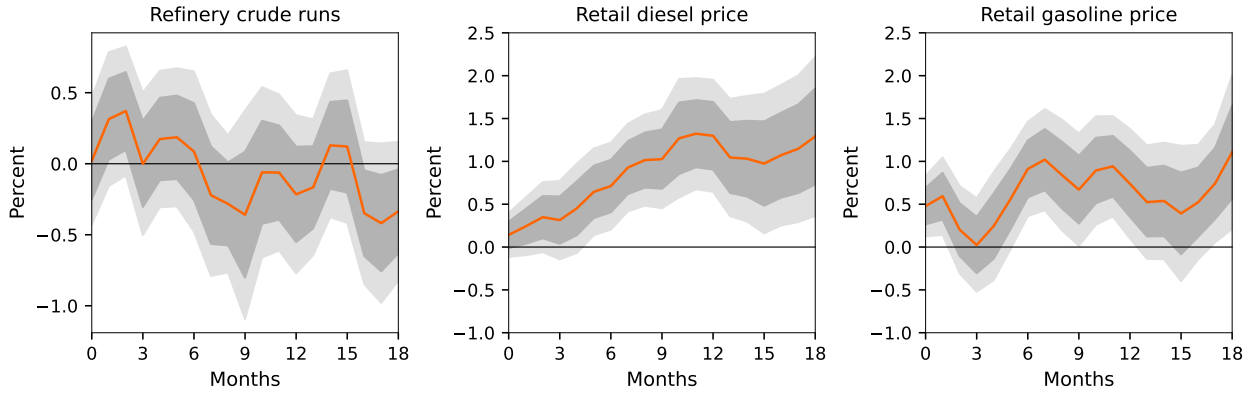
where $\alpha_{p,h}$ denotes regional fixed effects and D_t^{Covid} equals one from March through August 2020. We include both the common shock and its exposure interaction, together with the exposure main

effect. The outcomes are refinery crude inputs and retail diesel and gasoline prices, seasonally adjusted with X-13ARIMA-SEATS and expressed as 100 log. Due to data limitations, the estimation starts in January 2011 instead of January 2010. The coefficient γ_h measures how the response varies with exposure. Thus, $0.10\gamma_h$ gives the difference in responses in percentage points between districts whose exposure differs by ten percentage points. As in the baseline, the shock raises the BDTI by approximately 10 percent on impact.

Figure 19 reports the responses of crude-oil inputs at U.S. refineries and blenders, retail diesel and gasoline prices. Refinery inputs show no clear decline on impact but fall at medium and longer horizons. This pattern is consistent with refiners initially using inventories to maintain activity despite disrupted deliveries. The lagged increase in retail diesel and gasoline prices is consistent with this gradual adjustment in refinery activities.

We report pointwise 68 and 90 percent confidence bands based on Driscoll–Kraay standard errors. Figure 20 shows larger increases in diesel and gasoline prices in more exposed districts. At twelve months, a ten-percentage-point higher exposure is associated with increases of approximately 1.30 percent in diesel prices and 0.74 percent in gasoline prices. Refinery-input point estimates turn negative after the initial months, but are less precisely estimated. Appendix Figure D7 replaces the trailing average with exposure in the preceding month, holding the observations and estimation method fixed. The price responses remain similar, while refinery-input declines are more pronounced and better estimated with this exposure measure.

Figure 20: Regional exposure: differential response to an oil-shipping-cost shock



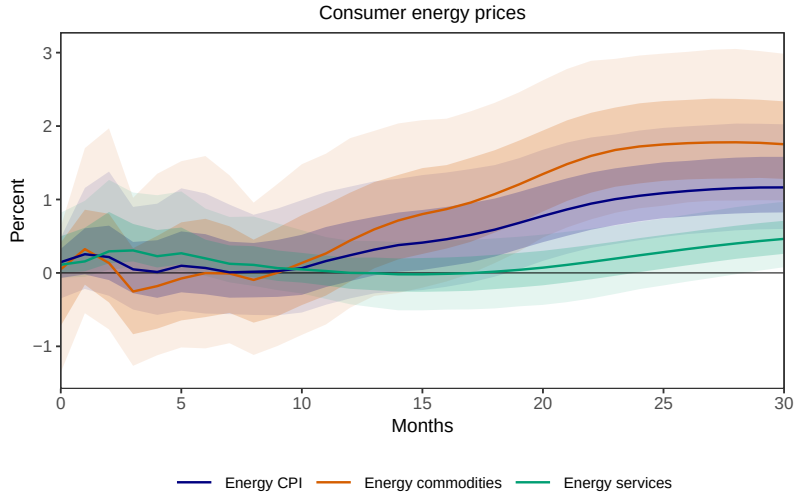
Notes: Estimates of $0.10\gamma_h$ in equation (13): differential responses for ten percentage points higher exposure to Persian Gulf crude, measured as the average import-to-refinery-input ratio over months $t-12$ through $t-1$, using unadjusted inputs. Outcomes are seasonally adjusted with X-13ARIMA-SEATS. The recovered baseline VAR shock is normalized to raise the BDTI by approximately 10 percent on impact. The regressions include regional fixed effects, the shock, its exposure interaction, the exposure main effect and a March–August 2020 dummy. Orange lines are point estimates; dark and light grey shaded areas are pointwise 68 and 90 percent degrees-of-freedom-adjusted Driscoll–Kraay intervals, with Bartlett weights and maximum lag $h+1$.

Energy prices and household consumption. We finally turn to households and examine how consumer energy prices respond at the national level. Figure 21 reports the responses of the energy CPI and its two components, namely energy commodities (mainly gasoline and fuel oil) and energy

services (electricity and gas). The energy CPI rises gradually, reaching approximately 1 percent after two years. The increase is larger for energy commodities and follows a similar timing to the rise in retail gasoline and diesel prices documented above. However, energy services prices increase only slightly and at longer horizons. This contrast is consistent with the more direct exposure of petroleum products to disruptions in crude-oil transportation.

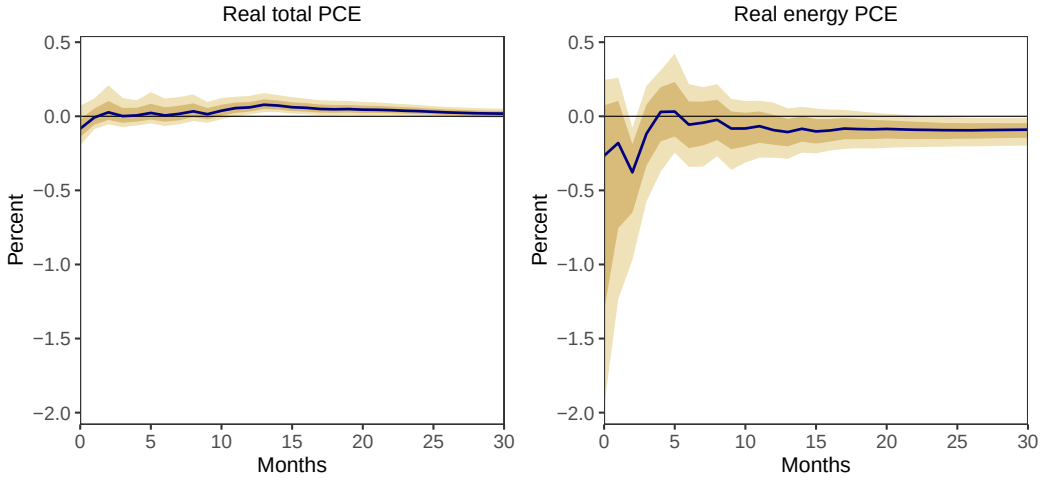
We complement this analysis by looking at real personal consumption expenditures (PCE) to assess how household consumption volumes respond. Figure 22 shows a persistent decline in real energy consumption. Coupled with the increase in consumer energy prices, this response is consistent with households consuming less energy as it becomes more expensive. By contrast, real total PCE falls modestly on impact (0.08 percent) and then rises after two months, but its response is imprecisely estimated.

Figure 21: The energy-trade channel: consumer energy prices



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are obtained by augmenting the baseline VAR with each variable of interest separately. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

Figure 22: Real personal consumption expenditures



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are obtained by augmenting the baseline VAR with each variable of interest separately. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

5.2.2 Manufacturing adjustment

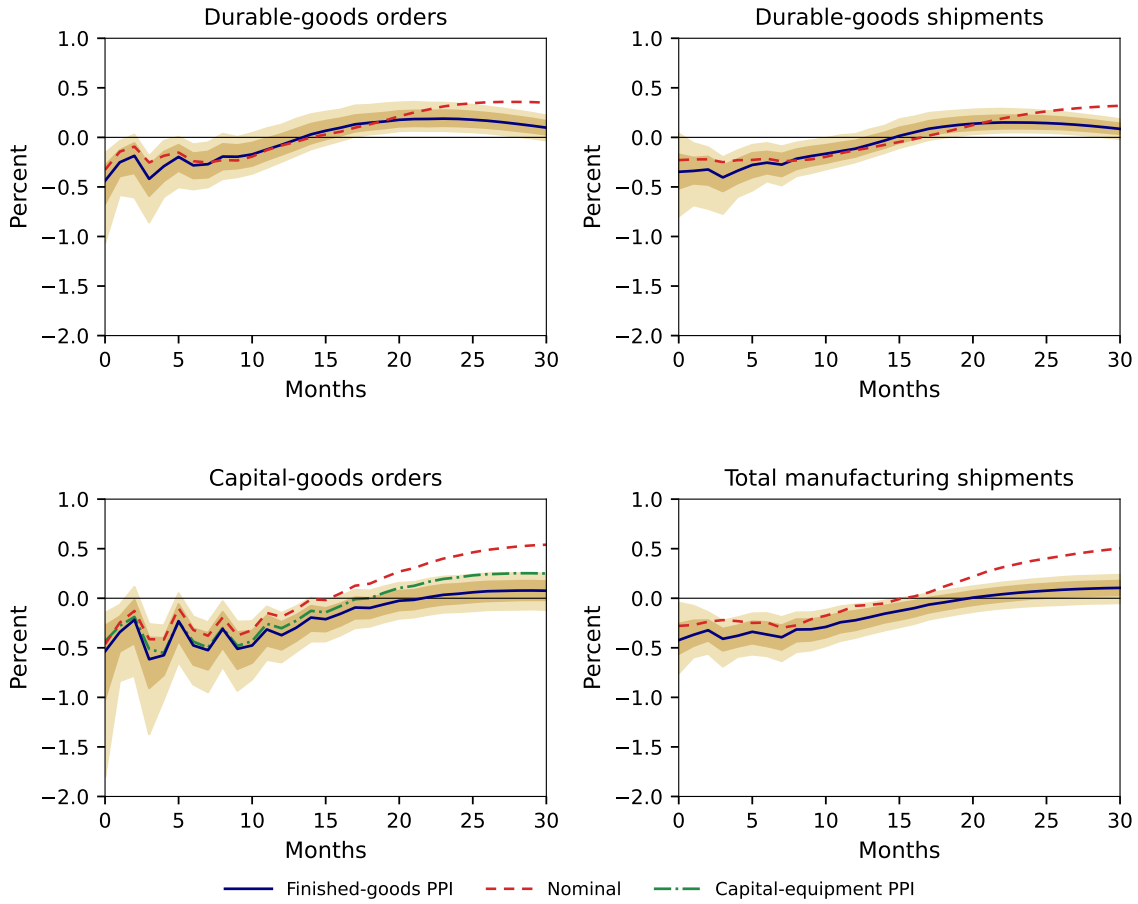
Another important transmission channel operates through adjustments in manufacturing activity following an oil-shipping-cost shock. We study the margins through which firms can adjust production rapidly, helping explain the immediate decline in industrial production documented in Section 5.1.

Manufacturing orders and shipments. We examine new orders and shipments to assess how manufacturing activity adjusts following the shock. We consider durable goods excluding transportation equipment, limiting the influence of large, irregular orders, particularly for aircraft. Likewise, to distinguish changes in activity from changes in prices, we deflate nominal values by the finished-goods PPI, giving us proxies for changes in volumes.

Figure 23 shows that durable-goods orders and shipments decline on impact by approximately 0.44 and 0.35 percent, respectively. Nondefense capital-goods orders excluding aircraft and total manufacturing shipments also fall, by approximately 0.54 and 0.42 percent. The dashed lines show similar declines in nominal values, indicating that the results are not driven by the price adjustment.[16] These responses are consistent with a precautionary adjustment by firms. One plausible channel is that heightened concerns about future oil deliveries may lead firms to postpone purchases and investment. The initial increase in oil-related geopolitical risk is consistent with this interpretation. The resulting decline in orders contributes to the immediate contraction in industrial production.

[16]Using the capital-equipment PPI to deflate capital-goods orders yields a similar impact decline of 0.43 percent, shown by the green line.

Figure 23: Manufacturing orders and shipments



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are obtained by augmenting the baseline VAR with each variable of interest separately. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. Navy lines show responses deflated by the finished-goods PPI; red dashed lines show nominal responses. The green line uses the capital-equipment PPI for capital-goods orders. Durable-goods orders and shipments exclude transportation equipment; capital-goods orders exclude defense and aircraft.

5.3 Global effects

5.3.1 Responses of global variables

In order to assess the external validity of our findings, we examine whether oil-shipping-cost shocks generate similar responses across a broader set of economies. We estimate a global SVAR-IV containing the BDTI, an OECD crude-inventory proxy, world crude oil production, real WTI prices, industrial production, consumer prices and oil-related geopolitical risk. Industrial production and consumer prices cover the OECD and six major non-member economies, and WTI prices are deflated by this broader consumer price index.[17] We keep a specification similar to that of the U.S.

[17]The inventory proxy equals OECD petroleum stocks multiplied by the U.S. crude-to-petroleum stock ratio and therefore does not measure worldwide crude inventories. The consumer price index combines BIS national CPIs using lagged annual nominal GDP weights from the World Bank. Its coverage comprises the July 2009 OECD membership plus

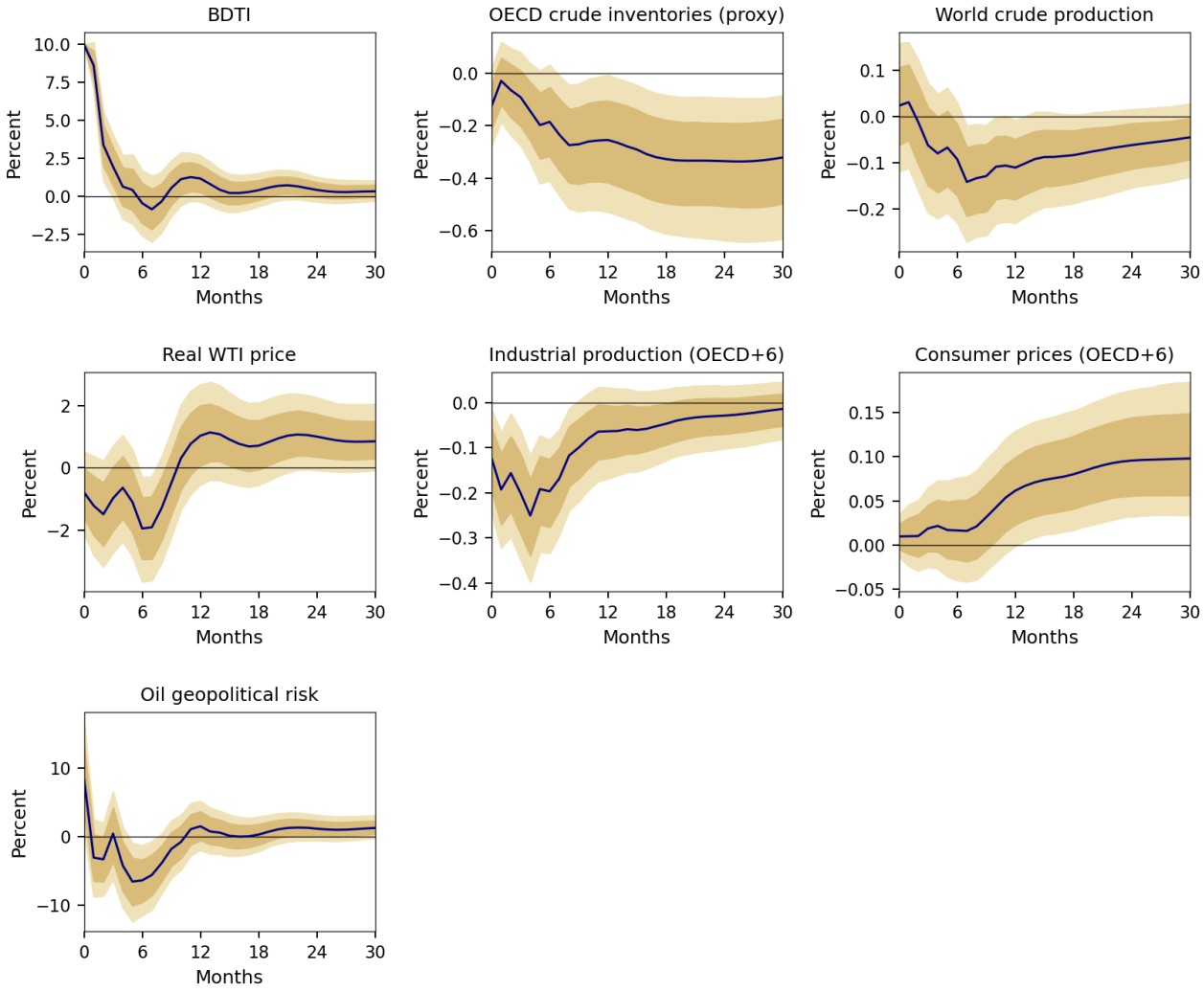
exercise, with all variables in log levels, six lags, a constant, a linear trend and the pandemic variance adjustment of Hamilton (Forthcoming). We also use the same instrument and data period, July 2009–February 2026. For inference, we use a wild bootstrap to accommodate heteroskedasticity (Gonçalves and Kilian, 2004; Bauer et al., 2026), with 5,000 replications as it leads to more precise estimates compared to the moving-block bootstrap for this exercise.[18]

Figure 24 shows a stagflationary response similar to the U.S. baseline. Industrial production falls by approximately 0.12 percent on impact and reaches a trough of about 0.20 percent after three to six months before gradually recovering. Consumer prices rise by about 0.06 percent after one year and 0.10 percent after two years. These responses tend to be smaller than in the United States. This difference likely reflects variation in exposure to oil-shipping disruptions across the economies included in the aggregates. Moreover, the oil-market responses also resemble those in the U.S. baseline. Indeed, the OECD crude-inventory proxy declines persistently, consistent with inventories cushioning disruptions. Reassuringly, world crude production shows no significant response on impact and declines by about 0.1 percent after one year, providing no evidence of an offsetting expansion or a large immediate decline in global production. Likewise, the real oil price rises after about one year and not on impact, and the increase is significant only at the 68 percent level. Finally, oil-related geopolitical risk rises on impact but quickly returns to zero and even falls below baseline after a few months. As in the U.S. results, this shows that the effects on activity and prices persist beyond the initial increase in geopolitical risk. One possible explanation for the subsequent decline is that military responses, particularly naval deployments, help restore security along affected shipping routes (Hacaga, 2020).

Brazil, China, India, Indonesia, Russia and South Africa. The appendix provides data sources and seasonal-adjustment procedures.

[18] In each replication, we multiply the reduced-form residuals and the instrument at each date by the same Rademacher draw and re-estimate the pandemic variance scale and the VAR.

Figure 24: Responses in the global VAR



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by approximately 10 percent on impact. Responses are obtained from the seven-variable global VAR. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively, based on a wild bootstrap with 5,000 replications.

5.3.2 Quantitative importance

How important is the oil-shipping shock for the macroeconomy? To answer this question, we compute the forecast error variance decomposition (FEVD) based on our world model. Using the notation of equations (4)–(6), the forecast error at horizon h, with $h = 0$ denoting impact, is $\mathbf{y}_{t+h} - \mathbb{E}_{t-1}\mathbf{y}_{t+h} = \sum_{k=0}^{h} \mathbf{B}_k \mathbf{S} \boldsymbol{\varepsilon}_{t+h-k}$, with $\mathbf{B}_0 = \mathbf{I}$. The share of the forecast error variance of variable i at horizon h explained by the shipping-cost shock $\varepsilon_{1,t}$ is

$$\text{FEVD}_i(h) \;=\; \frac{\sum_{k=0}^{h} \left(\mathbf{e}_i' \mathbf{B}_k \mathbf{s}_1\right)^2}{\sum_{k=0}^{h} \mathbf{e}_i' \mathbf{B}_k \boldsymbol{\Sigma} \mathbf{B}_k' \mathbf{e}_i}, \tag{14}$$

where $\mathbf{e}_i$ is the i-th column of the identity matrix, $\mathbf{s}_1$ is the first column of $\mathbf{S}$, i.e. the impact vector identified by the proxy and scaled such that $\text{Var}(\varepsilon_{1,t}) = 1$, and the denominator is the total mean squared error of variable i, since $\mathbf{\Sigma} = \mathbf{S}\mathbf{S}'$. Since (14) only requires $\mathbf{s}_1$ and $\mathbf{\Sigma}$, the contribution of $\varepsilon_{1,t}$ is identified without identifying the remaining structural shocks. Table 1 reports the estimated shares and their 90 percent confidence intervals. We compare these estimates with those reported for oil supply news shocks by Känzig (2021, Table 2). Differences in sample periods and variable coverage imply, however, that this comparison provides a benchmark for the magnitude of the contributions rather than a direct test of their differences.

Table 1: Forecast Error Variance Decomposition (World VAR)

Panel A: Shipping, oil market and global activity

Horizon	BDTI	Oil price	Oil inventories	Oil production	World IP
0	0.90	0.02	0.04	0.00	0.06
	[0.78, 0.97]	[0.00, 0.10]	[0.00, 0.16]	[0.00, 0.01]	[0.00, 0.21]
6	0.51	0.05	0.04	0.02	0.21
	[0.37, 0.65]	[0.01, 0.16]	[0.01, 0.16]	[0.00, 0.06]	[0.06, 0.43]
12	0.44	0.05	0.07	0.06	0.21
	[0.31, 0.57]	[0.02, 0.13]	[0.01, 0.27]	[0.01, 0.18]	[0.06, 0.42]
24	0.42	0.06	0.12	0.08	0.19
	[0.29, 0.55]	[0.02, 0.14]	[0.01, 0.37]	[0.02, 0.23]	[0.06, 0.39]
30	0.41	0.07	0.14	0.08	0.18
	[0.29, 0.54]	[0.02, 0.16]	[0.02, 0.39]	[0.02, 0.23]	[0.06, 0.38]

Panel B: Consumer prices and geopolitical risk

Horizon	World CPI	Oil GPR
0	0.01	0.06
	[0.00, 0.06]	[0.00, 0.18]
6	0.01	0.08
	[0.00, 0.05]	[0.04, 0.16]
12	0.03	0.09
	[0.00, 0.12]	[0.04, 0.17]
24	0.10	0.09
	[0.01, 0.30]	[0.04, 0.16]
30	0.12	0.09
	[0.02, 0.35]	[0.04, 0.16]

Notes: Entries are shares of forecast error variance explained by the oil-shipping shock. Horizons are in months. Brackets report 90 percent confidence intervals from a wild bootstrap with 5,000 replications.

Freight rates and the oil market. The oil-shipping-cost shock accounts for a large share of the dirty tanker freight-rate fluctuations. Indeed, it explains 90% of the BDTI forecast error variance on

impact, with a 90% confidence interval of 78%–97%, and 41% after thirty months. For comparison, Känzig (2021, Table 2) finds that oil supply news shocks explain 68% of oil-price forecast error variance on impact and 32%–39% at longer horizons. Our shock contributes considerably less to real oil-price fluctuations. This contrast is consistent with its interpretation as a disruption originating in oil transportation and not a production shock. The contribution to other variables builds more gradually. After one year, the shock explains 7% of the variability of crude inventories at the point estimate, but up to 27% at the upper end of the 90 percent confidence interval. Since these shares are averages over the whole sample, the shock can contribute much more to crude inventories in larger episodes of attacks on oil tankers. Its contribution to consumer prices reaches 12% after thirty months, consistent with the gradual price response documented above.

World industrial production. The shock also accounts for a substantial share of fluctuations in industrial activity. It explains 21% of the forecast error variance of world industrial production after six and twelve months and 19% after twenty-four months. These point estimates exceed those reported for oil supply news shocks by Känzig (2021, Table 2), although differences in samples and variable coverage limit a direct comparison. The contribution to industrial production increases over the first six months, consistent with shipping disruptions having effects that extend beyond their immediate impact Känzig and Raghavan (2026). One possible explanation for the difference relative to the oil supply news shock of Känzig (2021) is how the two shocks affect oil importers and exporters. Higher oil prices can benefit net exporters while hurting net importers, partly offsetting their effects on aggregate activity. Disruptions to oil transportation can instead impede deliveries for importers and access to markets for exporters. This mechanism could help explain the larger estimated contribution to world industrial production.

6 Sensitivity Analysis

This section finally assesses the interpretation of the identified shock and the robustness of our findings. We first compare attacks on oil tankers with attacks on other vessels, then examine predictability, the choice of event window, sensitivity to individual events and the treatment of the pandemic. We conclude with alternative samples and model specifications. Additional results are reported in Appendix E.

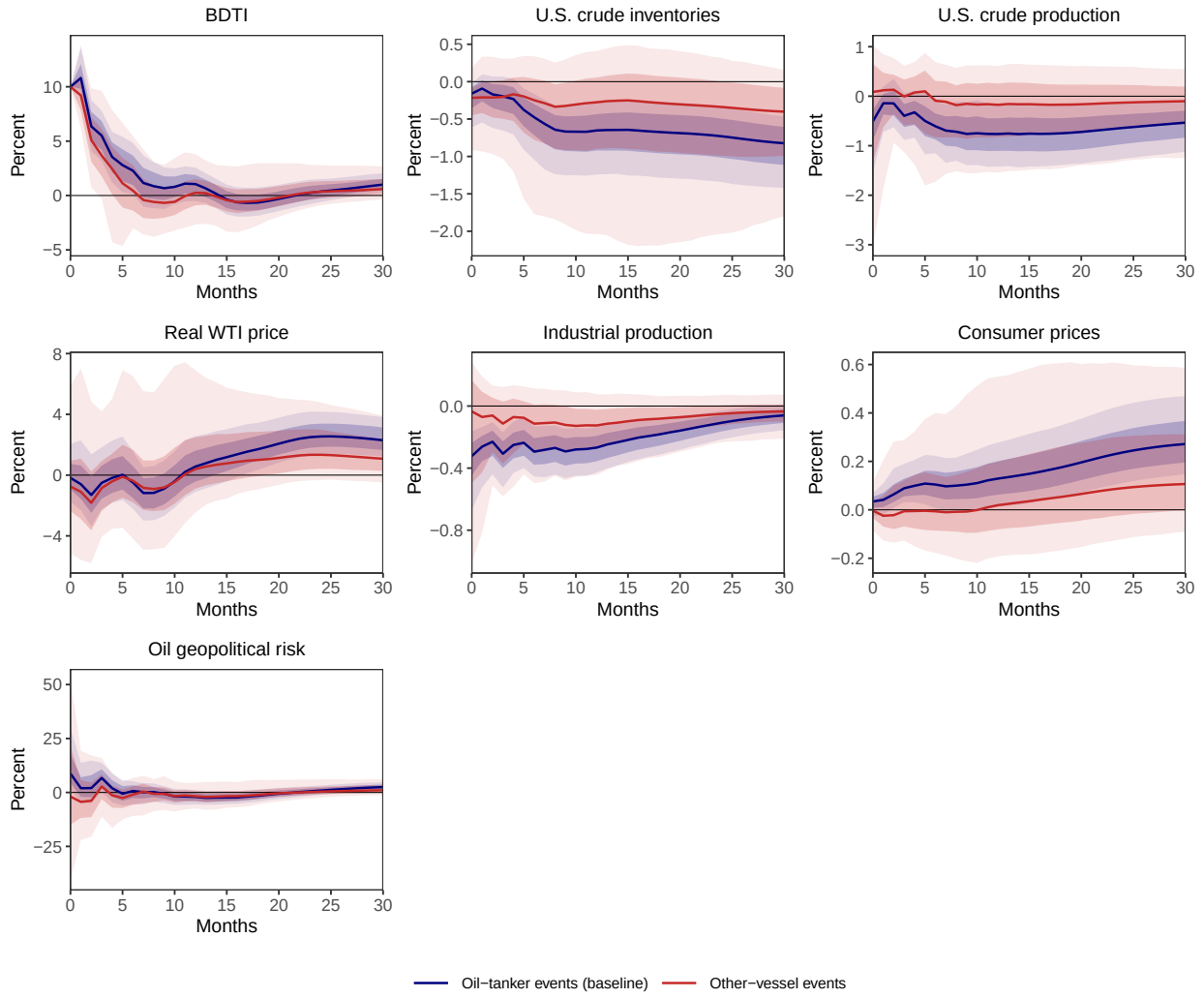
6.1 Attacks on Other Vessels

We first examine whether the results are specific to attacks on oil tankers or also arise following attacks on other vessels along the same shipping routes. Indeed, attacks on other vessel types provide a useful comparison as they occur in overlapping shipping regions and can also raise tanker freight rates by increasing concerns about route safety. The comparison sample contains 131 incidents involving other vessels over 2010–2025. Its narrative screen follows exactly the one presented in Section 3.1, except that we now focus on vessels other than crude-oil and heavy-fuel-oil tankers.

The sample hence includes general cargo ships, dry-bulk carriers, tankers carrying products, chemicals or gas, and other vessel types. As in the baseline sample, incidents are concentrated in the Somali piracy wave of the early 2010s and the Houthi campaign of 2023–2024.

For each event, we construct the BDTI surprise and monthly instrument using equations (2) and (3). We then estimate the same seven-variable proxy-SVAR, keeping the estimation sample, lag length, pandemic adjustment and bootstrap procedure unchanged. Both specifications are normalized to increase the BDTI by 10 percent on impact. The robust first-stage F-statistic is 48.9 for the other-vessel instrument, compared with 71.04 for the baseline instrument. Although the other-vessel instrument is relevant for the BDTI, Figure 25 emphasizes that the estimated oil-market and macroeconomic responses are mainly not statistically significant. In particular, only the responses of the WTI price, industrial production and crude inventories are significant at the 68 percent confidence level, mostly at long horizons. Moreover, they are all lower in magnitude than the baseline responses. These results hence suggest that broader maritime insecurity alone does not fully explain the responses to attacks on oil tankers.

Figure 25: Attacks on oil tankers and other vessels



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Blue denotes identification using attacks on oil tankers; red denotes identification using attacks on other vessels. Solid lines are point estimates; dark and light shaded areas indicate 68 and 90 percent confidence intervals, respectively.

6.2 Are the Surprises Predictable?

High-frequency surprises may reflect information available before the event, which can complicate their interpretation as shocks (Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023). Following Bauer and Swanson (2023), we assess predictability by estimating the event-level regression

$$\text{SCS}_d = \alpha + \boldsymbol{\beta}' \mathbf{X}_{d-} + \eta_d, \tag{15}$$

where SCS_d is the shipping-cost surprise associated with attack d and $\mathbf{X}_{d-}$ contains information dated strictly before the attack. We estimate these event-level regressions by OLS and report heteroskedasticity-robust standard errors. Table 2 reports four nested specifications. Column (a)

includes the surprise components of the latest U.S. industrial production, ISM manufacturing and trade balance releases. Column (b) adds changes in equity prices and the yield curve slope, and column (c) adds changes in WTI prices and oil-price volatility. Column (d) also includes geopolitical risk measures. Financial, commodity and geopolitical predictors are measured as three-month changes, with transformations specified in the table. No individual predictor is statistically significant. The WTI coefficient is negative, with $p = 0.113$ in column (c) and $p = 0.276$ in column (d). The R^2 rises from 0.096 to 0.253 when WTI prices and oil-price volatility are added in column (c), but adding WTI alone to column (b) raises the R^2 to 0.229, hence accounting for most of this increase. Consequently, as an additional check, we regress the event-level surprises on a constant and the pre-event three-month WTI log change, then aggregate the residuals using the same monthly weights as in the baseline. The oil-price-purged event-level series has a correlation of 0.95 with the raw surprises (Appendix Figure E1). Reassuringly, using the oil-price-purged instrument yields responses similar to those in the baseline (Appendix Figure E2), although the responses of consumer prices and U.S. crude production become less precisely estimated. These results suggest that the main findings are not driven by the component of the surprises associated with prior WTI movements. Finally, our event-selection criteria aim to exclude incidents tied to broader geopolitical conflict. As an additional check, we purge the event-level surprises of the pre-event three-month change in oil-related geopolitical risk, using the same method. As in the exercise with oil prices, the resulting impulse responses remain broadly unchanged (Appendix Figure E3), suggesting that our results are not driven by prior changes in oil-related geopolitical risk.

Table 2: Predictability of the shipping-cost surprises

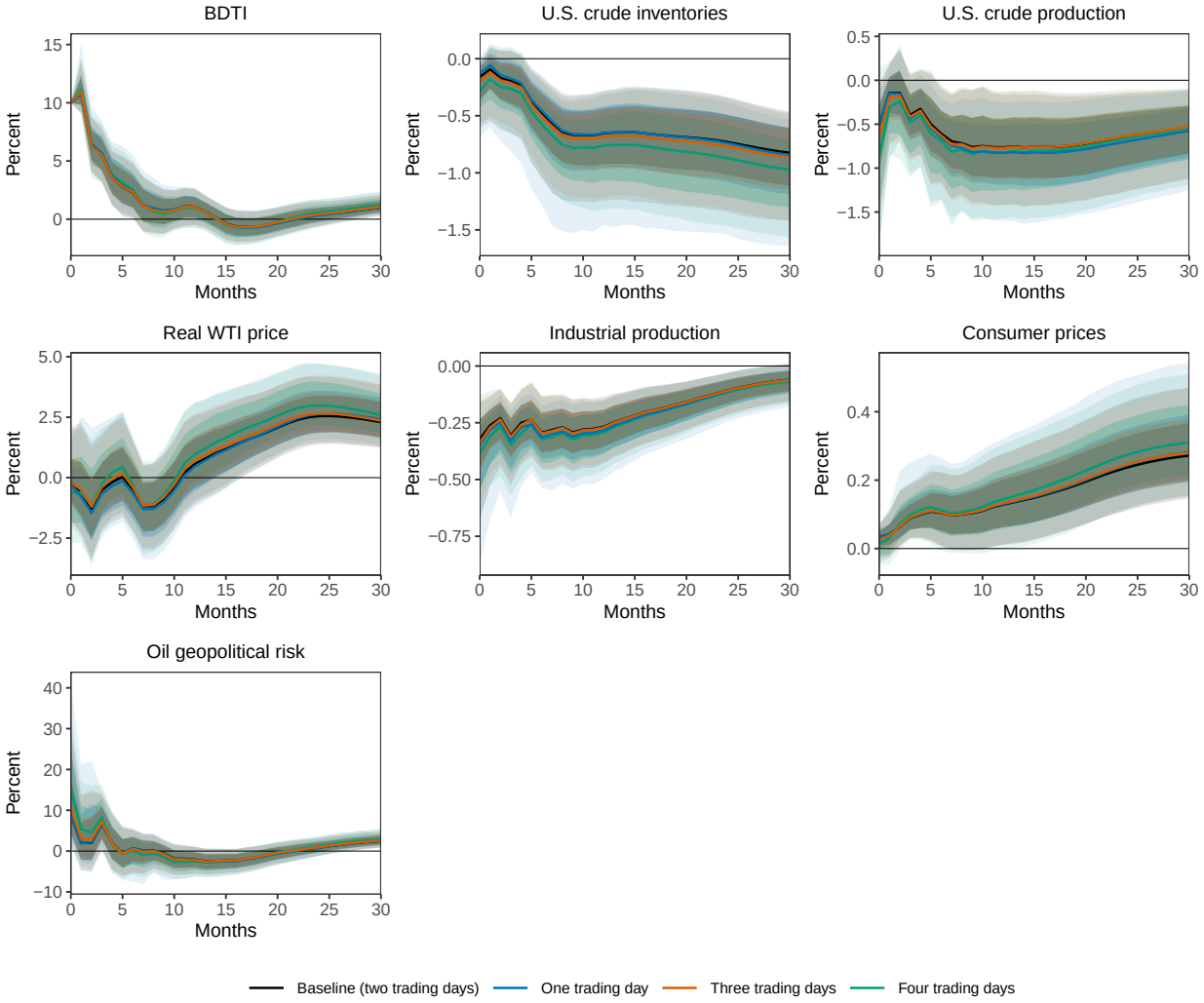
Shipping-cost surprise:	(a) Macro news	(b) Financials	(c) Commodities	(d) Other
IP surprise	2.637	4.568	2.910	2.672
	(2.741)	(3.305)	(3.174)	(3.099)
ISM surprise	-0.004	-0.003	-0.005	-0.004
	(0.005)	(0.005)	(0.005)	(0.006)
Trade balance surprise	0.002	0.003	0.008	0.006
	(0.003)	(0.003)	(0.005)	(0.006)
S&P 500 (3M log change)		0.218	0.301	0.266
		(0.274)	(0.247)	(0.261)
Yield curve slope (3M change)		-0.036	-0.053	-0.045
		(0.041)	(0.048)	(0.054)
WTI price (3M log change)			-0.188	-0.179
			(0.113)	(0.159)
OVX (3M log change)			0.041	0.019
			(0.032)	(0.038)
Oil GPR (3M change)				-3.1e-05
				(4.02e-05)
GPR (3M log change)				0.026
				(0.024)
Middle East GPR (3M change)				3.65e-05
				(0.0001)
R^2	0.062	0.096	0.253	0.278
Adj. R^2	-0.046	-0.093	0.016	-0.102
N	30	30	30	30

Notes: Estimated coefficients from predictive regressions (15) of the daily shipping-cost surprises on information available before each attack. Column (a) includes the Bloomberg surprise components of the most recent U.S. industrial production, ISM manufacturing and trade balance releases; column (b) adds the three-month log change in the S&P 500 and the three-month change in the yield curve slope; column (c) adds the three-month log changes in the WTI price and in the OVX; column (d) adds the three-month log change in geopolitical risk (GPR) and the three-month level changes in oil-related and Middle East geopolitical risk. These three measures are from the AI-based indices of Iacoviello and Tong (2026). HC1 robust standard errors in parentheses. ***, **, * denote significance at the 1, 5 and 10 percent levels. $N = 30$ events.

6.3 Choice of Event Window

The event window must allow enough time for news of an attack to be reflected in the BDTI while limiting the influence of unrelated developments (Pinchetti, 2026; Verduzco-Bustos and Zanetti, 2026). We assess whether our results are sensitive to the baseline window of two trading days by reconstructing the instrument using alternative windows and re-estimating the baseline SVAR-IV. Figure 26 shows that the main response patterns remain similar across the shorter or longer window specifications. We also assess sensitivity to monthly aggregation. The main impulse responses remain similar when the instrument is constructed as the unweighted sum of daily surprises rather than their linearly weighted sum (Appendix Figure E16).

Figure 26: Impulse responses by the width of the surprise window

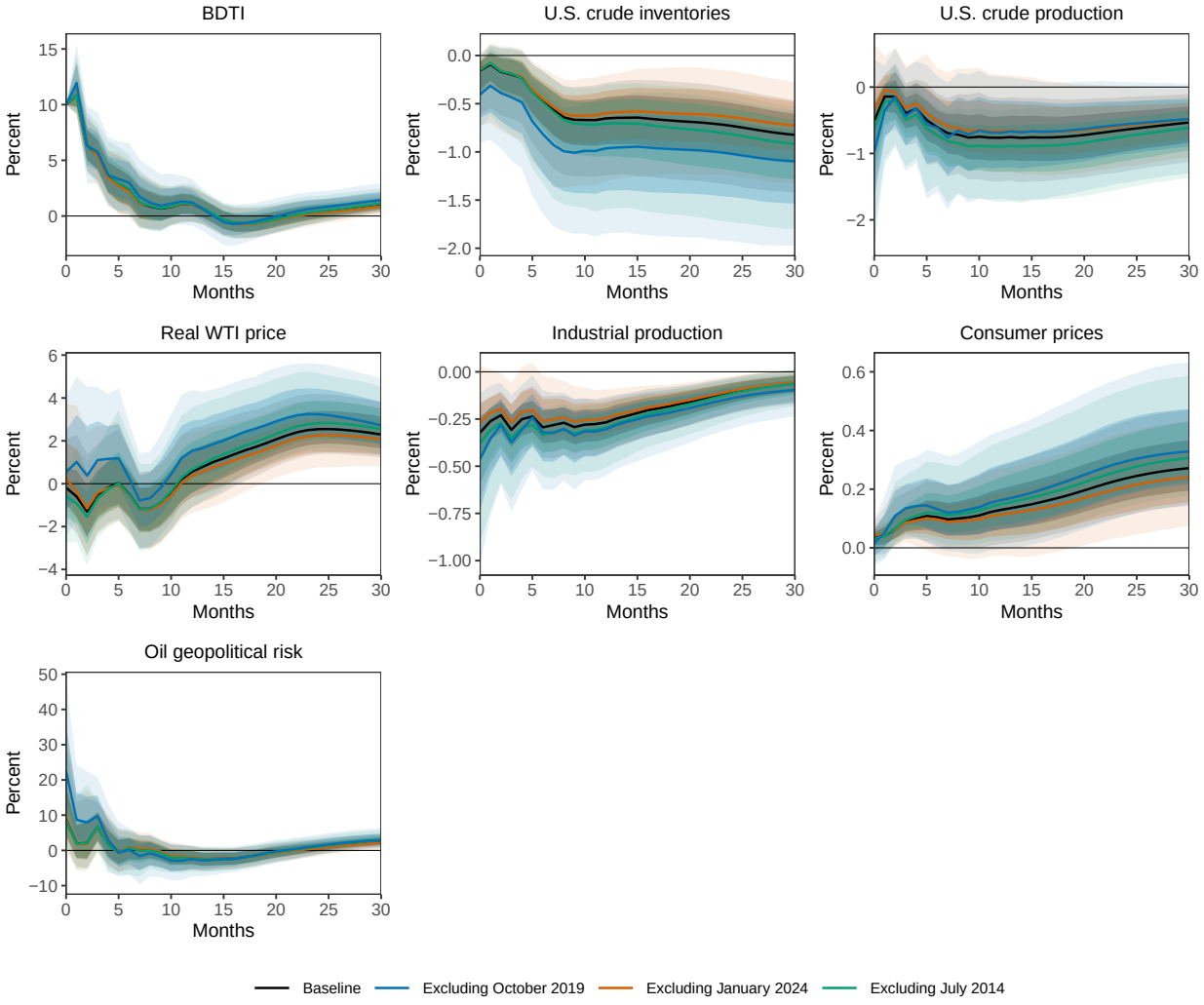


Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Surprises are measured as $\log P^{\text{BDTI}}_{d+W} - \log P^{\text{BDTI}}_{d-1}$ for $W = 0, 1, 2, 3$ forward trading days; $W = 1$ is the baseline. The solid black line reports the baseline; the other colors distinguish the alternative event windows. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. Bands are not robust to weak instruments.

6.4 Is a Single Event Driving the Results?

To assess sensitivity to individual events, we set each of the three largest monthly surprises to zero, one at a time, and re-estimate the model. These observations are October 2019, January 2024 and July 2014. Figure 27 compares the resulting responses with the baseline. Reassuringly, the response patterns remain similar, with the confidence bands largely overlapping, which suggests that no single event drives our findings.

Figure 27: Excluding the three largest surprises, one at a time



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The black line reports the baseline; colored lines report estimates after setting the monthly instrument to zero in October 2019, January 2024 or July 2014, one at a time. Dark and light shaded areas are 68 and 90 percent confidence bands for each alternative specification, respectively.

As a separate diagnostic, we draw 1,000 samples of 30 trading dates, excluding windows around the 30 retained attacks and March–May 2020. For each sample, we construct daily surprises and aggregate them into a monthly instrument using the baseline procedure. We then use each placebo instrument in our baseline VAR. The median heteroskedasticity-robust first-stage F-statistic is 3.00. Among these instruments, only 1.5 percent reach or exceed the baseline value of 71.04 (Appendix Figure E4) and the vast majority of them produce a first-stage below 10. These results indicate that the selected attack dates provide stronger first-stage relevance than virtually all randomly selected dates. We also estimate impulse responses using each placebo instrument, normalizing the impact on the BDTI to 10 percent. Appendix Figure E5 reports the pointwise 5th–95th percentile range across these responses. The placebo responses are widely dispersed, consistent

with the generally weak first stages obtained from randomly selected dates (Känzig, Forthcoming).These results provide reassurance that the strong relationship between our instrument and shipping costs is specific to the selected attack dates rather than a typical feature of randomly chosen dates.

6.5 Is the Pandemic Driving the Results?

The pandemic raises two main concerns. The first is that the exceptional volatility of 2020 may affect the VAR estimates. The baseline addresses this concern through the heteroskedasticity adjustment described in Section 4.2, which downweights observations from March–August 2020 using an estimated variance scaling factor. We assess sensitivity to both the adjustment and the period over which it is applied. First, we replace it with separate monthly dummies for January–June or January–December 2020 (Appendix Figure E6). The responses remain close to our baseline results under both specifications, with some small differences in magnitude when the dummies extend through December. Second, we retain the variance adjustment but apply it to January–June or January–December 2020 instead of March–August (Appendix Figure E7). These responses also remain close to what we find in our baseline model. Overall, these results indicate that our main findings are robust to changes in the adjustment window.

The second concern is that news about the pandemic may contaminate the instrument around attack dates. To limit this risk, our event-selection procedure excludes incidents during the onset of the pandemic in March–May 2020. This restriction complements the treatment of pandemic volatility in the VAR.

6.6 Alternative Samples and Model Specifications

Sample period. We first assess sensitivity to the sample period. Appendix Figure E8 reports estimates for January 2010–December 2023, excluding the retained attacks from the 2023–2024 Houthi campaign and subsequent periods. The responses remain similar to the baseline. Additionally, Appendix Figure E9 shows estimates for a sample starting in January 2016, which excludes the early Somali piracy episode. The main response patterns are again preserved, except for oil-related geopolitical risk, whose response is close to zero. However, estimates in this shorter samples tend to be less precise (i.e., they have wider confidence bands), consistent with fewer observations and identifying events.

Model specification and estimation. We next examine whether the results depend on the lag length, estimation method or variables included in the model. Allowing for richer dynamics by increasing the lag length to twelve preserves the main response patterns (Appendix Figure E10). To assess sensitivity to estimation in a relatively short sample (approximately 16 years), we also estimate a Bayesian VAR-IV in the spirit of Miranda-Agrippino et al. (2026), with a Normal-Wishart prior and Minnesota-type shrinkage, which regularizes the VAR coefficients. This specification preserves the main response patterns, although the inventory and CPI responses are smaller and their 90 percent posterior credible bands include zero at longer horizons (Appendix Figure E11).

Moreover, a VAR also extrapolates long-horizon responses from a finite number of lags, which can induce lag-truncation bias. To relax this restriction, we estimate a local projection in which each variable at horizon h is regressed on the structural shock recovered from the baseline SVAR-IV and six lags of the dependent variable, after residualizing each series on a constant and a dummy for March–August 2020. We use heteroskedasticity-robust standard errors. We project on the recovered shock rather than on the instrument because estimating a separate LP-IV regression at each horizon would be difficult given the relatively short instrument window and hence would severely reduce the effective sample size at long horizons. The local projections are consistent with our baseline results (Appendix Figure E12) and as expected, the LP results display wider bands, consistent with the larger sampling variability of this estimator.

Financial variables. We then examine whether explicitly including financial conditions changes the estimated responses. Following Mori and Peersman (2026), we add the three-month Treasury bill rate, the S&P 500, the VIX and the oil volatility index (OVX), one at a time. These variables capture short-term interest rates, equity prices and implied volatility in equity and oil markets. Appendix Figures C2, C3, E13 and C4 show that the main response patterns remain similar, even though some confidence bands widen, consistent with the additional parameters estimated in these larger models.

Freight measures. We also assess sensitivity to the freight index by replacing the BDTI with alternative measures while retaining our instrument based on attacks on oil tankers. The Baltic Clean Tanker Index tracks freight rates for refined petroleum products. Using this index yields a robust first-stage F-statistic of 19.54 and preserves the main response patterns, although confidence bands are wider (Appendix Figure E14). The Baltic Dry Index instead tracks freight rates for dry-bulk commodities and yields a much lower F-statistic of 2.36. This weak relationship is consistent with our instrument's focus on oil transportation and not dry-bulk shipping. For completeness, we report the responses in Appendix Figure E15. We find no clear evidence of a response, although the weak first stage limits the interpretation of these estimates.

7 Conclusion

In this paper, we study the macroeconomic effects of attacks on oil tankers using a narrative record of events and high-frequency changes in the shipping costs of crude oil. We find that these attacks have stagflationary effects on the U.S. economy as well as on the global economy. Declining crude inventories and the responses of imports, refinery activity and petroleum-product prices point to an energy-trade channel. Moreover, immediate declines in manufacturing orders and shipments are consistent with a precautionary adjustment in firms' purchases and investment plans. Importantly, these responses differ from those associated with conventional oil shocks and dry-bulk shipping shocks. Finally, we show that attacks on other vessels along the same shipping routes produce macroeconomic responses that are generally not statistically distinguishable from zero. This suggests that the effects of tanker attacks reflect disruptions specific to oil transportation, rather than

concerns about route safety alone.

Our findings have important implications for energy security policy. While greater domestic oil production can make the U.S. economy more resilient to conventional oil shocks, our results show that it remains vulnerable to disruptions in oil transportation through its dependence on imported crude grades. Assessing this vulnerability therefore requires considering refinery requirements, the availability of substitute supplies and the reliability of maritime transportation. Moreover, the ability of non-state actors to disrupt shipping highlights a dimension of geopolitical risk that receives less attention in debates centered on relations between states. Addressing these risks requires policy responses adapted to their incentives and constraints, which differ from those of governments. Recent developments in the Red Sea underscore the continuing threat that non-state actors pose to the security of oil transportation. Moreover, the global results suggest that the economic costs extend beyond the United States. One possible explanation we put forward is that shipping disruptions impede both deliveries to importers and exporters' access to markets. Their effects may therefore offset each other less than those of conventional oil shocks, for which higher oil prices can benefit net exporters while hurting net importers. Examining these differences across countries is an important direction for future research.

Ongoing work extends the analysis in four directions. First, we are expanding the narrative record by identifying additional attacks and extending the sample period. Second, we investigate the transmission mechanisms in greater detail using micro data and AIS vessel-tracking data. Third, building on the global analysis, we extend the analysis to Europe and Japan to study whether greater dependence on imported oil makes economies more vulnerable to tanker attacks. Finally, we are developing a quantitative macroeconomic model to distinguish the roles of delivery delays, transport risk and substitution constraints, and to assess the welfare effects of policies that reduce exposure to maritime disruptions.

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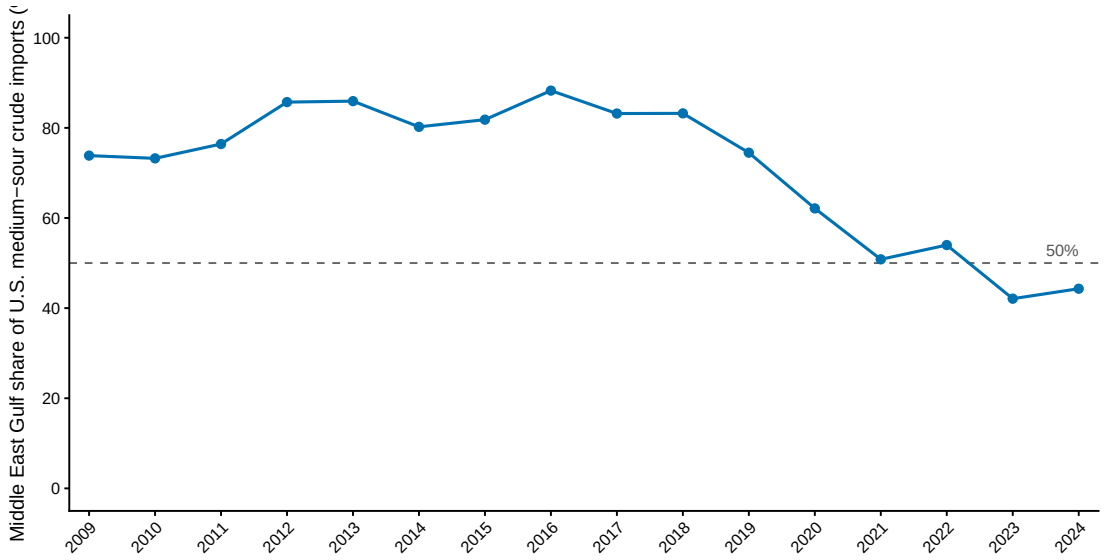
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A Data

Figure A1: Middle East Gulf share of U.S. medium sour crude oil imports



Notes: Annual share of U.S. medium sour crude imports supplied by the Middle East Gulf. The dashed line marks 50 percent. Source: EIA Form EIA-814, company-level imports (U.S. Energy Information Administration, 2026a).

Table A1: Data description, sources and transformations

Variable	Description and identifier	Source	Treatment
Instrument			
BDTI (daily)	Baltic Dirty Tanker Index, daily assessment	Baltic Exchange / Bloomberg	100 Δ log
Baseline VAR (monthly)			
BDTI	Baltic Dirty Tanker Index, monthly average of daily values	Baltic Exchange / Bloomberg	100 log
Inventories	U.S. crude oil stocks, excluding the Strategic Petroleum Reserve	EIA	100 log; X-13
Production	U.S. field production of crude oil	EIA	100 log; X-13
Real WTI	`WTISPLC` (NSA) deflated by the seasonally adjusted U.S. CPI (`CPIAUCSL`)	FRED	100 log ratio
GPR oil	Oil-related geopolitical risk index	Iacoviello and Tong (2026)	100 log
IP	U.S. industrial production, total index	FRED	100 log; SA
CPI	U.S. consumer price index, all items (`CPIAUCSL`)	FRED	100 log; SA
Sectoral production and employment (monthly)			
Manufacturing production	`IPMAN`	FRED	100 log; SA
Mining production	`IPMINE`	FRED	100 log; SA
Utilities production	`IPUTIL`	FRED	100 log; SA
Services employment	`SRVPRD`	BLS via FRED	100 log; SA
Durable manufacturing	`IPDMAN`	FRED	100 log; SA
Nondurable manufacturing	`IPNMAN`	FRED	100 log; SA
Food	`IPG311S`	FRED	100 log; SA
Petroleum and coal products	`IPG324S`	FRED	100 log; SA
Chemicals	`IPG325S`	FRED	100 log; SA
Plastics and rubber	`IPG326S`	FRED	100 log; SA
Fabricated metals	`IPG332S`	FRED	100 log; SA
Machinery	`IPG333S`	FRED	100 log; SA
Computers and electronics	`IPG334S`	FRED	100 log; SA
Motor vehicles and parts	`IPG3361T3S`	FRED	100 log; SA
Paper	`IPG322S`	FRED	100 log; SA
Defense and space equipment	`IPB52300S`; defense and space market group, not aerospace manufacturing alone	FRED	100 log; SA
Production by product group	Consumer goods; intermediate products; business equipment; construction supplies. Constant-price production series	FRED	100 log; SA
Manufacturing orders, shipments and deflators (monthly)			
Durable-goods orders	`ADXTNO`; Excluding transportation equipment; nominal values in millions of dollars	Census M3 via FRED	100 log nominal or PPI-deflated; SA
Durable-goods shipments	`ADXTVS`; Excluding transportation equipment; nominal values in millions of dollars	Census M3 via FRED	100 log nominal or PPI-deflated; SA

Continued on next page

Table A1 continued

Variable	Description and identifier	Source	Treatment
Capital-goods orders	`NEWORDER`; Nondefense capital goods excluding aircraft; nominal values in millions of dollars	Census M3 via FRED	100 log nominal or PPI-deflated; SA
Manufacturing shipments	`AMTMVS`; Total manufacturing; nominal values in millions of dollars	Census M3 via FRED	100 log nominal or PPI-deflated; SA
Finished-goods PPI	`WPSFD49207`; common deflator for orders and shipments	BLS via FRED	SA; used as deflator
Private capital-equipment PPI	`WPSFD41312`; alternative deflator for capital-goods orders	BLS via FRED	SA; used as deflator
Oil deliveries, refinery activity and retail fuel prices (monthly)			
Imports by mutually exclusive origin	Saudi Arabia; other OPEC suppliers; Canada; all remaining suppliers. Constructed from crude-oil import volumes, converted to thousand barrels per day before adjustment	EIA; own calculations	100 log; X-13
U.S. refinery and blender inputs	`MCRRIUS2`; crude-oil inputs, thousand barrels per day	EIA	100 log; X-13
U.S. retail diesel price	`EMD_EPD2D_PTE_NUS_DPG`; No. 2 diesel, dollars per gallon	EIA	100 log; X-13
U.S. retail gasoline price	`EMM_EPM0_PTE_NUS_DPG`; all grades and formulations, dollars per gallon	EIA	100 log; X-13
PADD refinery inputs	PADDs 1–5; refinery and blender crude inputs, converted from monthly barrels to barrels per day	EIA	100 log; X-13
PADD retail fuel prices	PADDs 1–5; gasoline (all grades and formulations) and diesel, monthly averages of weekly dollar-per-gallon prices	EIA	100 log; X-13
PADD Persian Gulf exposure	Persian Gulf crude imports divided by refinery crude inputs in common daily units; previous month or trailing 12-month mean ending in the previous month	EIA; own calculations	Ratio from NSA inputs; no log
Household consumption and geopolitical risk (monthly)			
Energy CPI	CPI energy, energy commodities and energy services	FRED	100 log; SA
Energy shortages (U.S. extension)	Energy component of the newspaper-based shortage index	Caldara et al. (2026)	Level; NSA
Real total PCE	`PCEC96`; real household consumption expenditure	BEA via FRED	100 log; SA
Real energy PCE	`DNRGRX1M020SBEA`; real energy consumption expenditure	BEA via FRED	100 log; SA
Oil threats and oil acts	`GPR_OIL_THREATS` and `GPR_OIL_ACTS`; components replace total oil GPR in separate VARs	Iacoviello and Tong (2026)	100 log; no seasonal adjustment
Global VAR extension (monthly)			
World crude production	Datastream "Crude Oil Prod. World MBBL/DAY – TURNOVER BY VOLUME"; includes condensate	EIA via Datastream	100 log; X-13
OECD crude-inventory proxy	OECD petroleum stocks times the U.S. crude-to-petroleum stock ratio: EIA1976 × EIA1533 / EIA1541	EIA via Datastream; own calculations	100 log; X-13
OECD+6 industrial production	Published aggregate for OECD economies and six major non-member economies	Baumeister–Hamilton dataset	100 log; supplied SA

Continued on next page

Table A1 continued

Variable	Description and identifier	Source	Treatment
World CPI proxy	Chain-linked CPI for the July 2009 OECD membership plus Brazil, China, India, Indonesia, Russia and South Africa; lagged annual nominal GDP weights	BIS; World Bank; own calculations	100 log; X-11
Real WTI in global VAR	WTISPLC deflated by the constructed world CPI; both components seasonally adjusted	FRED; BIS; World Bank	100 log ratio
Energy shortages in global VAR	Energy component of the news-based shortage index; frozen vintage also used in the earlier U.S. exercise	Caldara et al. (2026)	Level, index points; X-11
Financial variables and robustness controls (monthly)			
Treasury bill rate	`TB3MS`; three-month U.S. Treasury bill rate	FRED	Level, percentage points
Shadow rate	Wu–Xia shadow rate through June 2023; effective federal funds rate thereafter	Wu and Xia (2016); FRED	Level, percentage points
S&P 500	Monthly equity-price index	Market data	100 log
VIX and OVX	Equity-market and crude-oil implied volatility indices	Cboe via FRED	100 log
Clean-tanker freight index	Baltic Clean Tanker Index (BCTI); alternative freight-price measure, using the adjusted series	Baltic Exchange / Bloomberg	100 log; seasonally adjusted
Oil-supply shock	Published monthly oil-supply shock, added to the baseline VAR	Baumeister and Hamilton (2019)	Source-series units
Dry-bulk shipping shock	Published monthly shipping shock; common sample ends in December 2022	Känzig and Raghavan (2026)	Source-series units
Monthly diagnostic series			
Supply-chain pressure	Global Supply Chain Pressure Index	Federal Reserve Bank of New York	Published index
Policy uncertainty	U.S. EPU and global EPU	Baker et al. (2016); Davis (2016)	Published indices
WTI expectations	Three-month WTI expectation series	Baumeister (2023)	Source-series units
Other comparison shocks	Oil, gas, monetary policy, climate, shipping and geopolitical shocks used in the correlation table	Sources listed in Table B2	Published shock series

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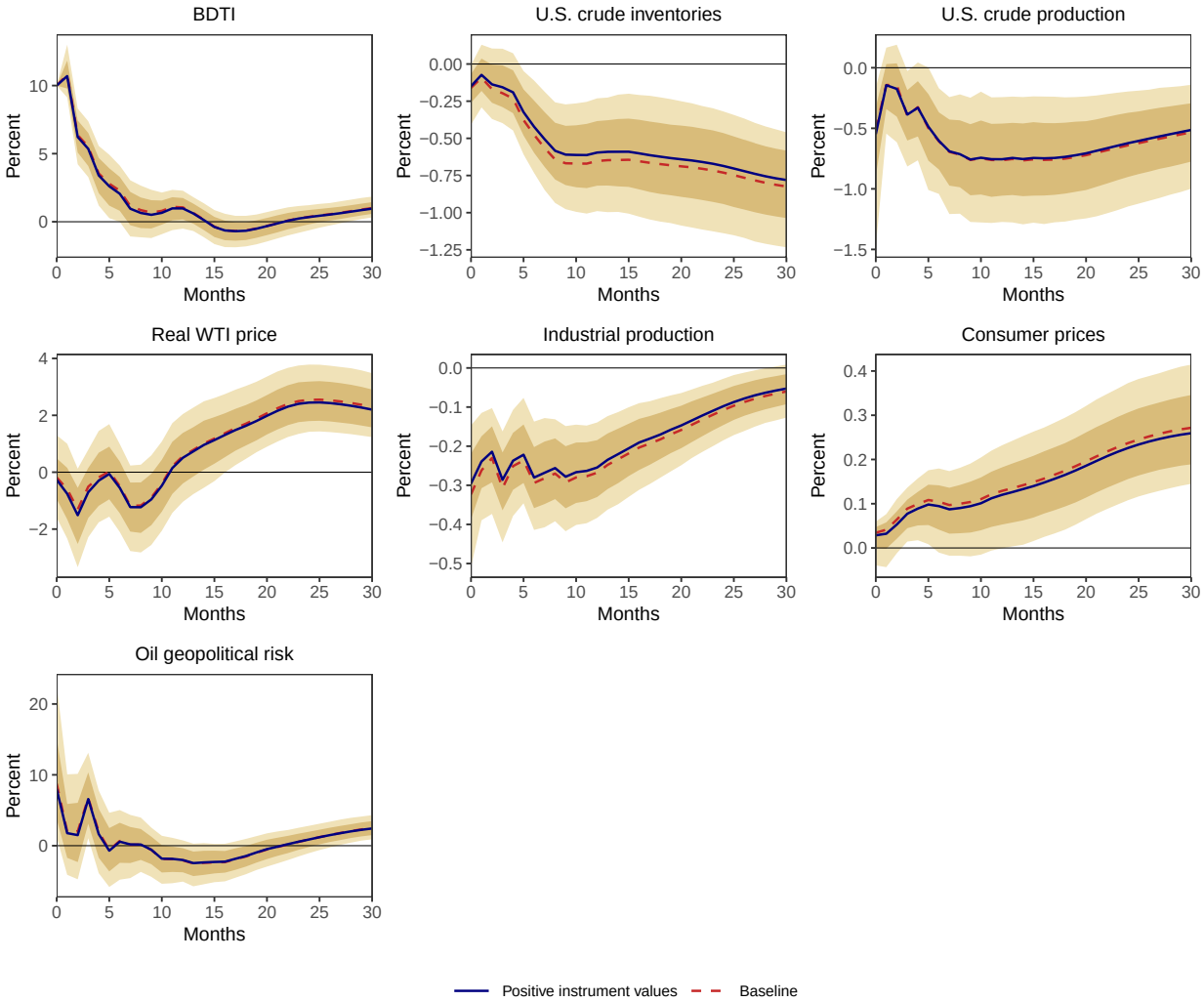
Table A1 continued

Variable	Description and identifier	Source	Treatment
Other retained outcomes and port measures			
Manufacturing unemployment	Monthly `LNU04032232`; previous industry of unemployed workers	BLS via FRED	Level; X-13
Mining, oil and gas unemployment	Monthly `LNU04032230`; previous industry of unemployed workers	BLS via FRED	Level; X-13
Transportation/utilities unemployment	Monthly `LNU04032236`; previous industry of unemployed workers	BLS via FRED	Level; X-13
Vessel dwell time	Monthly tanker/liquid-bulk vessel dwell times at U.S. ports; January 2019–June 2023	U.S. Bureau of Transportation Statistics	Level, hours
Port congestion components	Monthly congestion rates, waiting hours and delayed visits for Jebel Ali, Jeddah, Salalah and Antwerp, from 2016; first principal component of congestion rates	Bai et al. (2026); own calculations	Adjusted levels; standardized PC1

Notes: SA and NSA denote seasonally adjusted and unadjusted data, respectively. X-13 and X-11 denote seasonal adjustment using X-13ARIMA-SEATS and Census X-11, respectively. Responses of variables entering as 100 log are reported as approximate percentage changes.

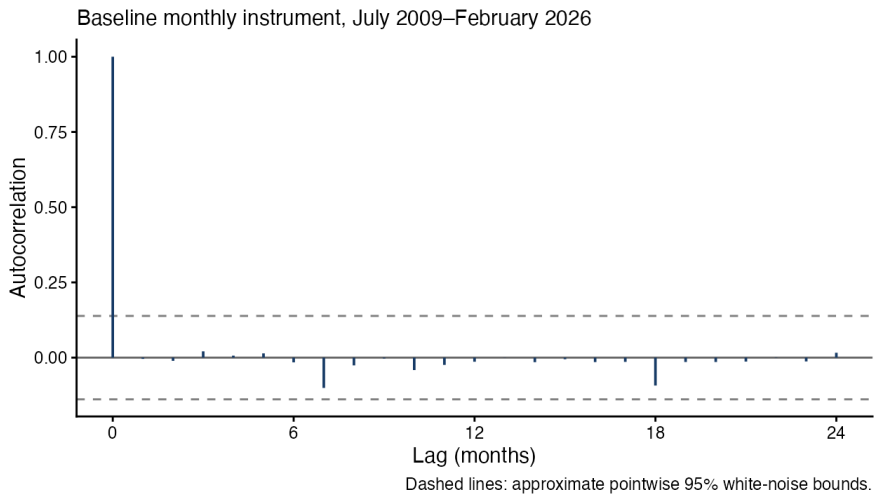
B Diagnostics of the Surprise Series

Figure B1: Censoring negative monthly instrument values



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Negative monthly instrument values are set to zero. The blue lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. The dashed red lines report the baseline proxy-SVAR responses.

Figure B2: Autocorrelation function of the monthly instrument



Notes: Sample autocorrelation function of the monthly instrument z_t with 95 percent confidence bounds.

Table B1.1: Granger causality tests: U.S. baseline

Variable	p-value
BDTI	0.7734
Geopolitical risk: oil	0.5484
VIX	0.9202
Crude Oil Volatility Index	0.9728
U.S. CPI	0.9966
Real WTI price	0.9852
WTI expectation, 3-month	0.8999
Global Supply Chain Pressure Index	0.7626
U.S. industrial production	0.7406
U.S. crude oil production	0.7518
U.S. crude oil inventories	0.6168
Global Economic Policy Uncertainty	0.2260
Energy Shortages	0.8840
US T-bill Rate 3 Months	0.8103
Joint test: core predictor set	0.9884

Notes: This table shows the p-values of Granger causality tests of the null that each variable does not Granger-cause the monthly instrument. Variables are first-differenced when an augmented Dickey–Fuller test does not reject a unit root. All tests include six lags and a constant. The final row reports a joint test for the variables marked with an asterisk, using their common sample.

Table B1.2: Granger causality tests: global VAR

Variable	p-value
BDTI	0.7734
OECD crude inventories (proxy)	0.4103
World crude production	0.7588
Real WTI price	0.9827
Industrial production (OECD+6)	0.9815
Consumer prices (OECD+6)	0.9945
Geopolitical risk: oil	0.3937
Joint test: global VAR variables	0.9871

Notes: p-values of Granger causality tests of the null hypothesis that the variable in the first column does not Granger-cause the monthly instrument, for the variables of the global VAR. The specification is that of Table B1.1: variables are first-differenced when an augmented Dickey–Fuller test does not reject a unit root; lag order six; a constant is included. The input sample matches the global VAR, July 2009–February 2026, using the BIS/World Bank consumer price index and the corresponding real WTI series. After allowing for six lags, the individual tests use 194 observations for variables in levels and 193 for first-differenced variables. The joint test uses 193 observations, February 2010–February 2026. The last row jointly tests the six lags of the six global-VAR variables other than the BDTI.

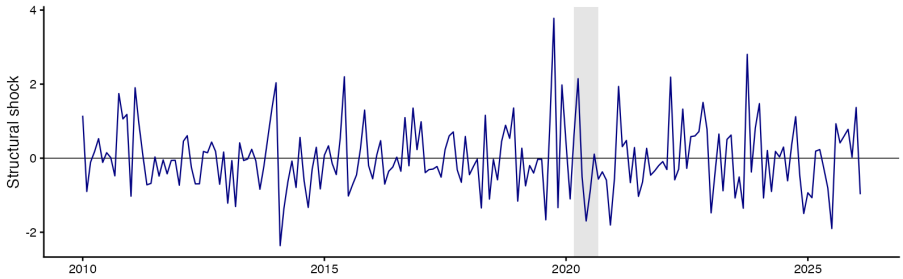
Table B2: Correlation with other shock measures

Shock measure	ρ	p-value	N	Sample
Energy Shocks				
Oil supply shock (Baumeister and Hamilton, 2019)	0.11	0.16	166	2009M07–2023M04
Natural gas demand shock (Colombo and Toni, 2026)	0.09	0.26	174	2009M07–2023M12
Natural gas supply shock (Colombo and Toni, 2026)	0.05	0.53	174	2009M07–2023M12
Oil supply news shock (Känzig, 2021)	0.05	0.52	198	2009M07–2025M12
Oil price surprise, 1-month (Baumeister, 2023)	-0.05	0.50	166	2009M07–2023M04
Natural gas supply shock (Alessandri and Gazzani, 2025)	0.04	0.62	198	2009M07–2025M12
Oil consumption demand shock (Baumeister and Hamilton, 2019)	-0.04	0.63	166	2009M07–2023M04
Economic activity shock (Baumeister and Hamilton, 2019)	-0.04	0.64	166	2009M07–2023M04
Oil demand shock (Caldara et al., 2019)	-0.04	0.74	78	2009M07–2015M12
Oil supply shock (Caldara et al., 2019)	-0.03	0.82	78	2009M07–2015M12
Oil inventory demand shock (Baumeister and Hamilton, 2019)	0.01	0.88	166	2009M07–2023M04
Oil price expectation shock (Baumeister, 2023)	0.01	0.89	166	2009M07–2023M04
Monetary Policy Shock				
Monetary policy shock (Bauer and Swanson, 2023)	0.08	0.27	174	2009M07–2023M12
Other Uncertainty and Risk Measures				
Geopolitical risk: World, AI-based (Iacoviello and Tong, 2026)	0.09	0.22	200	2009M07–2026M02
Geopolitical risk: oil (Iacoviello and Tong, 2026)	0.09	0.21	200	2009M07–2026M02
Geopolitical risk: military conflict (Iacoviello and Tong, 2026)	0.09	0.21	200	2009M07–2026M02
VIX (Cboe Global Markets, 2026b)	-0.08	0.27	200	2009M07–2026M02
Supply chain shock (Känzig and Raghavan, 2026)	-0.07	0.40	162	2009M07–2022M12
Geopolitical risk: terrorism (Iacoviello and Tong, 2026)	-0.06	0.38	200	2009M07–2026M02
Economic Policy Uncertainty US (Baker et al., 2016)	-0.05	0.50	200	2009M07–2026M02
Climate policy uncertainty shock (Gavriilidis et al., 2026)	-0.04	0.63	126	2009M07–2019M12
Geopolitical risk: diplomatic tension (Iacoviello and Tong, 2026)	0.04	0.60	200	2009M07–2026M02
Global Supply Chain Pressure Index (Benigno et al., 2022)	-0.04	0.58	200	2009M07–2026M02
Geopolitical risk: coup (Iacoviello and Tong, 2026)	-0.03	0.64	200	2009M07–2026M02
Supply chain shock (Bini, 2025)	-0.03	0.72	186	2009M07–2024M12
Crude Oil Volatility Index (Cboe Global Markets, 2026a)	-0.02	0.77	200	2009M07–2026M02
Geopolitical risk: macro (Pinchetti, 2026)	-0.02	0.75	198	2009M07–2025M12
Global Economic Policy Uncertainty (Davis, 2016)	-0.01	0.91	200	2009M07–2026M02
Geopolitical risk: sanctions (Iacoviello and Tong, 2026)	-0.01	0.89	200	2009M07–2026M02
Geopolitical risk: surprise (Pinchetti, 2026)	-0.01	0.85	198	2009M07–2025M12
Geopolitical risk: energy (Pinchetti, 2026)	0.00	0.97	198	2009M07–2025M12

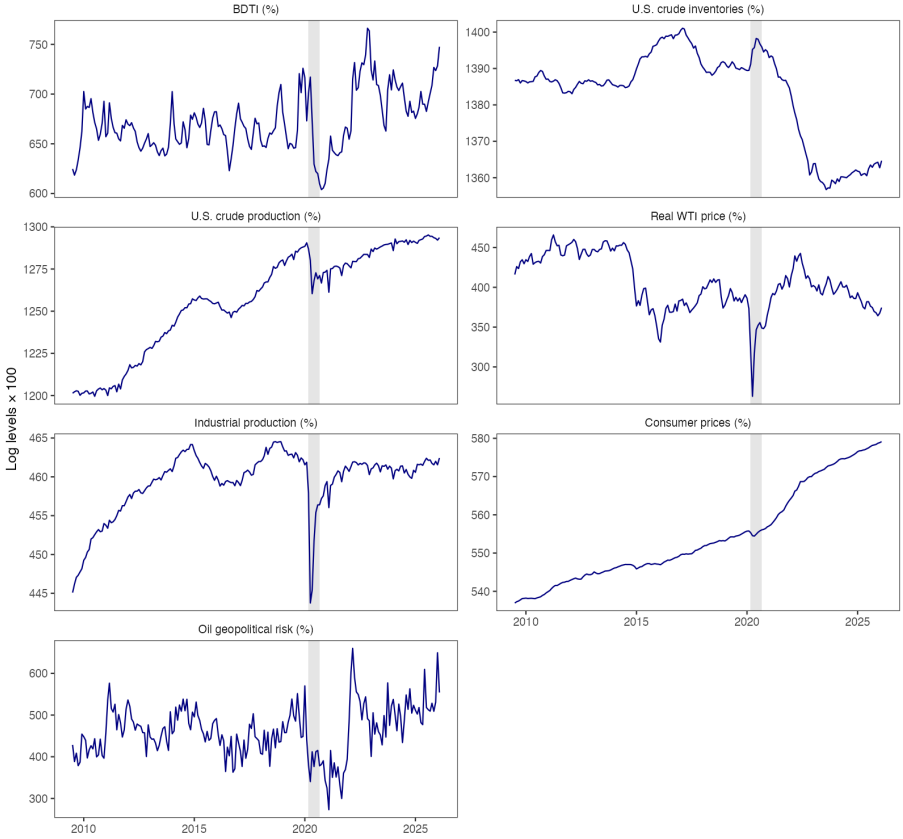
Notes: Correlation of the monthly instrument with structural shocks, supply-chain pressures, and uncertainty and risk measures from the literature. The GSCPI enters in levels and is not an identified structural shock. Measures are grouped by category and ordered by the absolute correlation within each group. ρ is the Pearson correlation, the p-value corresponds to the test that the correlation is zero, and n is the sample size. All measures shown are monthly. Each correlation uses the overlapping sample reported in the last column, excluding missing observations.

Figure B3: The structural shock and the VAR data

(a) Structural oil-shipping-cost shock extracted from the VAR



(b) Variables of the baseline VAR

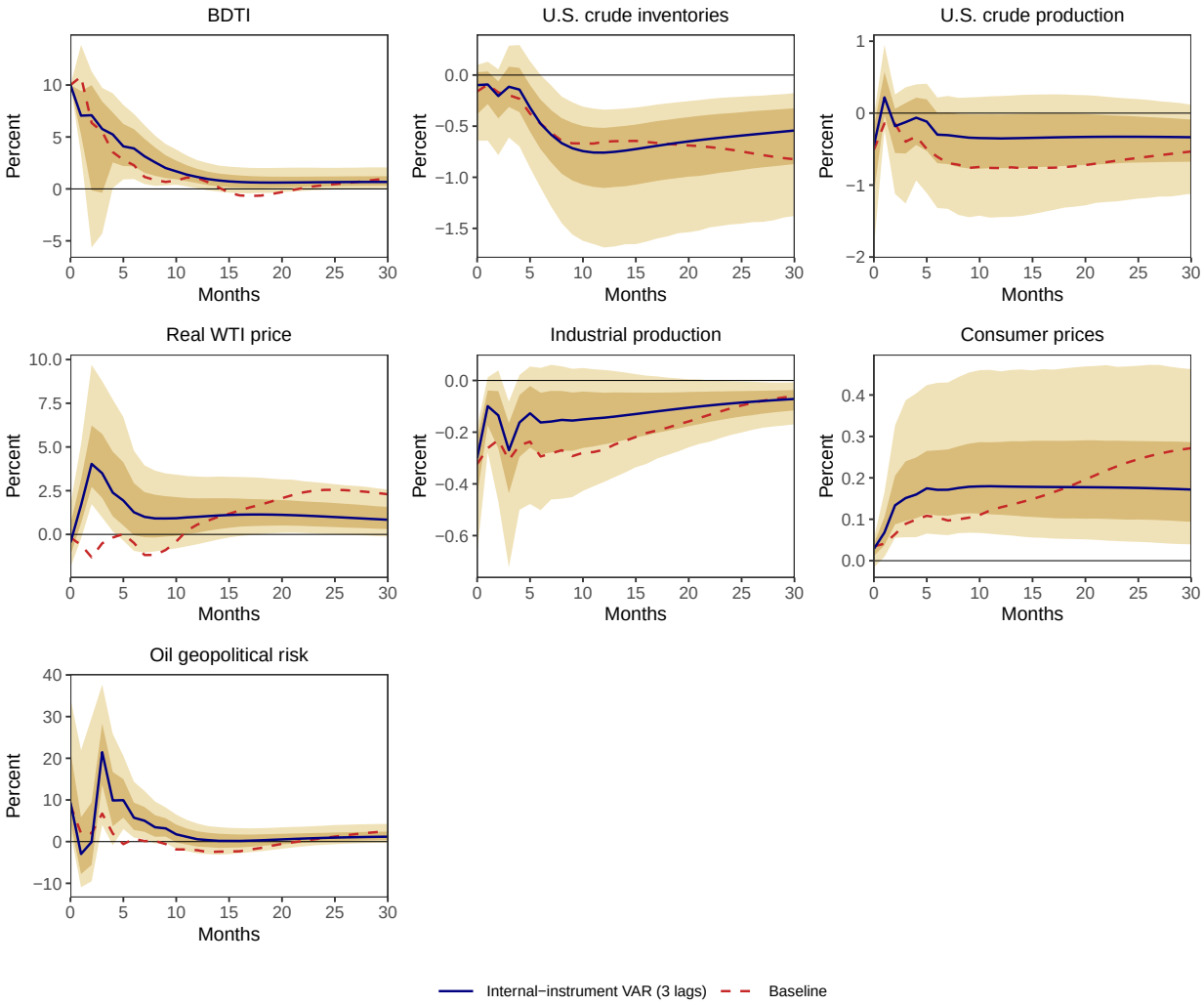


Notes: Panel (a) plots the structural shock $\hat{\varepsilon}_{1,t} = \mathbf{s}_1' \boldsymbol{\Sigma}^{-1} \hat{\mathbf{u}}_t$ extracted from the baseline VAR. Panel (b) plots the seven transformed series of the baseline VAR over 2009M7–2026M2. Grey shading marks March–August 2020, the pandemic variance-adjustment window.

C Additional Evidence on the Empirical Framework

C.1 Internal-Instrument Estimation

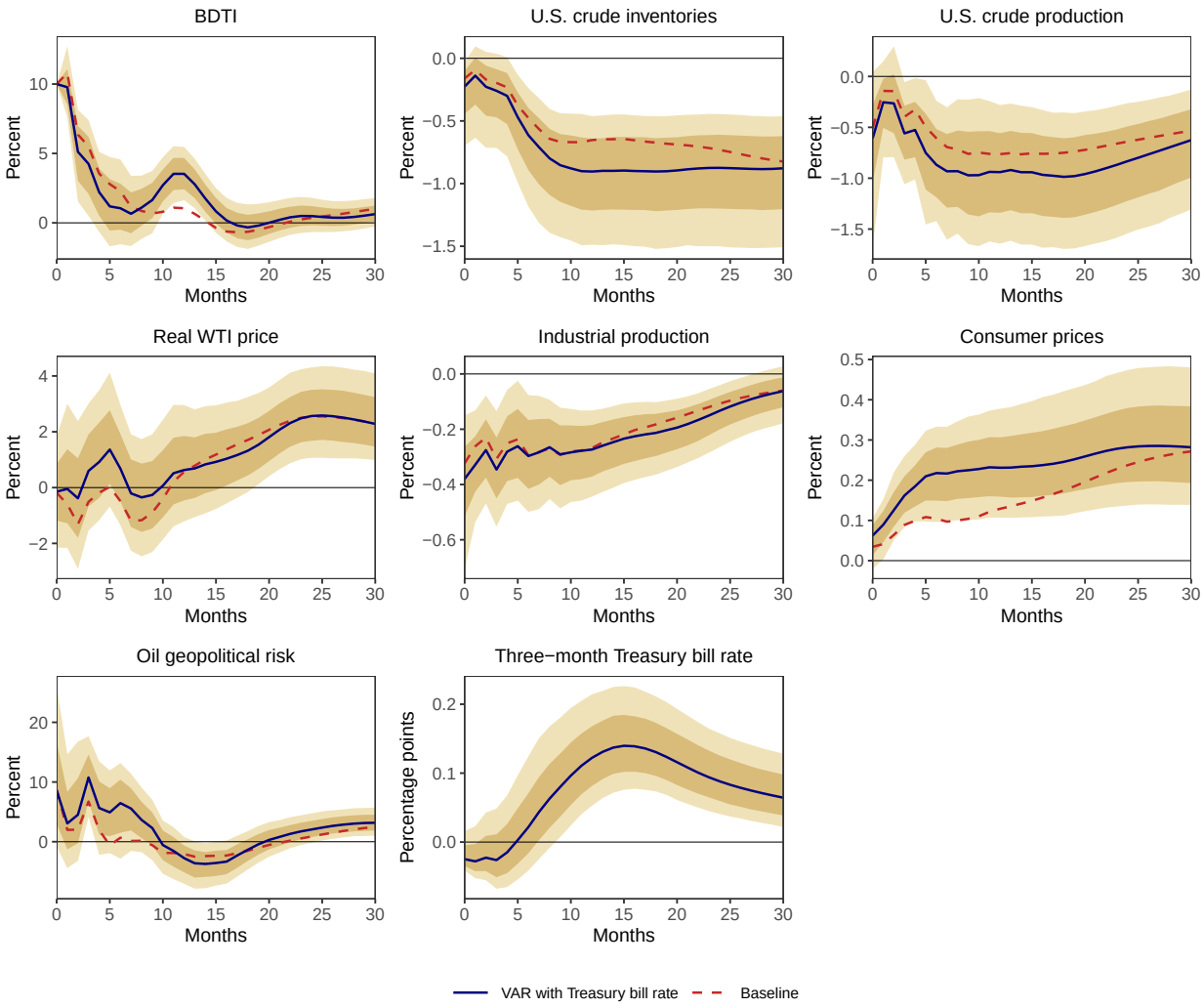
Figure C1: Internal-instrument VAR



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The original, unpurged instrument enters the VAR as an endogenous variable ordered first under recursive identification. The internal-instrument VAR uses three lags and the July 2009–February 2026 input sample, giving an effective estimation sample of October 2009–February 2026. The pandemic variance multiplier for March–August 2020 is fixed at the baseline estimate; confidence bands are based on 5,000 moving-block bootstrap replications and are conditional on that value. The blue lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. The dashed red lines report the six-lag baseline proxy-SVAR responses.

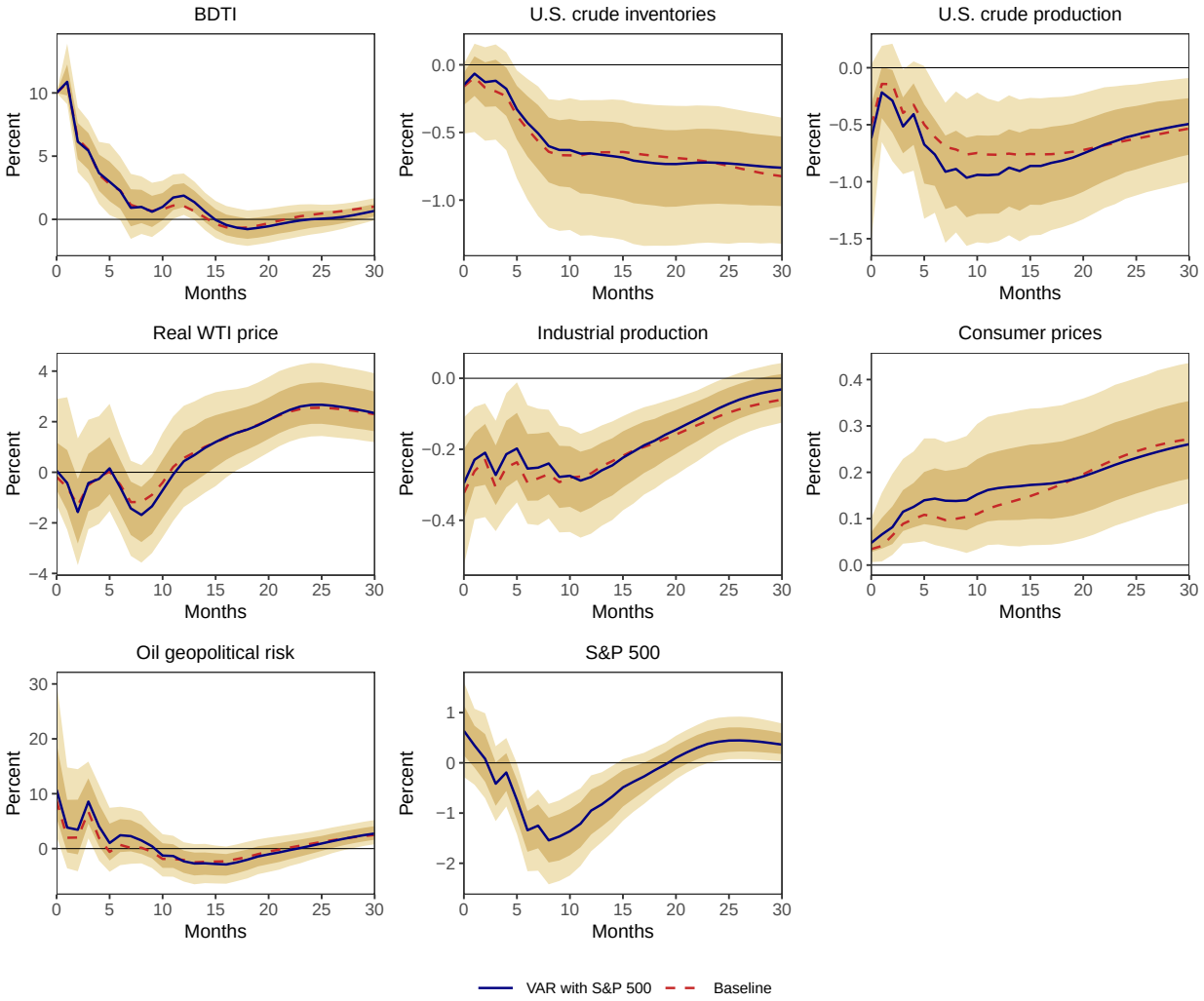
C.2 Financial Information

Figure C2: Augmenting the VAR with the short-term interest rate



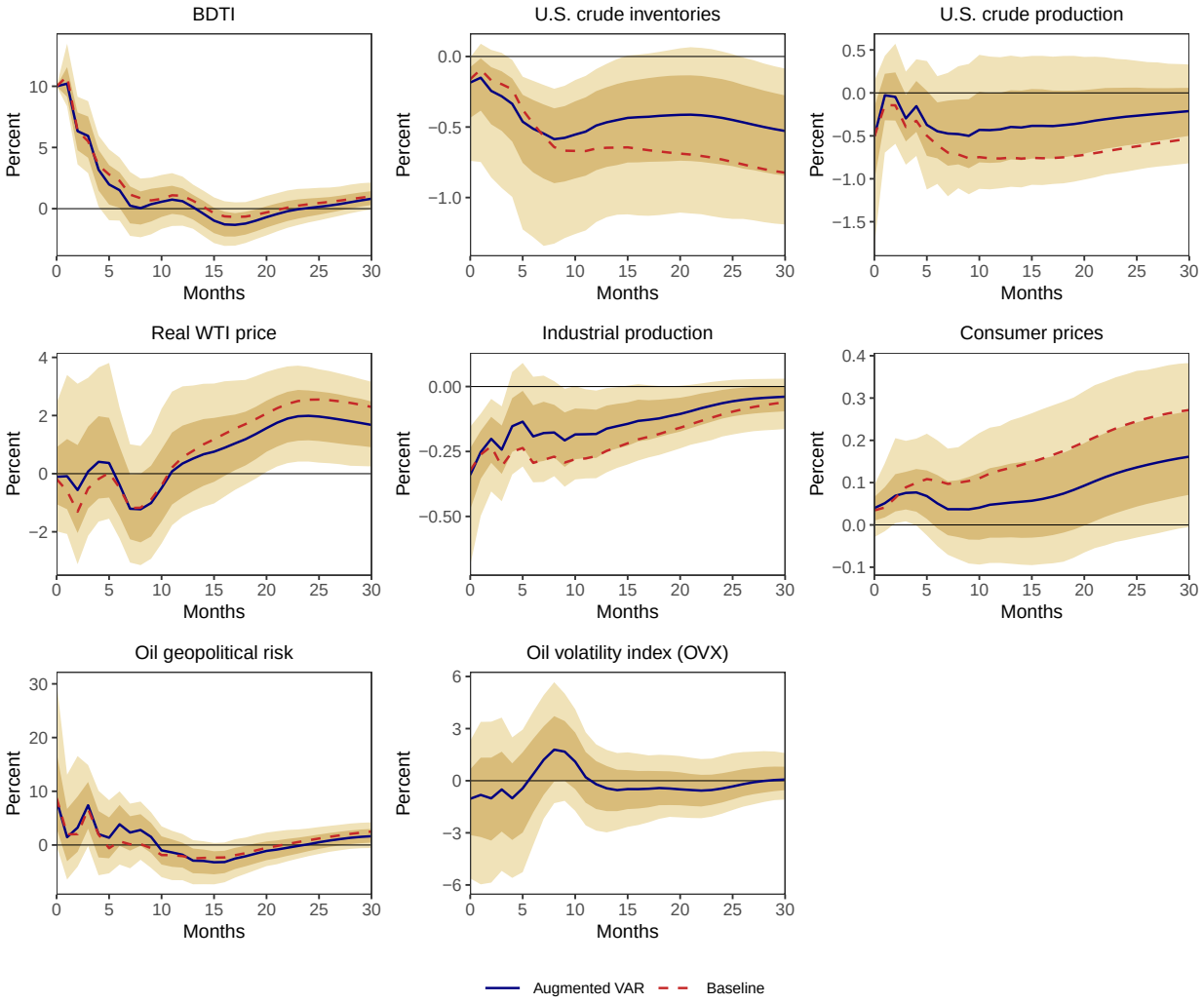
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid line reports the baseline VAR augmented with the three-month Treasury bill rate; the dashed red line reports the baseline. The interest-rate response is measured in percentage points. Dark and light shaded areas are 68 and 90 percent confidence bands for the augmented model, respectively.

Figure C3: Augmenting the VAR with equity prices



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid line reports the baseline VAR augmented with the S&P 500; the dashed red line reports the baseline. Dark and light shaded areas are 68 and 90 percent confidence bands for the augmented model, respectively.

Figure C4: Augmenting the VAR with oil-price volatility

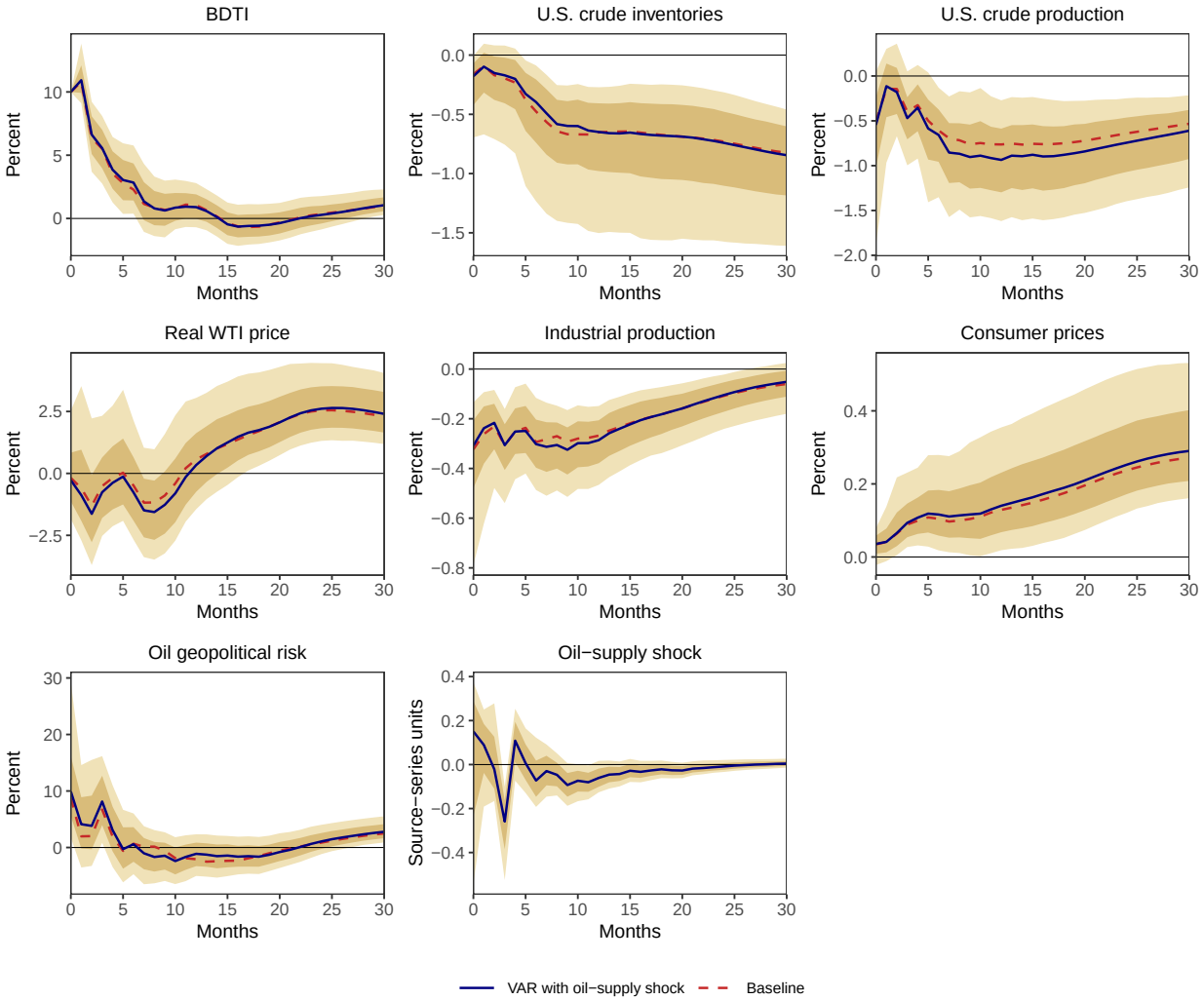


Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid blue lines report the baseline VAR augmented with the oil volatility index (OVX), retaining oil-related geopolitical risk. The dashed red lines report the baseline responses for its seven variables. Dark and light yellow shaded areas are 68 and 90 percent confidence bands for the augmented model, respectively. Responses are expressed as approximate percentage deviations.

D Additional Results

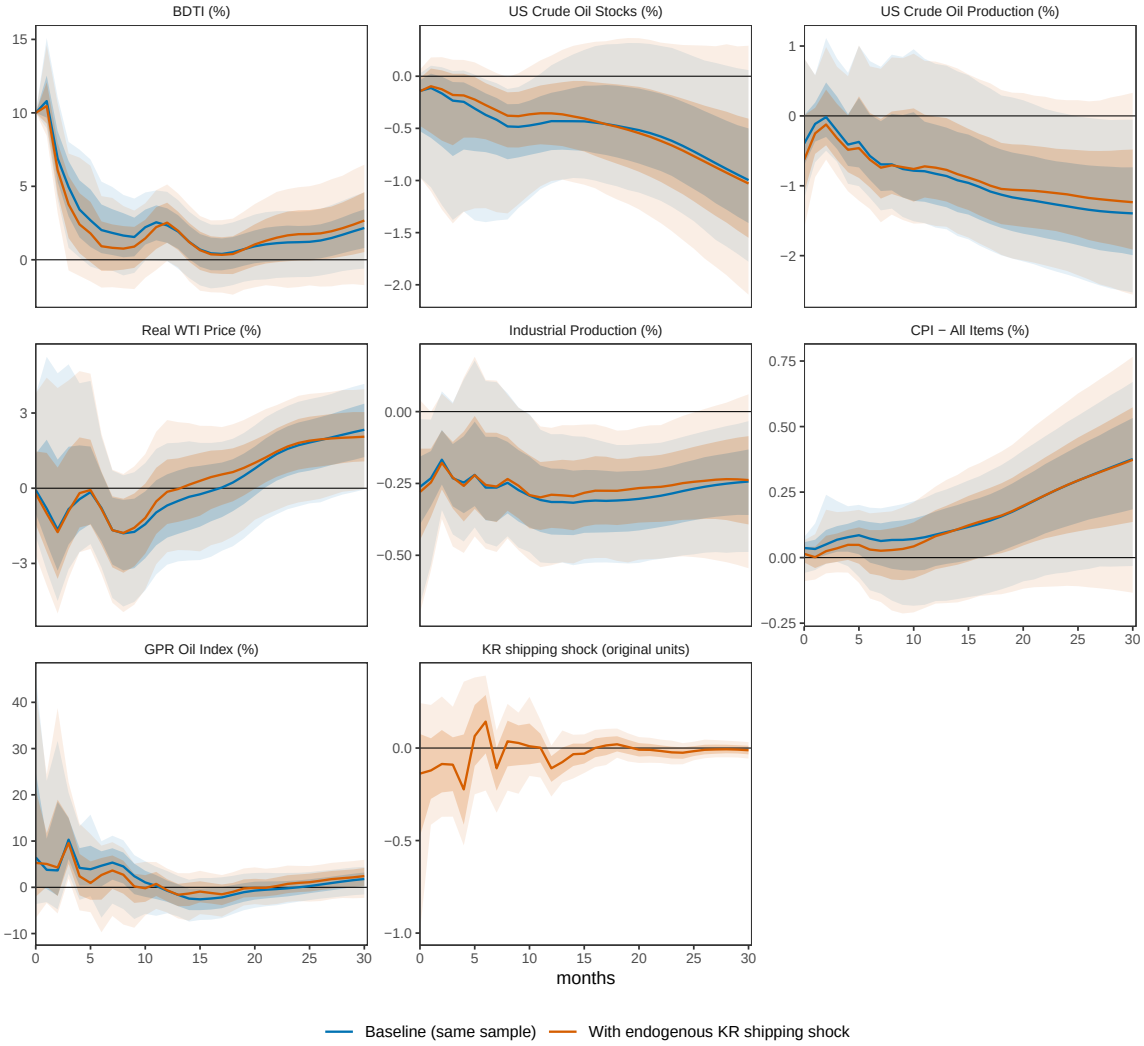
D.1 Comparison with Other Shocks

Figure D1: Augmenting the VAR with the Baumeister–Hamilton oil-supply shock



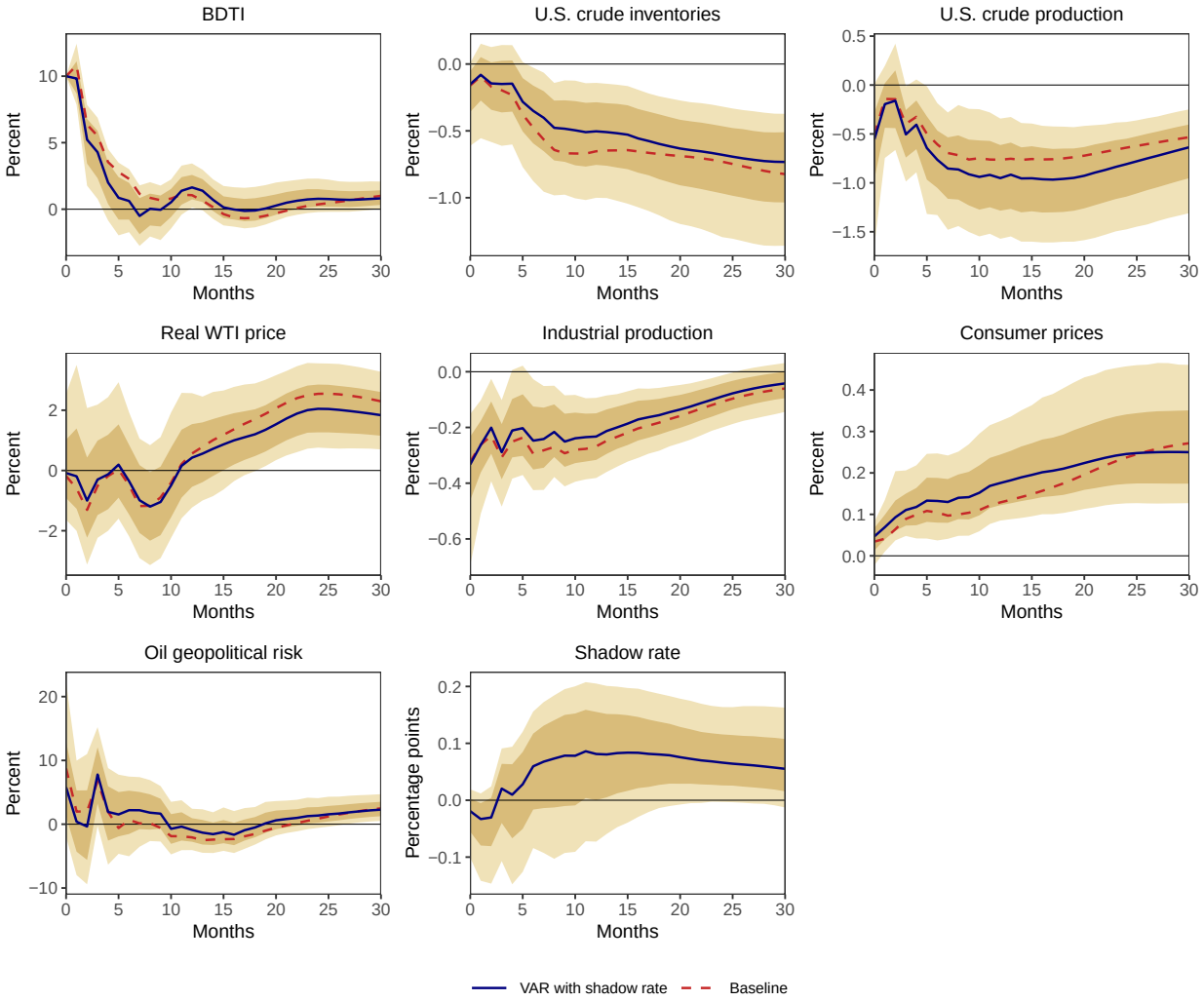
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid blue line reports the baseline VAR augmented with the oil-supply shock of Baumeister and Hamilton (2019); the dashed red line reports the baseline. Dark and light shaded areas are 68 and 90 percent confidence bands for the augmented model, respectively. The oil-supply shock response is in source-series units; inference conditions on the published shock series.

Figure D2: Augmenting the VAR with the dry-bulk shipping shock



Notes: Impulse responses to an oil-shipping-cost shock normalized to increase the BDTI by 10 percent on impact. Blue lines show the baseline; orange lines show the VAR augmented with the dry-bulk shipping shock of Känzig and Raghavan (2026). Both models use our tanker-attack instrument and data from July 2009 to December 2022. Dark and light shading indicates 68 and 90 percent confidence bands based on 5,000 moving-block bootstrap replications.

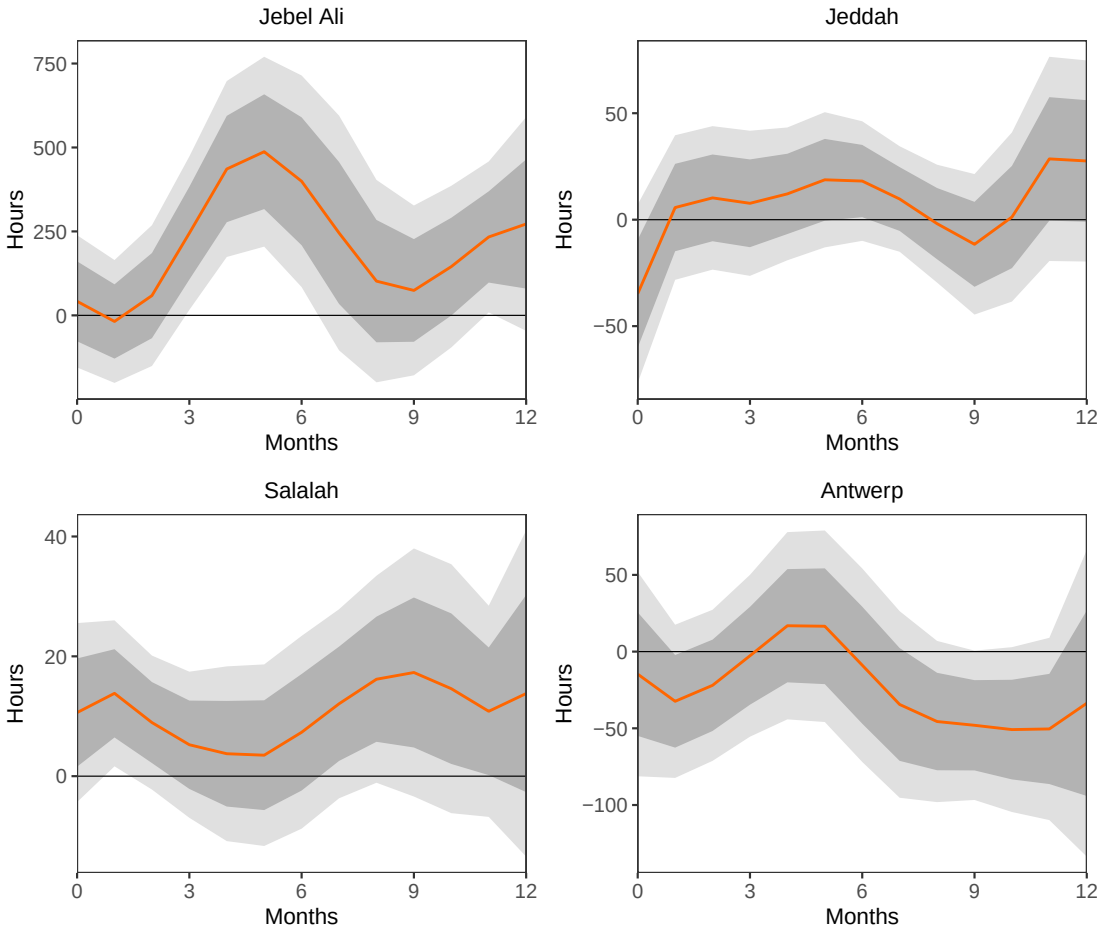
Figure D3: Augmenting the VAR with the shadow rate



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The baseline VAR is augmented with the shadow rate of Wu and Xia (2016) through June 2023, spliced with the effective federal funds rate thereafter. The blue lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands. Rate responses are in percentage points. The dashed red lines report the baseline VAR responses for its seven variables.

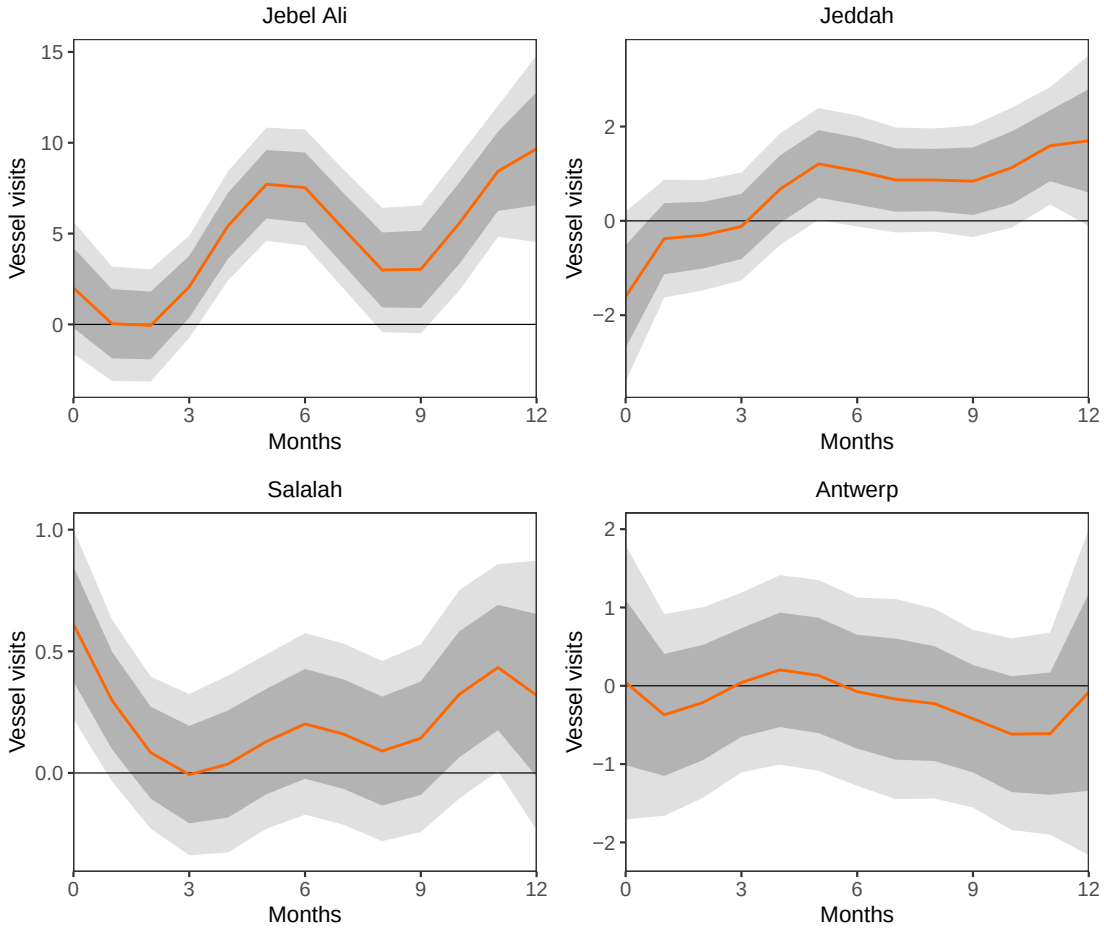
D.2 Port Congestion and Vessel Dwell Time

Figure D4: Port congestion: delay hours



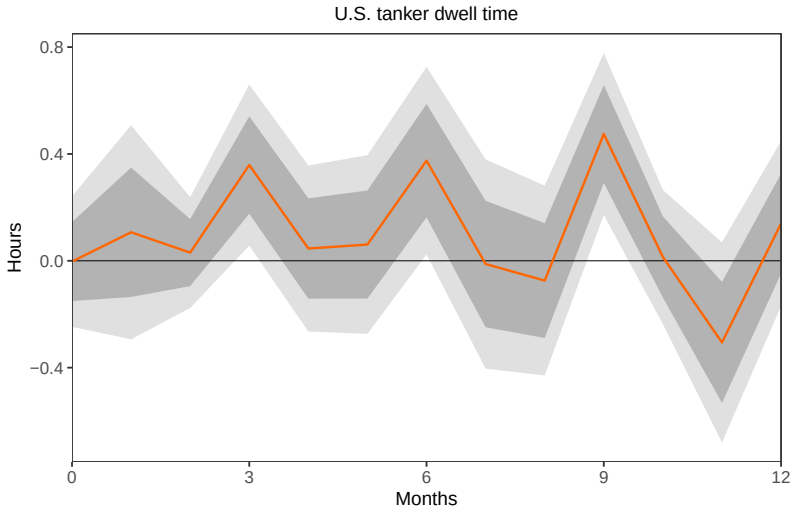
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are estimated by smooth local projections (11) on the structural shock extracted from the baseline SVAR-IV. The orange lines are point estimates, and the dark and light grey shaded areas are 68 and 90 percent confidence bands, respectively. Responses are measured in hours.

Figure D5: Port congestion: delayed vessel visits



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are estimated by smooth local projections (11) on the structural shock extracted from the baseline SVAR-IV. The orange lines are point estimates, and the dark and light grey shaded areas are 68 and 90 percent confidence bands, respectively. Responses are measured in vessel visits.

Figure D6: The energy-trade channel: tanker dwell times at U.S. ports

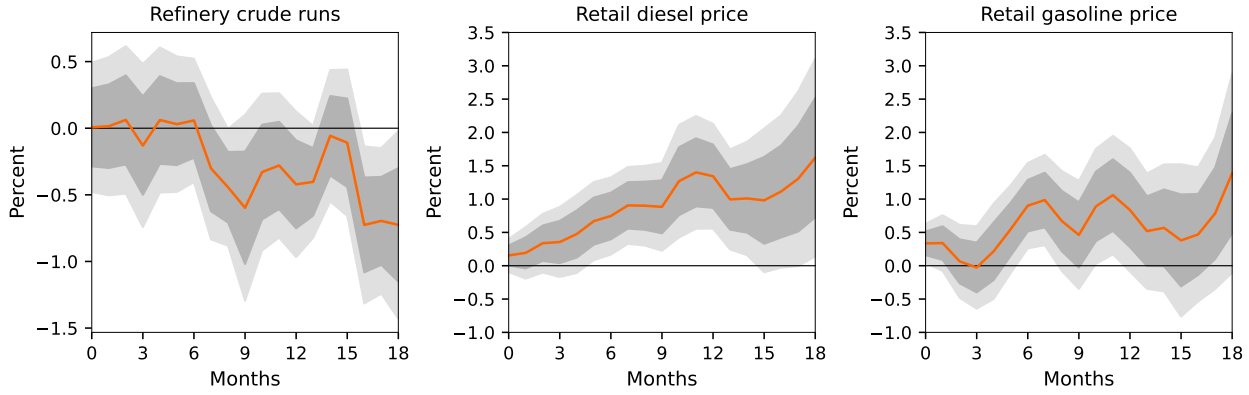


Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are estimated by local projections (11) on the structural shock extracted from the baseline SVAR-IV. The orange lines are point estimates, and the dark and light grey shaded areas are 68 and 90 percent confidence bands, respectively. Dwell time includes waiting and cargo handling for tanker/liquid-bulk vessels, not only crude tankers. Source: U.S. Bureau of Transportation Statistics, January 2019–June 2023. Responses are measured in hours.

D.3 Regional Exposure: Alternative Timing

We assess sensitivity to the exposure measure by replacing the trailing twelve-month average in equation (13) with the preceding month's Persian Gulf import-to-refinery-input ratio, $E_{p,t-1}$, both on its own and interacted with the recovered baseline VAR shock. The outcomes are seasonally adjusted as in the main specification. All other regressors and the inference method are unchanged. For each outcome and horizon, both specifications use the same observations, requiring availability of both exposure measures. Figure D7 reports the alternative responses. Diesel and gasoline price responses are similar to those in the main text; refinery-input declines are larger under the previous-month measure, indicating greater sensitivity of this outcome to the exposure definition.

Figure D7: Regional exposure: previous-month exposure



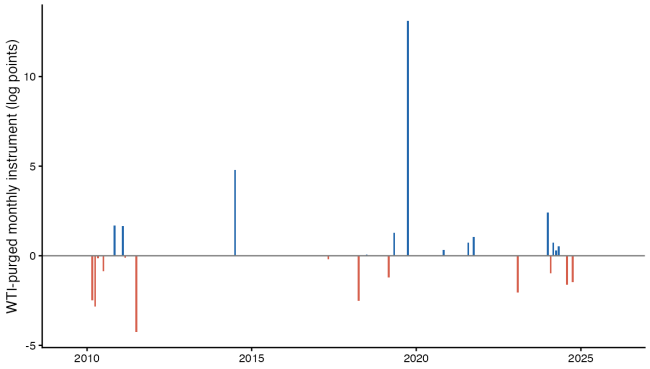
Notes: Differential responses for ten percentage points higher exposure to Persian Gulf crude, using $E_{p,t-1}$ instead of the trailing twelve-month average in equation (13). The original baseline VAR shock, sample at each outcome and horizon, regional fixed effects and March–August 2020 dummy are unchanged. Orange lines are point estimates; dark and light grey shaded areas are pointwise 68 and 90 percent Driscoll–Kraay confidence bands with Bartlett weights, maximum lag $h+1$ and degrees-of-freedom correction $N/(N-K_h)$. Inference conditions on the recovered shock and exposure. Diesel and gasoline prices share a vertical scale; refinery inputs use a separate scale.

E Additional Sensitivity Checks

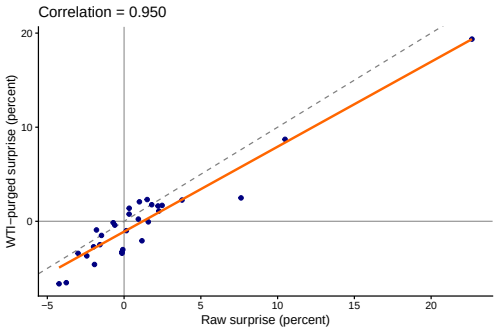
E.1 Predictability of the Instrument

Figure E1: Surprise series purged of oil-price information

(a) Purged monthly instrument

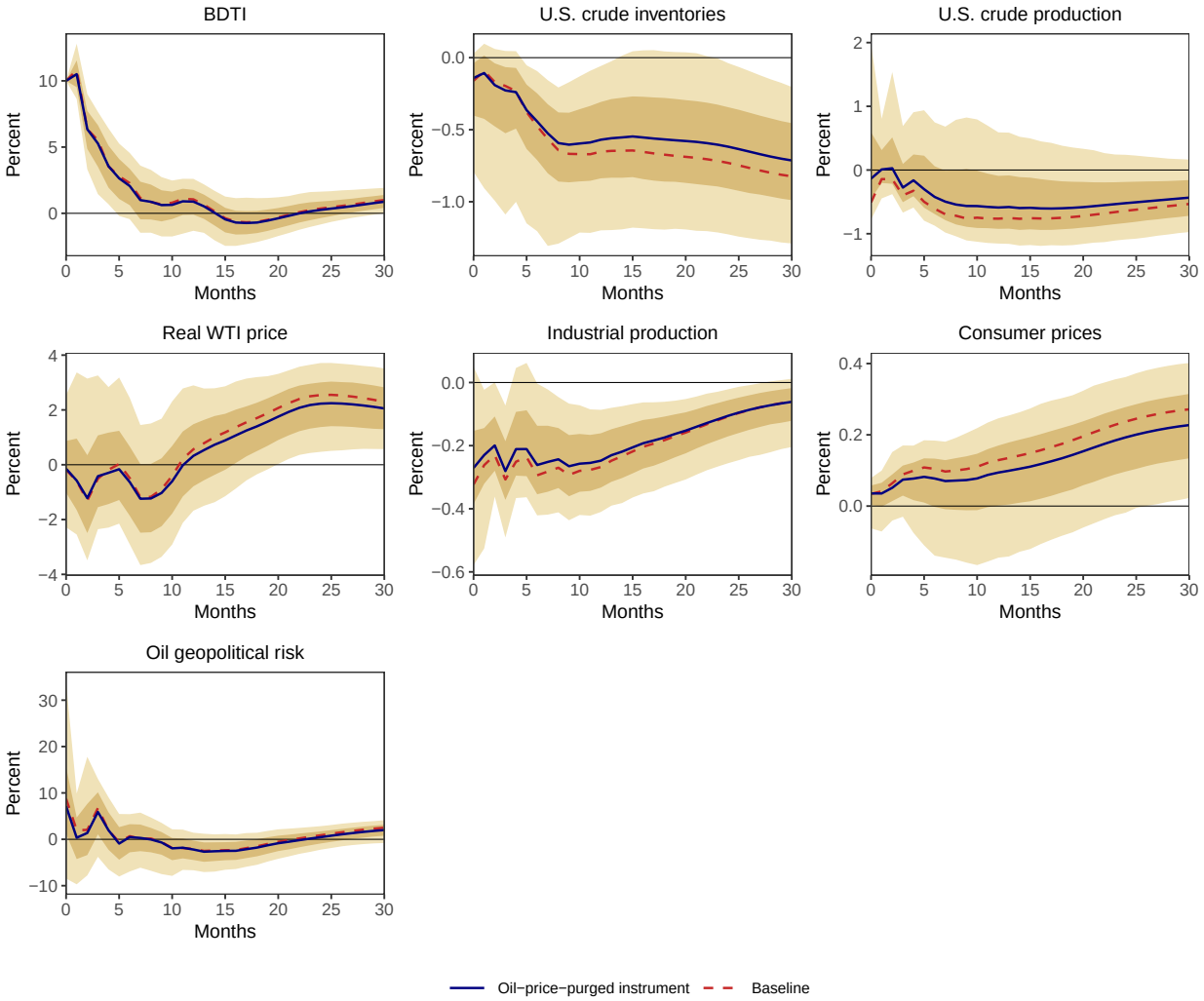


(b) Raw and purged surprises



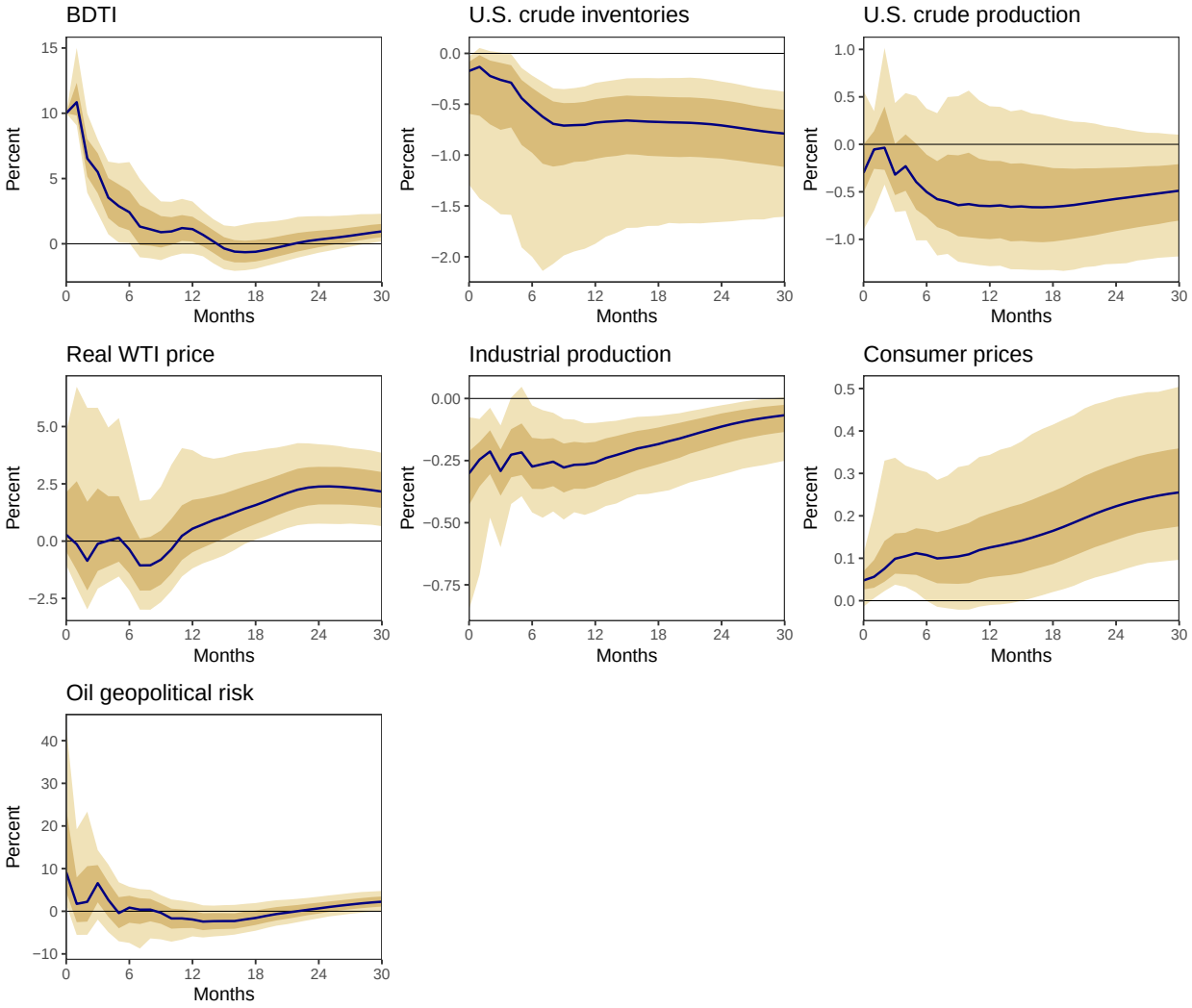
Notes: The purged event-level surprise is the OLS residual from a regression of the raw surprise on a constant and the latest available three-month WTI log change dated strictly before the event. Panel (a) plots its monthly linear-decay aggregation. Panel (b) plots the raw and purged event-level surprises. Their correlation is 0.95.

Figure E2: Surprises purged of prior oil-price changes



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The instrument is purged of the pre-event three-month WTI log change. The solid blue line reports this specification; the dashed red line reports the baseline. Dark and light shaded areas are 68 and 90 percent confidence bands for the purged specification, respectively.

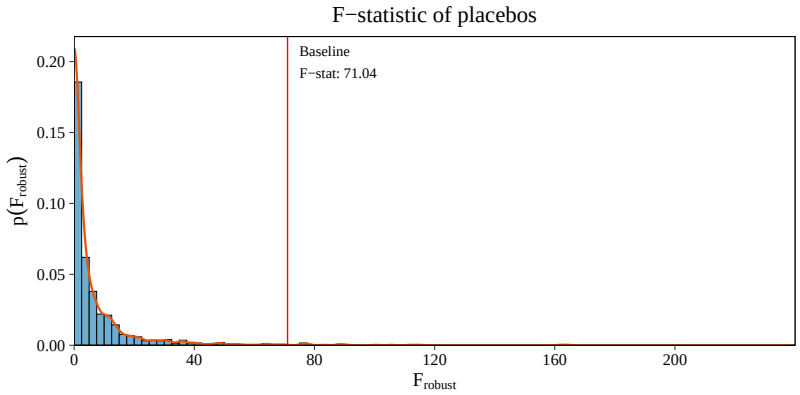
Figure E3: Surprises purged of prior oil-related geopolitical risk



Notes: Impulse responses to an oil-shipping-cost shock normalized to increase the BDTI by 10 percent on impact. The instrument uses residuals from a regression of daily surprises on a constant and the pre-event three-month change in oil-related geopolitical risk, aggregated using the baseline monthly weights. The VAR specification is unchanged. Blue lines show point estimates; dark and light shading indicates 68 and 90 percent confidence bands based on 5,000 bootstrap replications.

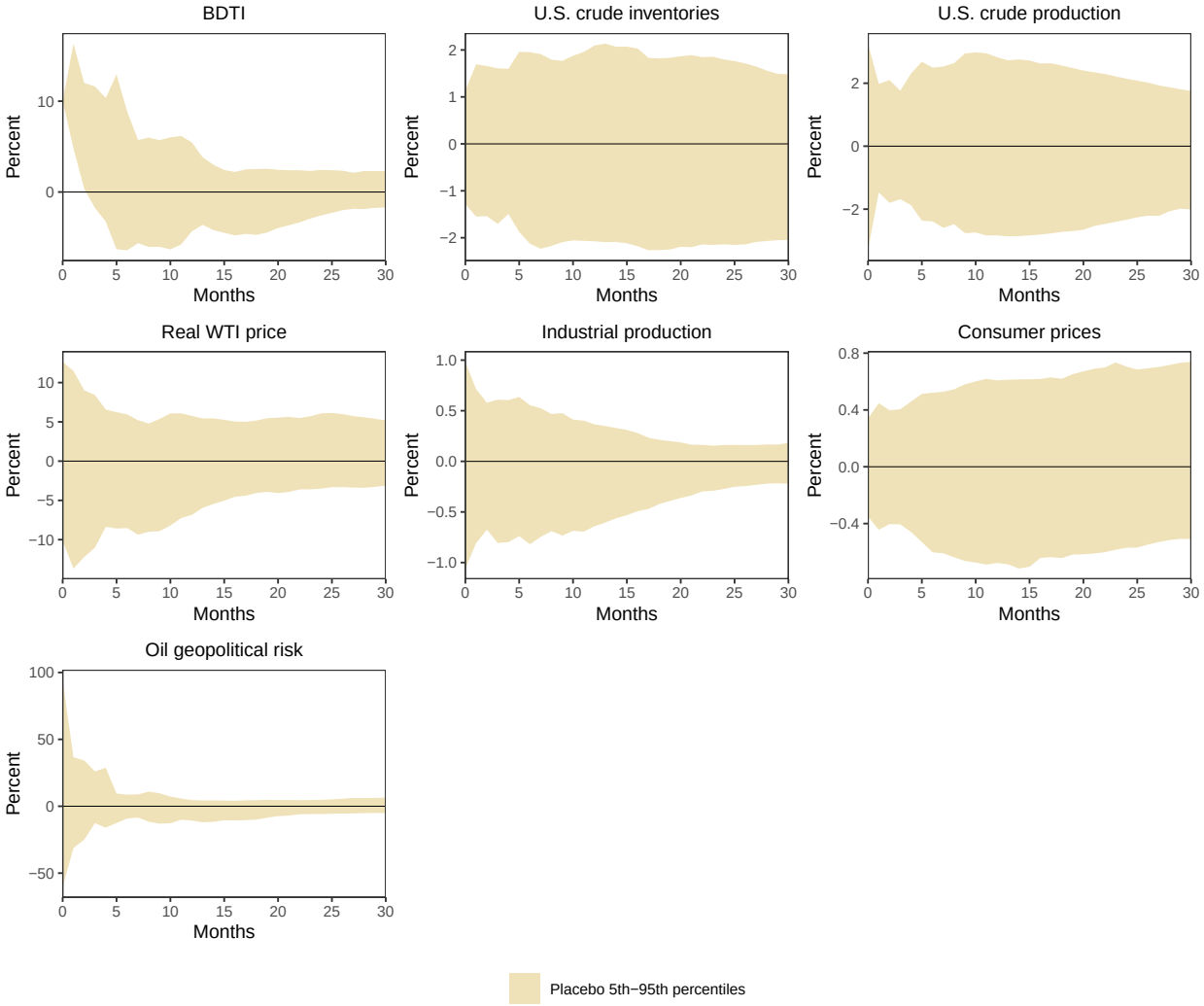
E.2 Random-Date Placebos

Figure E4: Instrument strength on random non-event dates



Notes: Distribution of robust first-stage *F*-statistics from 1,000 random samples of 30 distinct trading dates between January 2010 and December 2025. Draws exclude windows overlapping the 30 retained attacks or March–May 2020. Surprises and monthly instruments are constructed as in the baseline, keeping the estimated VAR fixed. The orange line shows a kernel density estimate; the red line marks the baseline *F*-statistic of 71.04.

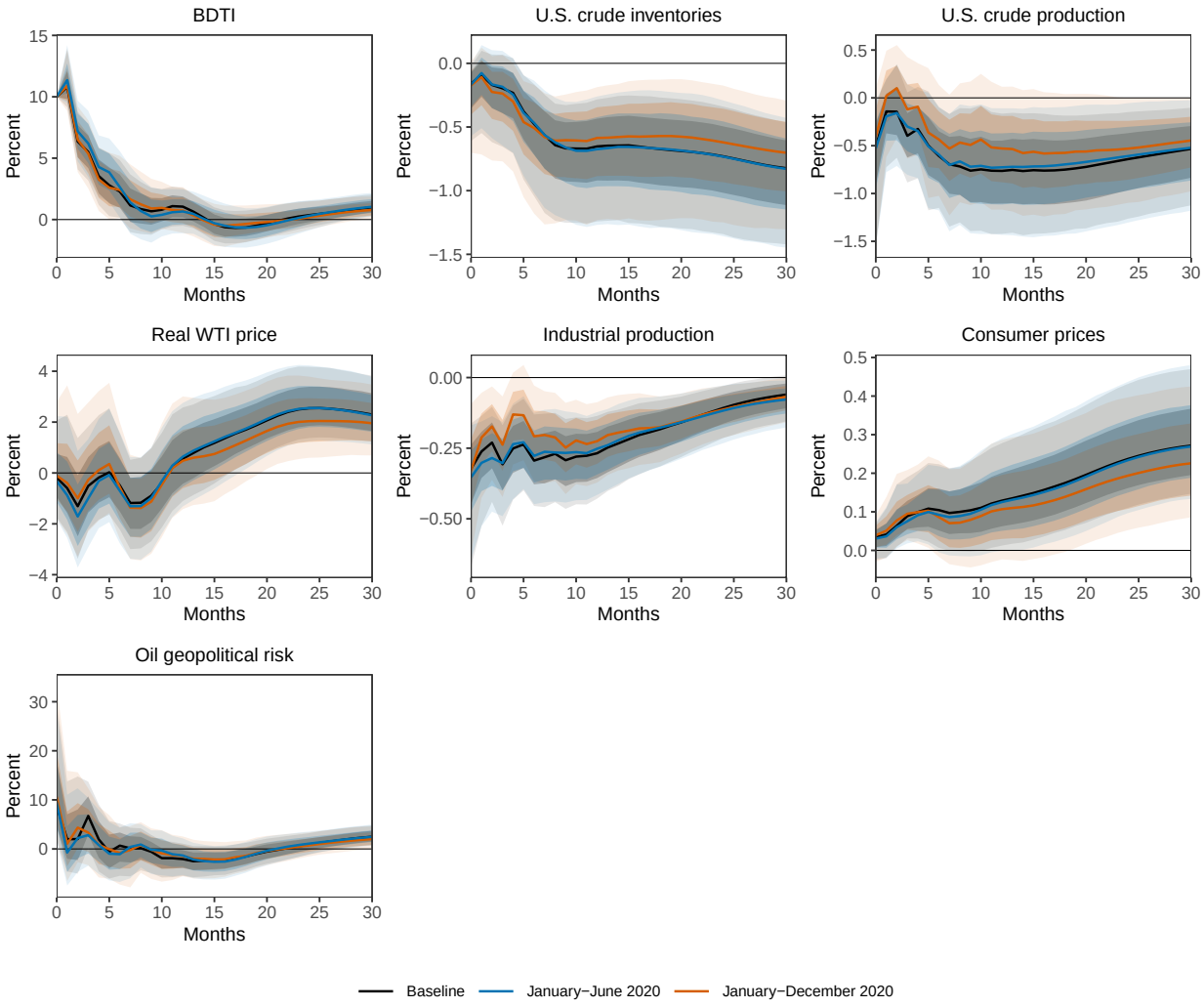
Figure E5: Impulse responses identified using random non-event dates



Notes: The shaded area shows the pointwise 5th–95th percentile range of impulse responses obtained using the 1,000 placebo instruments described in Figure E4. These ranges are percentiles across placebo estimates, not confidence bands. All responses are normalized to raise the BDTI by approximately 10 percent on impact, holding the reduced-form VAR fixed.

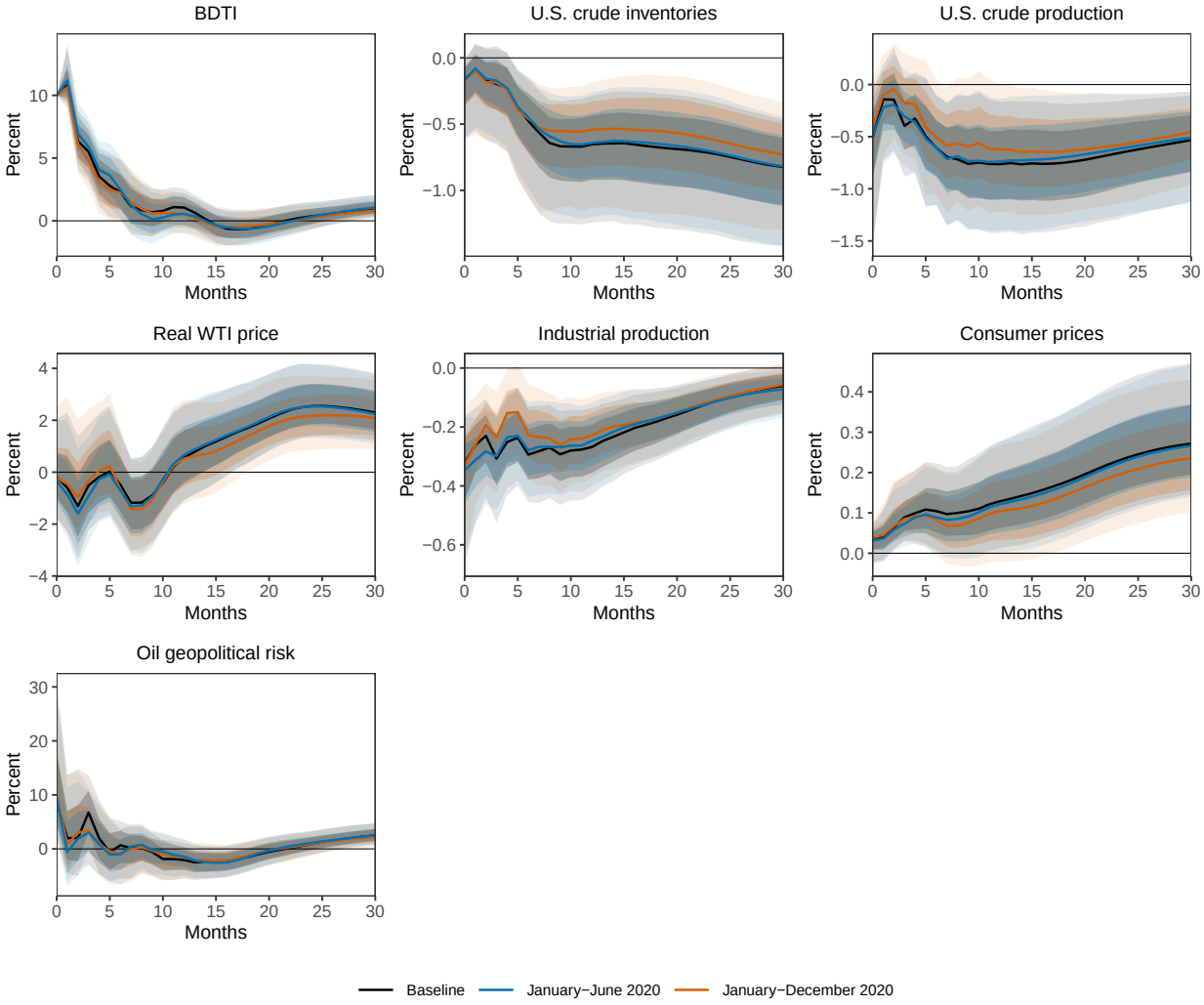
E.3 Pandemic Adjustment

Figure E6: Pandemic sensitivity: alternative dummy windows



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The black solid line reports the baseline. Blue and orange solid lines replace the baseline variance adjustment with monthly dummies for January–June and January–December 2020, respectively. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

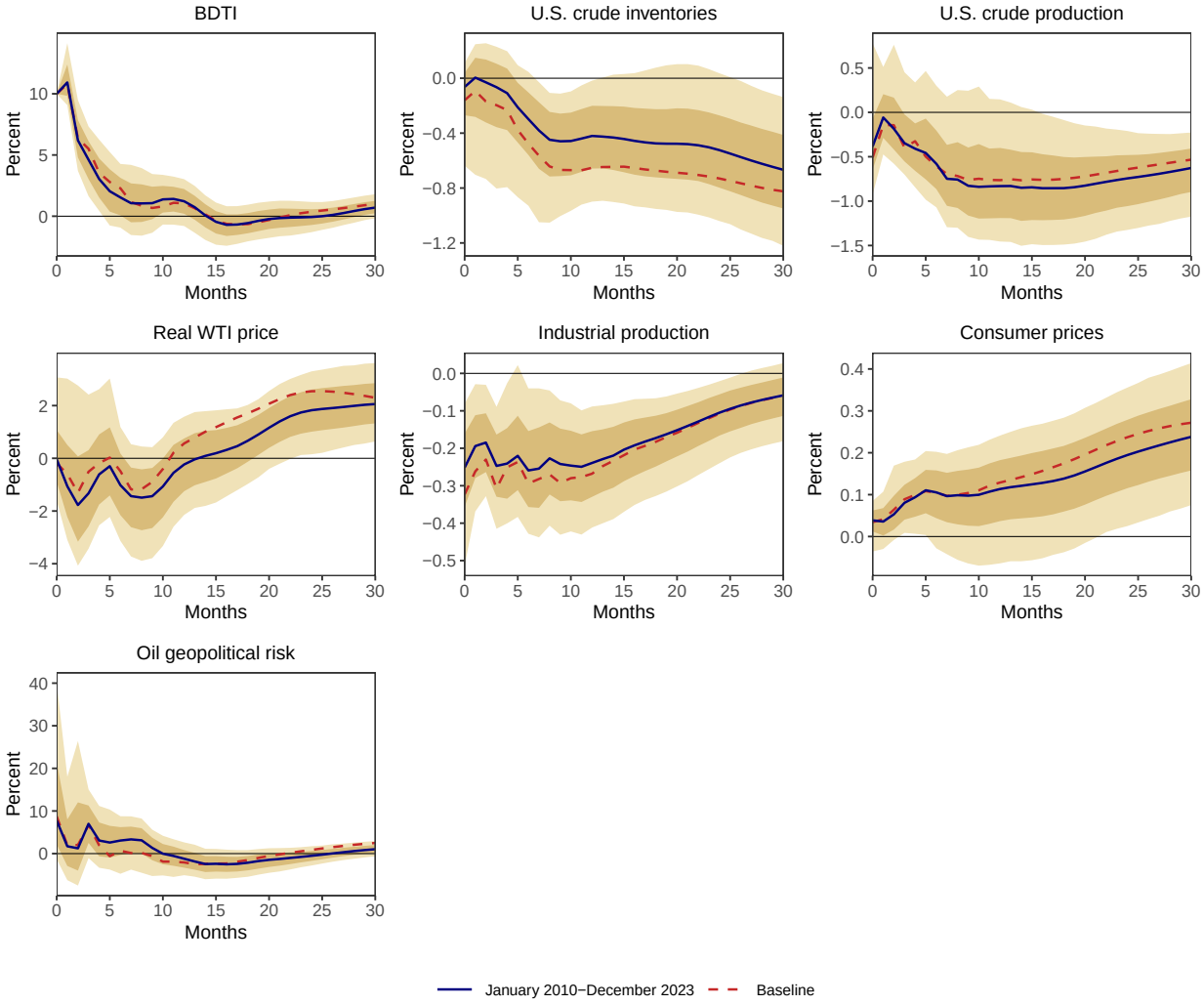
Figure E7: Pandemic sensitivity: alternative variance-adjustment windows



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The variance adjustment covers March–August 2020 in the baseline (black solid), January–June 2020 (blue solid) or January–December 2020 (orange solid). The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

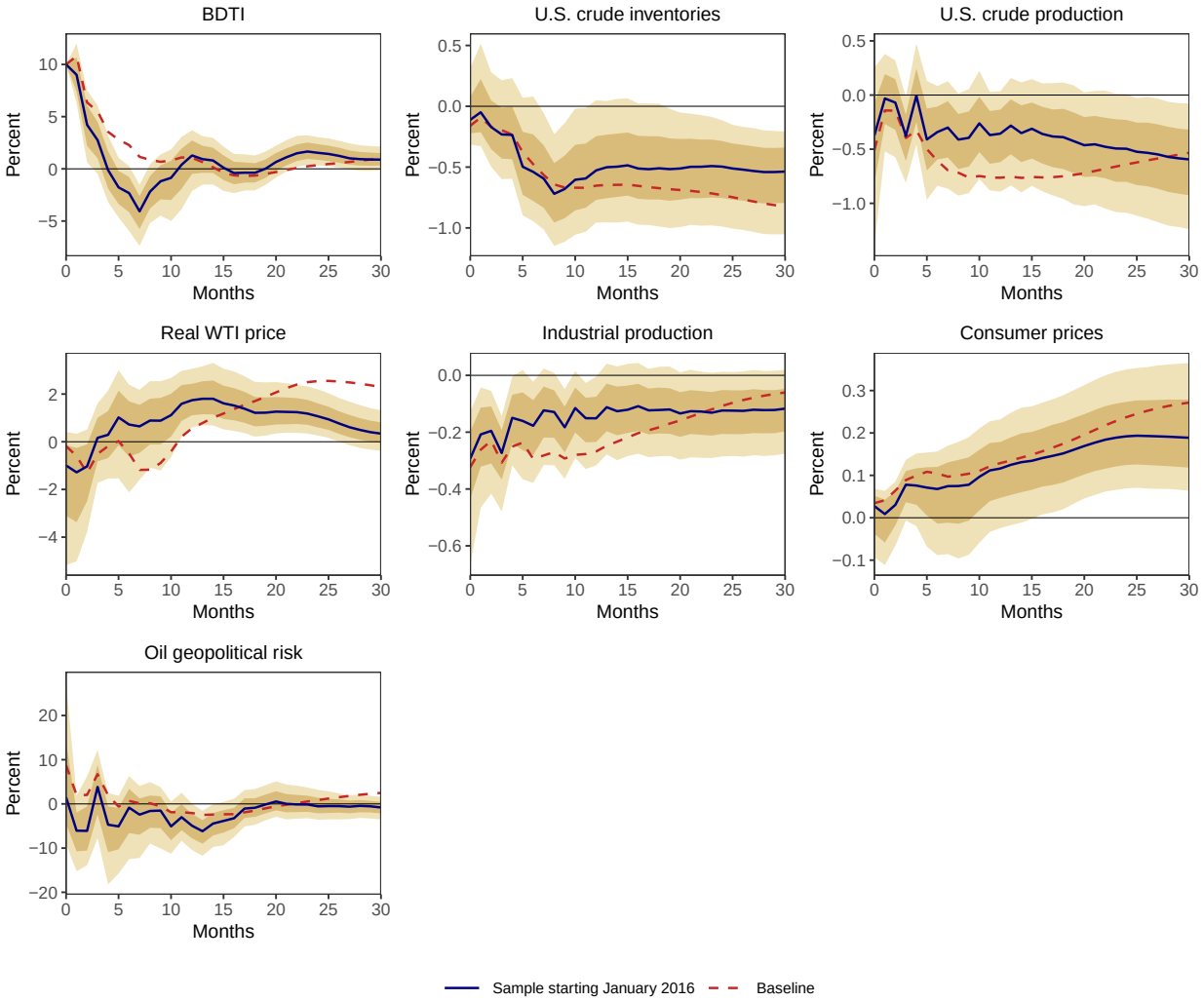
E.4 Sample Period

Figure E8: Sample sensitivity: January 2010–December 2023



Notes: Impulse responses estimated over January 2010–December 2023 to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The blue lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. The dashed red lines report the baseline proxy-SVAR responses.

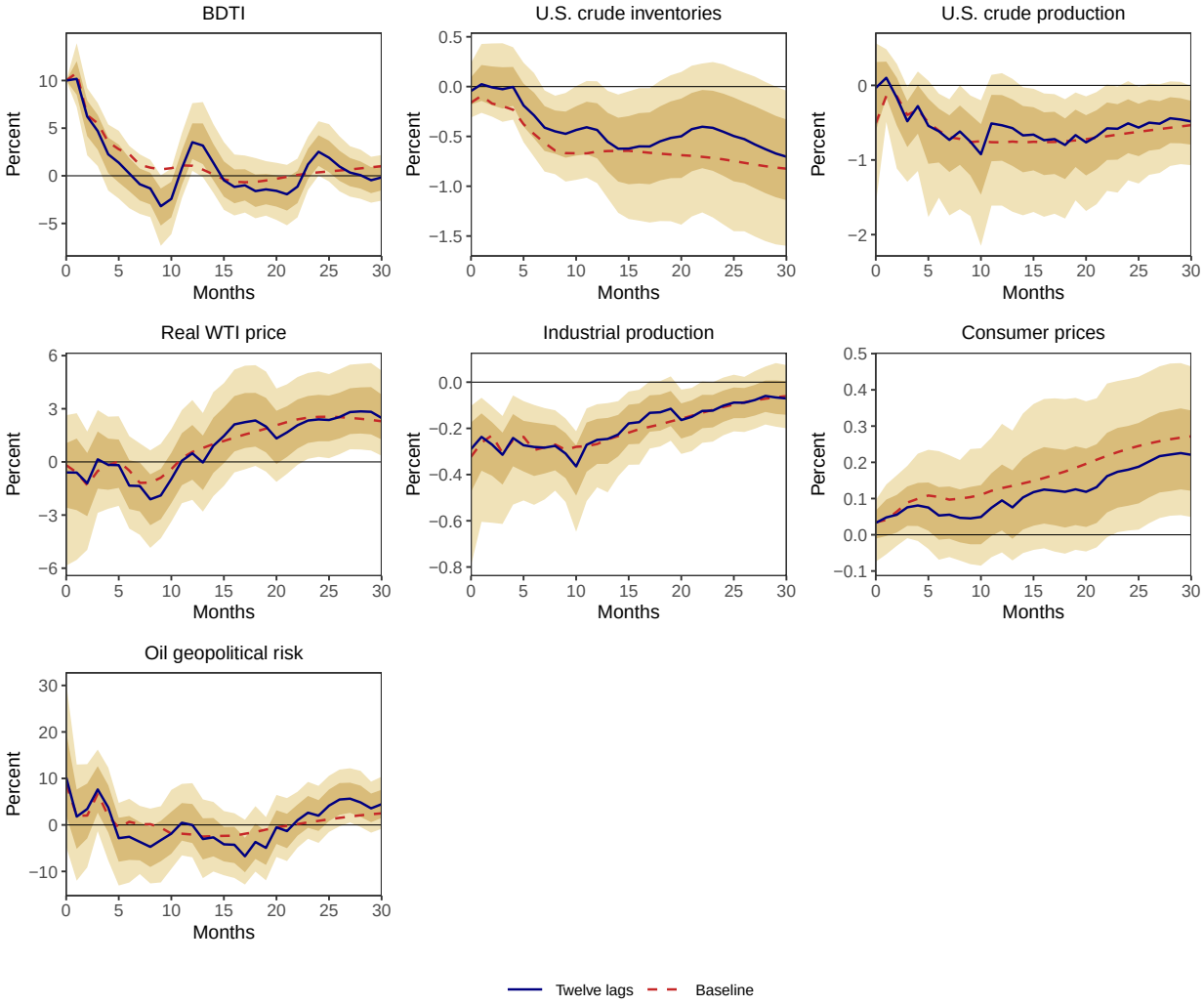
Figure E9: Sample sensitivity: starting in January 2016



Notes: Impulse responses estimated over January 2016–February 2026 to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The blue lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. The dashed red lines report the baseline proxy-SVAR responses.

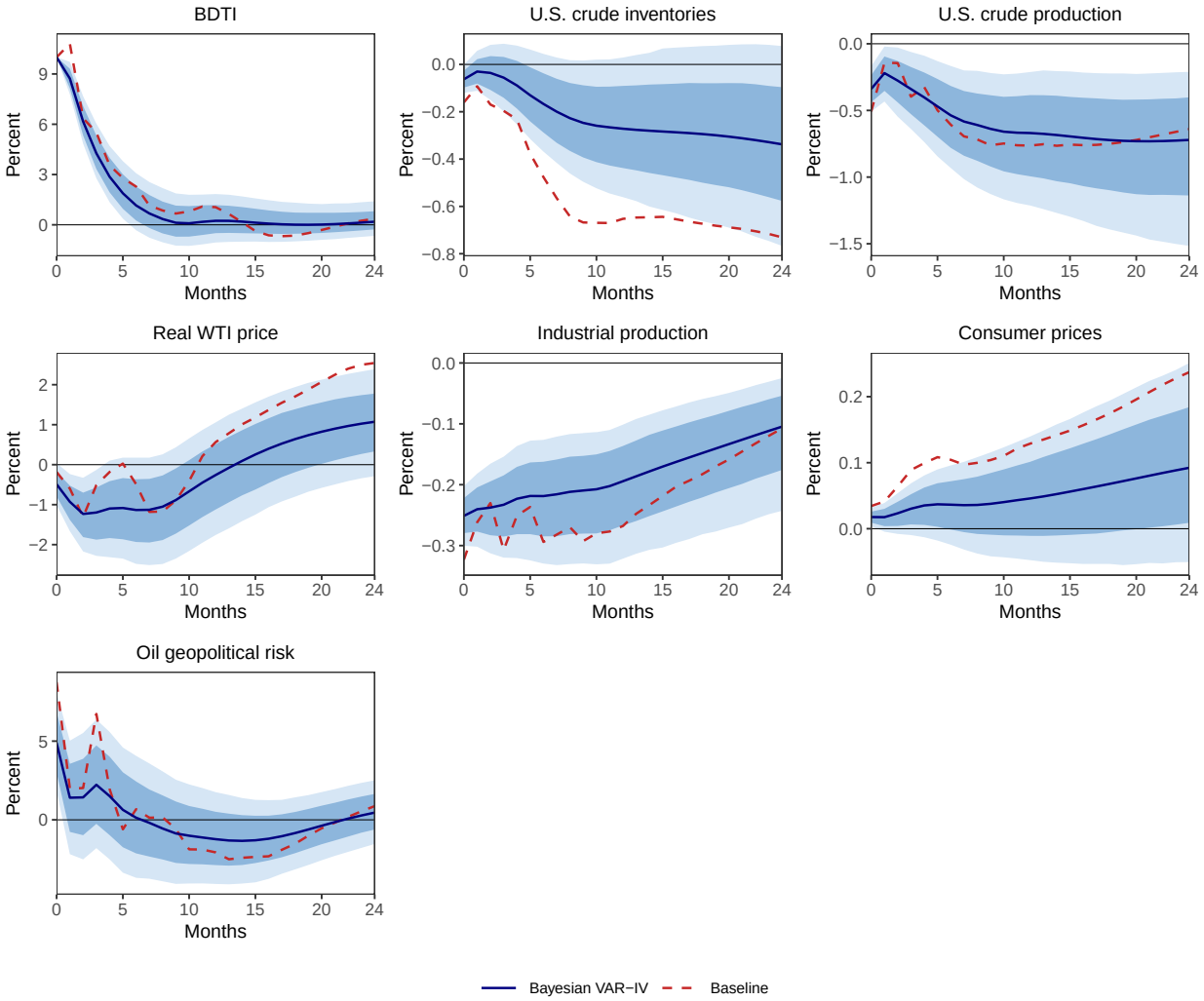
E.5 Model Specification

Figure E10: Sensitivity to the VAR lag order



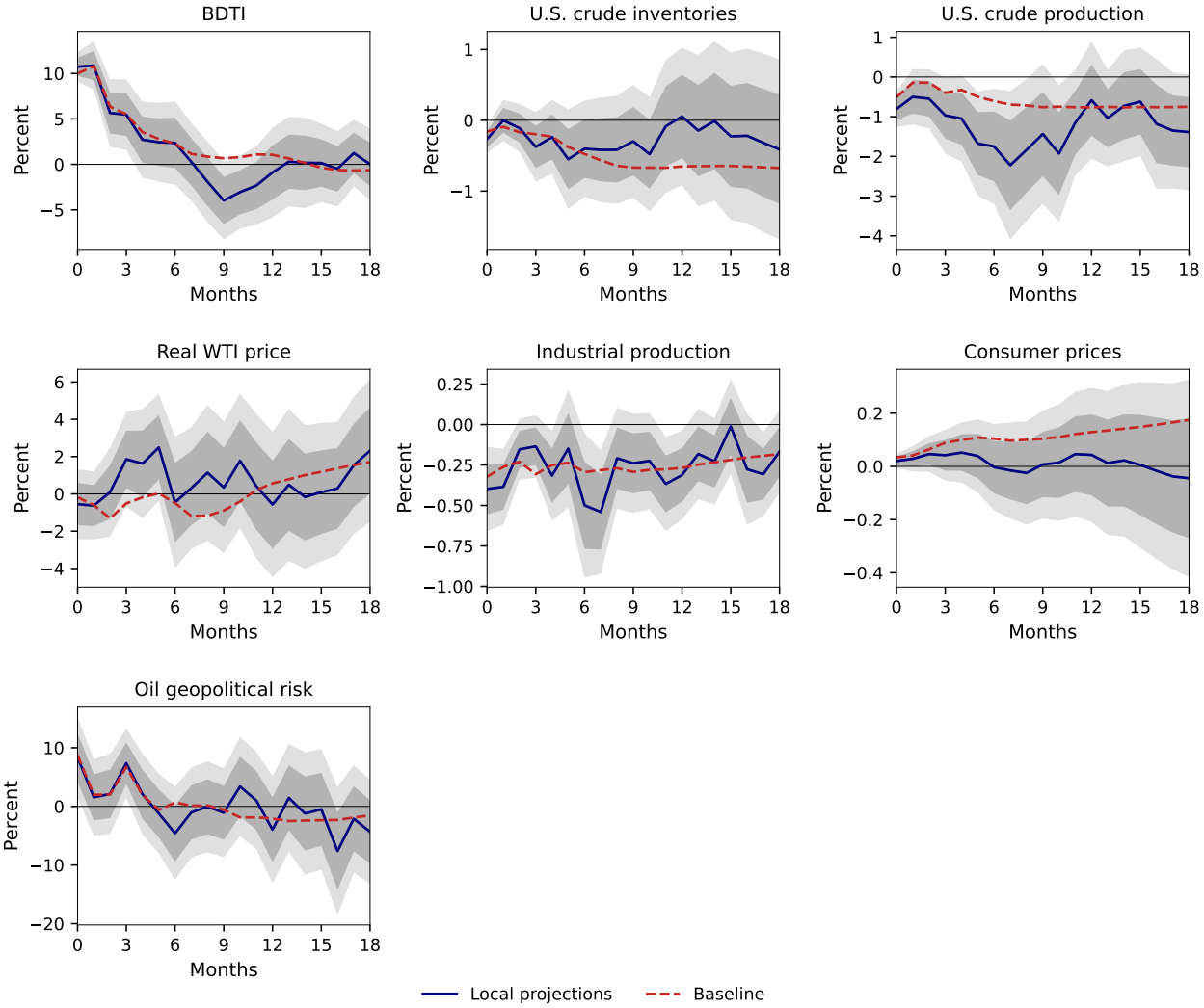
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid line reports the twelve-lag specification; the dashed red line reports the baseline. Dark and light shaded areas are 68 and 90 percent confidence bands for the twelve-lag specification, respectively.

Figure E11: Bayesian VAR-IV



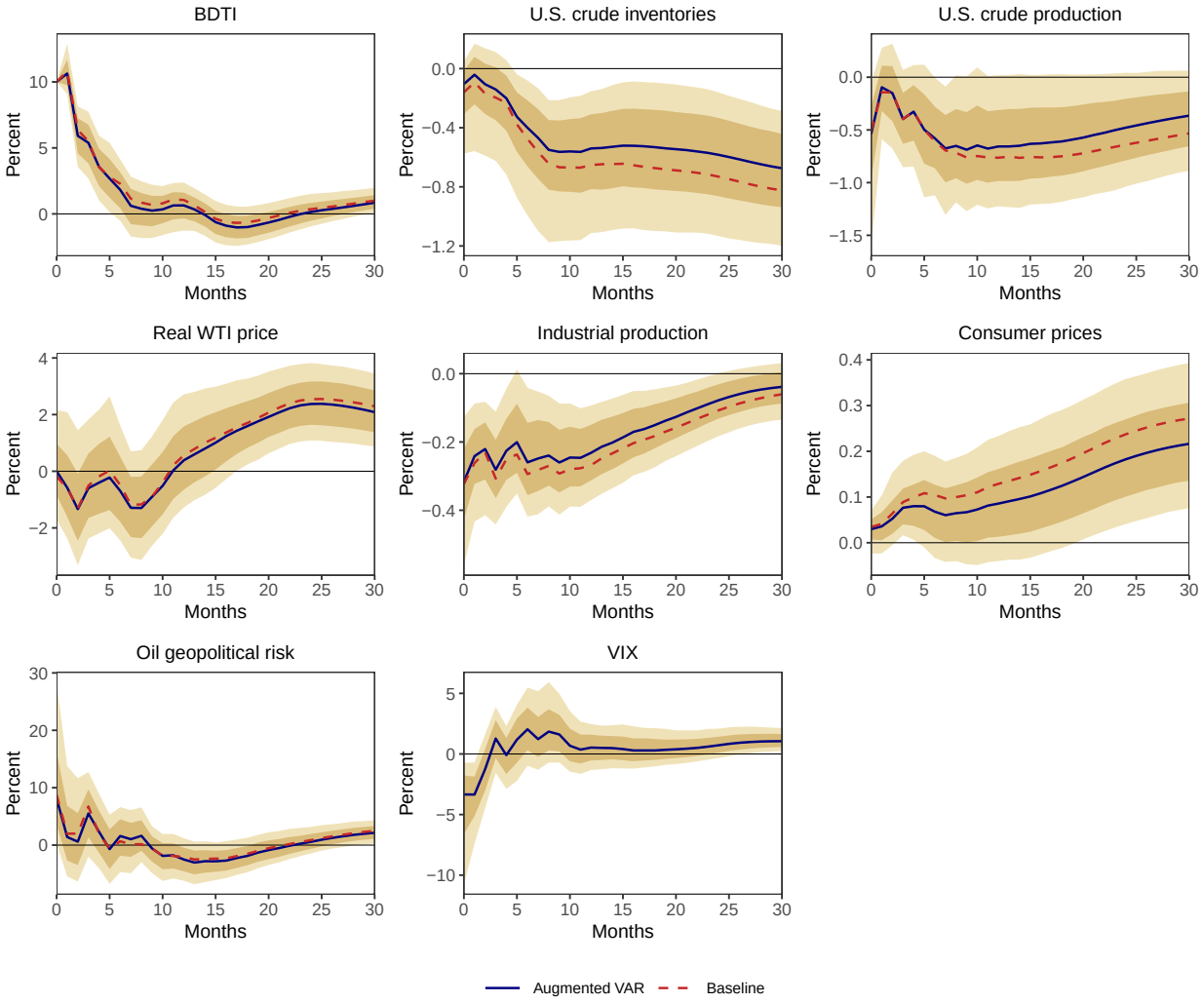
Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. Responses are estimated using a Bayesian VAR-IV. The solid blue lines are BVAR point estimates, and the dark and light blue shaded areas are 68 and 90 percent posterior credible bands, respectively. The dashed red lines report the baseline proxy-SVAR responses.

Figure E12: Local projections on the structural shock



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid line reports local projections of each variable at horizon h on the structural shock recovered from the baseline SVAR-IV, controlling for six lags of the dependent variable; each series is purged of a dummy for March–August 2020 and no trend is included. Dark and light shaded areas are 68 and 90 percent confidence bands based on heteroskedasticity-robust standard errors and normal critical values; inference treats the recovered shock as observed. The dashed red line reports the baseline SVAR-IV.

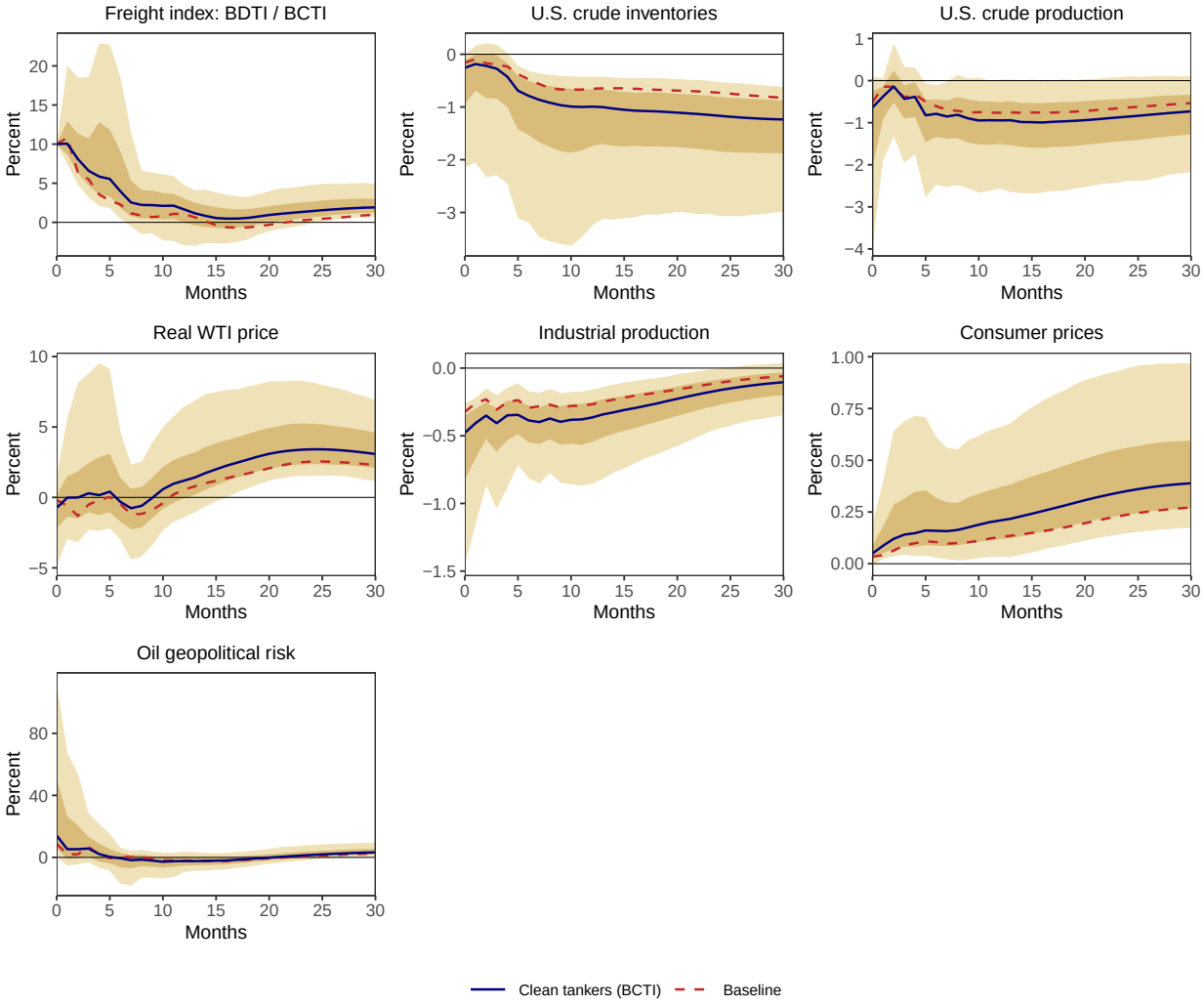
Figure E13: Augmenting the VAR with equity-market volatility



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid blue lines report the baseline VAR augmented with the VIX, retaining oil-related geopolitical risk. The dashed red lines report the baseline responses for its seven variables. Dark and light yellow shaded areas are 68 and 90 percent confidence bands for the augmented model, respectively. Responses are expressed as approximate percentage deviations.

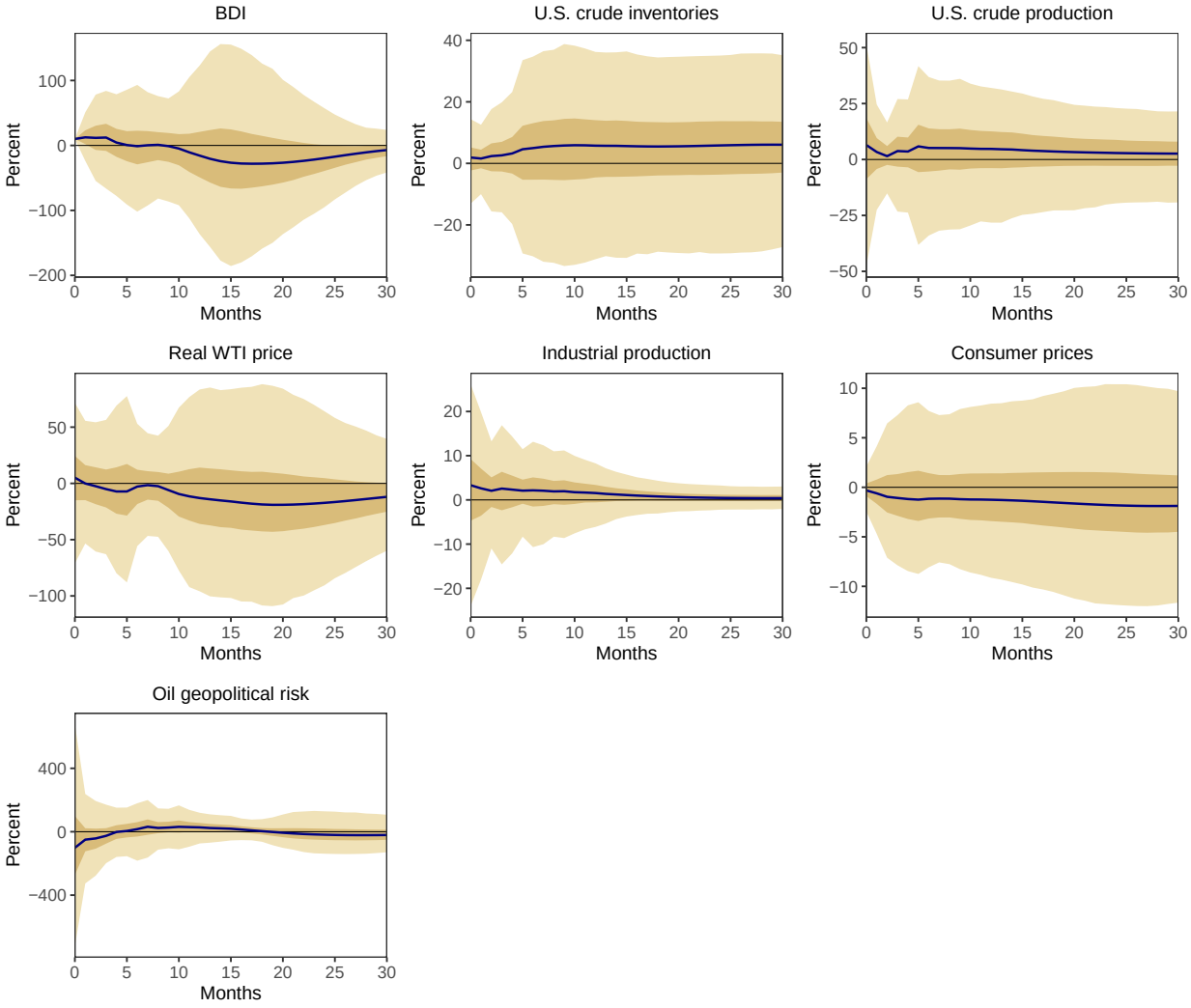
E.6 Alternative Freight Indices

Figure E14: Dirty versus clean tanker freight rates



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the respective freight index by 10 percent on impact. The solid blue line reports the specification using the BCTI; the dashed red line reports the baseline using the BDTI. Dark and light shaded areas are 68 and 90 percent confidence bands for the BCTI specification, respectively.

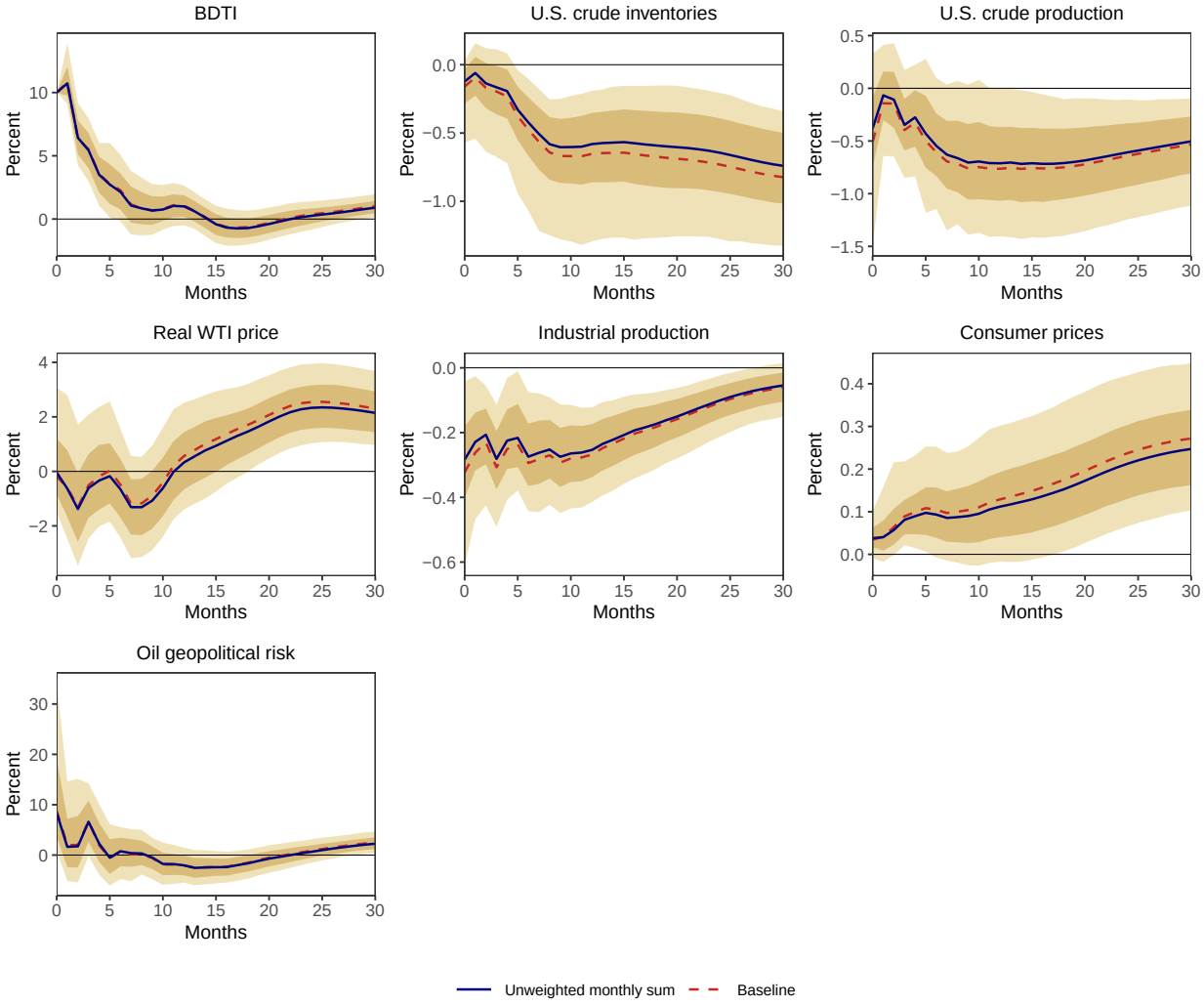
Figure E15: Replacing tanker freight with dry-bulk freight



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDI by 10 percent on impact. The BDI replaces the BDTI in the baseline VAR. The lines are point estimates, and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. The bands are not robust to weak instruments.

E.7 Monthly Aggregation

Figure E16: Sensitivity to monthly aggregation of the instrument



Notes: Impulse responses to an oil-shipping-cost shock, normalized to increase the BDTI by 10 percent on impact. The solid blue line uses the unweighted monthly sum of daily surprises; the dashed red line reports the baseline. Dark and light shaded areas are 68 and 90 percent confidence bands for the simple-sum specification, respectively.